

Topics in Quantitative Finance
Lecture Notes for the Graduate Course
Computational Methods in Finance, Code 21040

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Preface

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Part I

Discrete Financial Markets

Chapter 1

American and path dependent options

In this chapter we deal with the valuation of American and path-dependent options in a discrete-time framework. We use notations of [1] and refer to the same book for the basic notions of no-arbitrage evaluation. The discrete-time framework has the great advantage of allowing for a backward recursive formula for any American security. On the contrary, the continuous-time framework permits closed formulae for American derivatives in very few cases, such as perpetual call and put options or finite-maturity call options written on non-dividend paying stocks. Nevertheless the intuition we gain in the discrete-time framework is fundamental in order to understand the sophisticated numerical techniques used to price American options in continuous time.

1.1 Valuation of European Options

Here follows we consider a financial market that is complete and free of arbitrage opportunities. We denote by Q the unique risk neutral measure used to price derivatives and by $\mathcal{P}$ the information structure of investors (see [1] for the details). Due to market completeness, a European derivative X can be replicated by a self-financing strategy ϑ . The no-arbitrage price $V^X(t)$ of X and the value of the replicating strategy coincide and are given by

$$\frac{V^X(t)}{B(t)} = \frac{V_\vartheta(t)}{B(t)} = \mathbf{E}^Q \left[\frac{X(T)}{B(T)} \middle| \mathcal{P}_t \right]$$

for any $t = 0, \dots, T$.

In order to compute $V^X(0)$ we can either compute the expectation of $\frac{X(T)}{B(T)}$ under Q , or exploit the following *backward recursion in time*:

$$\frac{V^X(T)}{B(T)} = \frac{X(T)}{B(T)}$$

and for $t = T - 1, \dots, 0$

$$\frac{V^X(t)}{B(t)} = \mathbf{E}^Q \left[\frac{V^X(t+1)}{B(t+1)} \middle| \mathcal{P}_t \right]$$

The backward recursion exploits (and is equivalent to) the Q -martingality of the discounted value of the European derivative X . *Starting from the terminal value* $V^X(T) = X(T)$, we determine by backward induction $V^X(t)$ from the value of V^X at the step before, i.e. at $t + 1$, for $t = T - 1, \dots, 0$. This approach is precious when dealing with American options, because it can be generalized to account for the early exercise premium.

1.1.1 Computing the hedging strategy of European options

Suppose that X can be replicated with B and S_1 . For instance, the European option with terminal payoff:

$$X(T) = \left(\frac{1}{T} \sum_{t=1}^T S_1(t) - K \right)^+$$

is an arithmetic Asian option on S_1 that can be replicated with B and S_1 .

Since the replicating ϑ is self-financing,

$$\Delta V_\vartheta(t) = V_\vartheta(t+1) - V_\vartheta(t) = \vartheta_0(t)\Delta B(t) + \vartheta_1(t)\Delta S_1(t)$$

for $t = 0, \dots, T - 1$.

Denoting with a $\widetilde{(\cdot)}$ the discounted quantities, i.e.

$$\widetilde{S}_1(t) = \frac{S_1(t)}{B(t)}$$

and similarly for the other price processes, the variation of the discounted value of the self-financing replicating strategy is given by

$$\Delta \widetilde{V}_\vartheta(t) = \vartheta_1(t)\Delta \widetilde{S}_1(t)$$

for $t = 0, \dots, T - 1$.

The replicating strategy $\{\vartheta_0(t), \vartheta_1(t)\}$ can be computed by a backward recursion that involves the conditional covariance of the option value with the underlying S_1 .

The $\mathcal{P}_t$ -conditional covariance of two random variables Y and Z (under Q) is defined as

$$Cov^Q[Y, Z | \mathcal{P}_t] = \mathbf{E}^Q[Y \cdot Z | \mathcal{P}_t] - \mathbf{E}^Q[Y | \mathcal{P}_t] \cdot \mathbf{E}^Q[Z | \mathcal{P}_t]$$

(and therefore $\mathcal{P}_t$ -measurable random variables play the role of *constants* for $\mathcal{P}_t$ -conditional covariance).

The backward recursion for the replicating strategy is provided in the following proposition:

Proposition 1 *Suppose that the European option X is hedged by $\vartheta = (\vartheta_0, \vartheta_1)$. Then ϑ is determined by the following backward recursion:*

$$\tilde{V}^X(T) = \tilde{X}(T)$$

and for $t = T - 1, \dots, 0$

$$\begin{aligned} \tilde{V}^X(t) &= \mathbf{E}^Q \left[\tilde{V}^X(t+1) \middle| \mathcal{P}_t \right] \\ \vartheta_1(t) &= \frac{Cov^Q \left[\tilde{V}^X(t+1), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right]}{Var^Q \left[\Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right]} \\ \vartheta_0(t) &= \tilde{V}^X(t) - \vartheta_1(t) \tilde{S}_1(t) \end{aligned} \tag{1.1}$$

Proof. To show that (1.1) holds true, compute:

$$\begin{aligned} Cov^Q \left[\tilde{V}^X(t+1), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] &= Cov^Q \left[\tilde{V}_\vartheta(t+1), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] = \\ &= Cov^Q \left[\tilde{V}_\vartheta(t) + \Delta \tilde{V}_\vartheta(t), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] = \\ &= 0 + Cov^Q \left[\Delta \tilde{V}_\vartheta(t), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] = \\ &= Cov^Q \left[\vartheta_1(t) \Delta \tilde{S}_1(t), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] = \\ &= \vartheta_1(t) Cov^Q \left[\Delta \tilde{S}_1(t), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] \end{aligned}$$

Hence if the conditional variance of $\Delta\tilde{S}_1(t)$ is strictly positive we have

$$\vartheta_1(t) = \frac{Cov^Q \left[\tilde{V}^X(t+1), \Delta\tilde{S}_1(t) \middle| \mathcal{P}_t \right]}{Var^Q \left[\Delta\tilde{S}_1(t) \middle| \mathcal{P}_t \right]}$$

for $t = T-1, \dots, 0$ (Backward recursion for the hedging strategy). The number of units of the riskless bond $\vartheta_0(t)$ can be obtained for every $t = T-1, \dots, 0$ from the equality

$$\tilde{V}_\vartheta(t) = \tilde{V}^X(t) = \vartheta_0(t) + \vartheta_1(t)\tilde{S}_1(t)$$

as soon as $\vartheta_1(t)$ has been computed. Hence (1.1) is proved. ■

Remark 2 Observe that the coefficient $\vartheta_1(t)$ provided in (1.1), also called the Delta-greek of the option at time t , is the regression coefficient you obtain by regressing $\Delta\tilde{V}^X(t)$ against $\Delta\tilde{S}_1(t)$.

Remark 3 In the binomial framework the unique underlying asset $S = S_1$ has the following evolution

$$\begin{array}{rcl} t & & t+1 \\ & \nearrow & S_t u \quad \text{with prob } q = \frac{1+r-d}{u-d} \\ S_t & & \\ & \searrow & S_t d \quad \text{with prob } 1-q = \frac{u-(1+r)}{u-d} \end{array}$$

for any t . The coefficients $d < 1+r < u$ guarantee no-arbitrage and completeness of the market. The risk-neutral probability of an upwards movement of the underlying is $q = \frac{1+r-d}{u-d}$. Then the coefficient $\vartheta_1(t)$ provided in (1.1) becomes

$$\vartheta_1(t) = \frac{V^X(t+1, S_t u) - V^X(t+1, S_t d)}{S_t u - S_t d} \quad (1.2)$$

where $V^X(t+1, S_t u)$ is the time $t+1$ value of the derivative for $S(t+1) = S_t u$, and $V^X(t+1, S_t d)$ is the time $t+1$ value of the derivative for $S(t+1) = S_t d$.

Proof. To verify formula (1.2), we observe that (1.1) in the binomial

model can be rewritten as follows

$$\begin{aligned}
\vartheta_1(t) &= \frac{Cov^Q \left[\tilde{V}^X(t+1), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right]}{Var^Q \left[\Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right]} \\
&= \frac{Cov^Q \left[(1+r)^{-(t+1)} V^X(t+1), (1+r)^{-(t+1)} (S(t+1) - S(t)(1+r)) \middle| \mathcal{P}_t \right]}{Var^Q \left[(1+r)^{-(t+1)} (S(t+1) - S(t)(1+r)) \middle| \mathcal{P}_t \right]} \\
&= \frac{Cov^Q \left[V^X(t+1), (S(t+1) - S(t)(1+r)) \middle| \mathcal{P}_t \right]}{Var^Q \left[(S(t+1) - S(t)(1+r)) \middle| \mathcal{P}_t \right]} \\
&= \frac{\mathbb{E}_t^Q \left[V^X(t+1) \cdot (S(t+1) - S(t)(1+r)) \right] - \mathbb{E}_t^Q \left[V^X(t+1) \right] \cdot \mathbb{E}_t^Q \left[(S(t+1) - S(t)(1+r)) \right]}{Var^Q \left[(S(t+1) - S(t)(1+r)) \middle| \mathcal{P}_t \right]} \\
&= \frac{\mathbb{E}_t^Q \left[V^X(t+1) \cdot (S(t+1) - S(t)(1+r)) \right] - 0}{Var^Q \left[(S(t+1) - S(t)(1+r)) \middle| \mathcal{P}_t \right]} \text{ because } \mathbb{E}_t^Q \left[(S(t+1) - S(t)(1+r)) \right] = 0 \\
&= \frac{(V^X(t+1, S_t u) \cdot S_t (u - (1+r))) \cdot q + (V^X(t+1, S_t d) \cdot S_t (d - (1+r))) \cdot (1-q)}{(S_t (u - (1+r)))^2 \cdot q + (S_t (d - (1+r)))^2 \cdot (1-q)}
\end{aligned}$$

because $\mathbb{E}_t^Q \left[(S(t+1) - S(t)(1+r)) \right] = 0$, and thus the conditional variance at the denominator coincides with the second conditional moment of $S(t+1) - S(t)(1+r)$. Then, by substituting $(u - (1+r)) = (1-q)(u-d)$ and $d - (1+r) = -q(u-d)$ both in the numerator and in the denominator, we obtain

$$\begin{aligned}
\vartheta_1(t) &= \frac{(V^X(t+1, S_t u) \cdot S_t ((1-q)(u-d))) \cdot q + (V^X(t+1, S_t d) \cdot S_t (-q(u-d))) \cdot (1-q)}{(S_t ((1-q)(u-d)))^2 \cdot q + (S_t (-q(u-d)))^2 \cdot (1-q)} \\
&= \frac{(V^X(t+1, S_t u) \cdot S_t ((1-q)(u-d))) \cdot q + (V^X(t+1, S_t d) \cdot S_t (-q(u-d))) \cdot (1-q)}{(S_t ((1-q)(u-d)))^2 \cdot q + (S_t (q(u-d)))^2 \cdot (1-q)} \\
&= \frac{(1-q)(u-d)q \left[V^X(t+1, S_t u) + (V^X(t+1, S_t d) \cdot (-1)) \right]}{(1-q)(u-d)q \left[S_t (1-q)(u-d) + S_t (q(u-d)) \right]} \\
&= \frac{V^X(t+1, S_t u) + (V^X(t+1, S_t d) \cdot (-1))}{S_t (1-q)(u-d) + S_t q(u-d)} \\
&= \frac{V^X(t+1, S_t u) - V^X(t+1, S_t d)}{S_t (u-d) [(1-q) + q]}
\end{aligned}$$

which delivers the expression (1.2).

1.2 Valuation of American Options

Let X be an **American** option with maturity T . The option can be exercised *at every intermediate date*

$$t = 1, \dots, T - 1, T$$

If the holder exercises at the date τ , the time- t No-Arbitrage value of the discounted payoff he gets is

$$\mathbf{E}^Q \left[\frac{X(\tau)}{B(\tau)} \middle| \mathcal{P}_t \right]$$

If the option was European with maturity τ , this would be the value of the option at time t , that is the value of the holder position. By properly choosing the exercise time τ , the holder of the option can increase the time t value of his position. Hence, if the holder is smart enough, he will maximize his expected payoff by choosing the suitable τ . Hence the value of the American option at time t is

$$\tilde{V}(t) = \sup_{t \leq \tau \leq T} \mathbf{E}^Q \left[\frac{X(\tau)}{B(\tau)} \middle| \mathcal{P}_t \right] \quad (1.3)$$

where τ runs among all possible exercise policies, for $t = 0, \dots, T - 1$. At maturity, no time is left, and therefore $\tilde{V}(T) = \tilde{X}(T)$. The process $\tilde{V}$ is the fair price process of the American option. It is retrieved under the assumption that the holder of the option is able and willing to maximize the expected value of his position at any date during the life of the option. From an economic perspective, this is equivalent to assume that the holder is non-satiated. This hypothesis underlies the whole no-arbitrage theory in financial markets. However, there are circumstances in which an American option is not optimally early-exercised, due to practical reasons. In all these cases the writer of the option is allowed to *consume*, as we are going to see in the section 1.2.1.

As we have just seen, the selection of the exercise policy is crucial for American option and deserves some formal investigation. The holder selects the exercise time τ at time t exploiting the information he has at time t : hence the exercise time τ , that is random because it depends on the state of nature revealed at time t , must be adapted to the information structure.

Hence τ is a random variable, $\tau : \Omega \rightarrow T$ with the property that

$$\{\omega : \tau(\omega) = t\} \in \mathcal{P}_t$$

for every $t = 0, \dots, T$.

This property means that the holder decides to exercise the option at time t using the information $\mathcal{P}_t$.

A random variable τ with this property is called **stopping time** with respect to the information structure $\mathcal{P}$.

$\tilde{V}$ defined in (1.3) is called *Snell envelope* of the discounted payoff process $\tilde{X}$. The same definition extends also to the continuous-time framework¹.

The process $\tilde{V}$ defined in (1.3) has some important features that we list here follows (for details and proof see [5])

Properties of $\tilde{V}$ in (1.3) :

1. $\tilde{V}$ is a *super-martingale* under Q :

$$\tilde{V}(t_1) \geq \mathbf{E}^Q \left[\tilde{V}(t_2) \middle| \mathcal{P}_{t_1} \right] \quad (1.4)$$

for any $t_1 \leq t_2$.

This implies that $\tilde{V}$ is decreasing in mean (you may see this by simply taking the expectations from both sides of the inequality): this is so, because the early-exercise premium decreases, as time to maturity decreases. Property (1.4) has a clear financial interpretation: given $t_1 \leq t_2$, suppose that the exercise is forbidden from t_1 up to t_2 . In this case, the holder can exercise the option only at $t_2, t_2 + 1, \dots, T$. The discounted value at t_1 of the American option *with restricted exercise* is $\mathbf{E}^Q \left[\tilde{V}(t_2) \middle| \mathcal{P}_{t_1} \right]$. This discounted value is smaller than the discounted value of the original American option, where exercise is allowed at any date after t_1 . This is exactly what property (1.4) states.

2. $\tilde{V}$ is the *smallest* supermartingale greater or equal to the discounted payoff $\tilde{X}$.

This is important from the point of view of the writer, as we are going to see in few pages.

3. The optimal exercise policy is τ^* defined as

$$\tau^*(\omega) = \min \left\{ t : \tilde{V}(t)(\omega) = \tilde{X}(t)(\omega) \right\}$$

This means that the smart holder of the American option exercises as soon as the value of the American option equals the underlying payoff.

¹In the continuous-time framework the *supremum* is taken among a *continuum* of random variables and is therefore replaced by the *essential supremum* (see [3], for details).

Waiting in this case is not profitable and would decrease the value of the long position on the option.

4. The discounted value of the American option if optimally exercised is a Q -martingale:

$$\left\{ \tilde{V}((\tau^*(\omega) \wedge t)(\omega)) \right\}_{t=0}^T$$

(where $\tau^*(\omega) \wedge t = \min \{ \tau^*(\omega); t \}$) is a Q -martingale.

We conclude this section with the backward recursive formula for the value of the American option. A formal derivation of this formula can be found in [5]. We rely on its financial interpretation to justify it.

Proposition 4 (*Backward recursive formula for the American option*) *The discounted value of the American option $\tilde{V}$ defined in (1.3) is given by*

$$\tilde{V}(T) = \tilde{X}(T)$$

and

$$\tilde{V}(t) = \max \left\{ \tilde{X}(t); \mathbf{E}^Q \left[\tilde{V}(t+1) \middle| \mathcal{P}_t \right] \right\}$$

for $t = T-1, \dots, 0$

- $B(t) \cdot \mathbf{E}^Q \left[\tilde{V}(t+1) \middle| \mathcal{P}_t \right]$ is called *continuation value at time t* : it is the fair value in t of what the holder gains if he decides to wait till $t+1$.
- $X(t)$ is the *immediate payoff at time t* : what the holder gains if he exercises immediately.

The recursive formula is the simple application of the Law of One Price at maturity. At the intermediate dates the value of the option is the maximum of what you get if you exercise immediately (immediate payoff) and what you expect to get if you wait for the next period, based on the current information (continuation value). The comparison between immediate payoff and continuation value embeds step by step the early exercise premium in the value of the option V . Moreover, as one can see from the recursive formula, the discounted value of the American option is a Q -martingale in the continuation region (i.e. when early exercise is not optimal). This leads to the so called *free-boundary* formulation of the American pricing problem in the continuous-time framework.

1.2.1 Hedging the American option

Suppose (as in the European case) that the option can be hedged with B and S_1 .

The discounted value of the American option $\tilde{V}$ decreases in mean under Q (the EEP decreases as time to maturity decreases): hence a pure self-financing strategy is not suitable to replicate the option, because it provides zero intermediate cash-flows and its discounted value is a Q -martingale (with constant mean). On the contrary, an American option may (or not) produce intermediate cash-flows by early exercising and its discounted value is decreasing in mean. This suggests to allow for *money withdrawals* from self-financing strategies. More precisely:

Definition 5 A *portfolio-consumption pair* is given by a self-financing trading strategy $(\vartheta_0, \vartheta_1)$ and a consumption process C :

$$\phi = ((\vartheta_0, \vartheta_1), C)$$

The consumption $C(t)$ denotes the cumulative consumption in the interval $[0; t[$.

Therefore we assume that the discounted consumption

1. $0 = \tilde{C}(0) \leq \tilde{C}(1) \leq \dots \leq \tilde{C}(T-1)$
2. $\Delta\tilde{C}(t) = \tilde{C}(t+1) - \tilde{C}(t) =$ consumption in $[t; t+1[$ is measurable with respect to P_t (the writer decides at the beginning of the period how much to consume in the period)
3. The discounted value of the portfolio-consumption pair is the discounted value of the assets at time t

$$\tilde{V}_\phi(t) = \vartheta_0(t) + \vartheta_1(t)\tilde{S}_1(t) = \tilde{V}_\vartheta(t) \quad (1.5)$$

and its period-variation from t to $t+1$ is

$$\Delta\tilde{V}_\phi(t) = \vartheta_1(t)\Delta\tilde{S}_1(t) - \Delta\tilde{C}(t). \quad (1.6)$$

The last equality *extends* the usual budget constraint defining self-financing strategies from t to $t+1$ to the case of nonnegative possible consumption.

Equation (1.6) is indeed a *self-financing constraint after money withdrawal (if any)*. In fact, rearranging the terms we get

$$\begin{aligned}
\tilde{V}_\vartheta(t+1) - \tilde{V}_\vartheta(t) &= \tilde{V}_\phi(t+1) - \tilde{V}_\phi(t) \text{ by Eqn. (1.5)} \\
&= \Delta \tilde{V}_\phi(t) \\
&= \vartheta_1(t) \Delta \tilde{S}_1(t) - \Delta \tilde{C}(t) \text{ by Eqn. (1.6)} \\
&= \vartheta_1(t) \left(\tilde{S}_1(t+1) - \tilde{S}_1(t) \right) - \Delta \tilde{C}(t)
\end{aligned}$$

and therefore

$$\underbrace{\vartheta_0(t+1) + \vartheta_1(t+1) \tilde{S}_1(t+1)}_{\tilde{V}_\vartheta(t+1)} - \underbrace{\left(\vartheta_0(t) + \vartheta_1(t) \tilde{S}_1(t) \right)}_{\tilde{V}_\vartheta(t)} = \vartheta_1(t) \left(\tilde{S}_1(t+1) - \tilde{S}_1(t) \right) - \Delta \tilde{C}(t)$$

leading to

$$\underbrace{\vartheta_0(t+1) + \vartheta_1(t+1) \tilde{S}_1(t+1)}_{\tilde{V}_\vartheta(t+1)} = \underbrace{\vartheta_0(t) + \vartheta_1(t) \tilde{S}_1(t+1)}_{\text{disc. value at } t+1 \text{ of positions } \vartheta(t)} - \underbrace{\Delta \tilde{C}(t)}_{\text{disc. consumption from } t \text{ to } t+1}$$

As the reader can see, the previous equality reduces to the usual budget constraint for self-financing strategy when there is no consumption from t to $t+1$, i.e. for $\Delta \tilde{C}(t) = 0$.

A portfolio-consumption pair ϕ **super-replicates** the American contingent claim X if

$$V_\phi(t) \geq X(t)$$

for any $t = 1, \dots, T$.

The aim of the **writer** is to *find the super-replicating portfolio consumption pair ϕ^* whose cost is minimal*.

For attainable European contingent claims, we know that the value of any replicating strategy coincides with the risk-neutral expectation of the discounted terminal payoff. For American contingent claims, we are going to see that the *minimum cost* of its *super-replicating* portfolio consumption pair coincides with the *supremum over all exercise policies* of the risk-neutral expectation of the discounted (so exercised) payoff, i.e. the discounted value of the American option as defined in (1.3).

First of all, we observe that:

- for any ϕ , $\tilde{V}_\phi$ is a Q -supermartingale.

$$\begin{aligned} \text{In fact, } \mathbf{E}^Q \left[\Delta \tilde{V}_\phi(t) \middle| \mathcal{P}_t \right] &= \mathbf{E}^Q \left[\vartheta_1(t) \Delta \tilde{S}_1(t) - \Delta \tilde{C}(t) \middle| \mathcal{P}_t \right] = \\ &= \vartheta_1(t) \mathbf{E}^Q \left[\Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] - \mathbf{E}^Q \left[\Delta \tilde{C}(t) \middle| \mathcal{P}_t \right] = 0 - \Delta \tilde{C}(t) \leq 0 \end{aligned}$$

- Remember property 2 of $\tilde{V}$: $\tilde{V}$ is the *smallest* supermartingale greater or equal to the discounted payoff $\tilde{X}$

Hence, since by definition $\tilde{V}_\phi \geq \tilde{X}$, then the value of the super-replicating portfolio consumption pair ϕ^* whose cost is minimal must be equal to V :

$$\tilde{V}_{\phi^*}(t) = \tilde{V}(t) = \sup_{t \leq \tau \leq T} \mathbf{E}^Q \left[\frac{X(\tau)}{B(\tau)} \middle| \mathcal{P}_t \right].$$

Therefore the quantities:

$\tilde{V}_{\phi^*}(t)$ = the minimal discounted cost (for the writer) to hedge from time t to maturity the American option

and

$\tilde{V}(t)$ = the maximal discounted expected gain of the holder coincide!

Computing the hedging strategy for American options.

In this section we provide the backward recursion to compute the (super)replicating portfolio-consumption pair to hedge an American option written on S_1 . We use the *regression coefficient* employed also in the European case, that coincides with the *usual Delta* in the binomial case.

Proposition Suppose that the American option X is hedged by $\phi^* = (\vartheta^*, C^*)$ where $\vartheta^* = (\vartheta_0^*, \vartheta_1^*)$. Then ϕ^* is determined by the following backward recursion:

$$\tilde{V}(T) = \tilde{X}(T)$$

and for $t = T - 1, \dots, 0$

$$\begin{aligned} \tilde{V}(t) &= \max \left\{ \tilde{X}(t); \mathbf{E}^Q \left[\tilde{V}(t+1) \middle| \mathcal{P}_t \right] \right\} \\ \vartheta_1^*(t) &= \frac{\text{Cov}^Q \left[\tilde{V}(t+1), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right]}{\text{Var}^Q \left[\Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right]} \\ \vartheta_0^*(t) &= \tilde{V}(t) - \vartheta_1^*(t) \tilde{S}_1(t) \\ \Delta \tilde{C}^*(t) &= -\mathbf{E}^Q \left[\Delta \tilde{V}(t) \middle| \mathcal{P}_t \right] \end{aligned} \tag{1.7}$$

Proof. As for the European case we first compute

$$\begin{aligned}
Cov^Q \left[\tilde{V}(t+1), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] &= Cov^Q \left[\tilde{V}_{\phi^*}(t+1), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] = \\
&= Cov^Q \left[\tilde{V}_{\phi^*}(t) + \Delta \tilde{V}_{\phi^*}(t), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] = \\
&= 0 + Cov^Q \left[\Delta \tilde{V}_{\phi^*}(t), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] = \\
&= Cov^Q \left[\vartheta_1^*(t) \Delta \tilde{S}_1(t) - \Delta \tilde{C}^*(t), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] = \\
&= \vartheta_1^*(t) Cov^Q \left[\Delta \tilde{S}_1(t), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right] - 0
\end{aligned}$$

since $\Delta \tilde{C}^*(t)$ is $\mathcal{P}_t$ -measurable. Hence, if the conditional variance of $\Delta \tilde{S}_1(t)$ is strictly positive we have

$$\vartheta_1^*(t) = \frac{Cov^Q \left[\tilde{V}(t+1), \Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right]}{Var^Q \left[\Delta \tilde{S}_1(t) \middle| \mathcal{P}_t \right]}$$

for $t = T-1, \dots, 0$ (Backward recursion for the hedging strategy).

As in the European case, $\vartheta_0^*(t)$ can be obtained for every $t = T-1, \dots, 0$ from the equality

$$\tilde{V}_{\vartheta}(t) = \tilde{V}(t) = \vartheta_0^*(t) + \vartheta_1^*(t) \tilde{S}_1(t)$$

as soon as $\vartheta_1^*(t)$ has been computed.

The consumption $\tilde{C}^*$ can be determined from

$$\Delta \tilde{V}_{\phi^*}(t) = \vartheta_1^*(t) \Delta \tilde{S}_1(t) - \Delta \tilde{C}^*(t)$$

that implies

$$\begin{aligned}
\mathbf{E}^Q \left[\Delta \tilde{V}(t) \middle| \mathcal{P}_t \right] &= \mathbf{E}^Q \left[\Delta \tilde{V}_{\phi^*}(t) \middle| \mathcal{P}_t \right] = \mathbf{E}^Q \left[\vartheta_1^*(t) \Delta \tilde{S}_1(t) - \Delta \tilde{C}^*(t) \middle| \mathcal{P}_t \right] = \\
&= 0 - \mathbf{E}^Q \left[\Delta \tilde{C}^*(t) \middle| \mathcal{P}_t \right] = -\Delta \tilde{C}^*(t)
\end{aligned}$$

since $C^*(t+1) - C^*(t)$ is $\mathcal{P}_t$ -measurable. ■

If the option is optimally exercised, then no consumption is available for the writer. In fact, if the holder is smart, he does not exercise in the continuation region, where $\tilde{V}$ is a Q -martingale and where

$$\Delta \tilde{C}^*(t) = -\mathbf{E}^Q \left[\Delta \tilde{V}(t) \middle| \mathcal{P}_t \right]$$

is therefore equal to zero. Suppose now that early exercise becomes profitable at time t : in this case $\mathbf{E}^Q \left[\Delta \tilde{V}(t) \middle| \mathcal{P}_t \right] < 0$, but if the holder exercises, then the writer spends the whole amount $V(t)$ to hedge the position that is closed at t . The writer is therefore allowed to consume only when the holder does *not* exercise *optimally* the option. Such consumption corresponds to the early exercise premium that is lost by the holder in the period $[t; t+1[$. In fact, if the option is not optimally exercised at t , the writer needs $\vartheta_1^*(t)$ units of S as defined in (1.7) and

$$\vartheta_0(t) = \mathbf{E}^Q \left[\tilde{V}(t+1) \middle| \mathcal{P}_t \right] - \vartheta_1^*(t) \tilde{S}_1(t) < \tilde{V}(t) - \vartheta_1^*(t) \tilde{S}_1(t) = \vartheta_0^*(t)$$

units of B to hedge the option at the successive period. Hence the surplus of units of B

$$\begin{aligned} \vartheta_0^*(t) - \vartheta_0(t) &= \left(\tilde{V}(t) - \vartheta_1^*(t) \tilde{S}_1(t) \right) - \left(\mathbf{E}^Q \left[\tilde{V}(t+1) \middle| \mathcal{P}_t \right] - \vartheta_1^*(t) \tilde{S}_1(t) \right) \\ &= -\mathbf{E}^Q \left[\tilde{V}(t+1) - \tilde{V}(t) \middle| \mathcal{P}_t \right] = \Delta \tilde{C}^*(t) \end{aligned}$$

is available to the writer for consumption, if the option is not optimally exercised.

We conclude this section with a simple example to illustrate the backward pricing recursion, and the computation of the replicating strategy for an American put option. Consider a two-period binomial model with riskless interest rate $r = 10\%$, and a risky security S with initial value $S(0) = 10$. The risky security can have either an upwards movement with coefficient $u = 1.2$ or a downwards movement with coefficient $d = 0.8$. The risk-neutral probability of an upwards movement is therefore $q = \frac{1+r-d}{u-d} = 0.75$. Consider an American put option on S with maturity $T = 2$ and strike price $K = 10$. The immediate payoff process is $X(t) = (10 - S(t))^+$ for any $t = 0, 1, 2$. At maturity $T = 2$ the value of the American put option coincides with the terminal payoff, $V(2) = X(2) = (10 - S(2))^+$. At $t = 1$ we compute the *continuation value* $CV(1) = \mathbf{E}^Q \left[\frac{X(2)}{1+r} \middle| \mathcal{P}_1 \right]$, and we compare it with the immediate payoff $X(1)$ to obtain the value of the American put option $V(1)$. Early exercise is optimal at $t = 1$ if $S(1) = 8$ only, because in this case $V(1) = \max \{X(1), CV(1)\} = \max \{2, 1.09\} = 2 = X(1)$, as we can see from the tree for S, X, CV and V . At the initial date the *continuation value* $CV(0) = \mathbf{E}^Q \left[\frac{V(1)}{1+r} \right]$, and we compare it with the immediate payoff $X(0)$ to obtain the initial value of the American put option $V(0)$.

We now compute the portfolio-consumption pair to replicate the option.

At time $t = 1$, we select $\vartheta(1)$ such that $V_{\vartheta}(2) = X(2)$. This gives

$$\begin{bmatrix} \vartheta_0(1) \\ \vartheta_1(1) \end{bmatrix} = \begin{cases} \begin{bmatrix} 1.1^2 & 14.4 \\ 1.1^2 & 9.6 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.99174 \\ -8.3 \times 10^{-2} \end{bmatrix} & \text{if } S(1) = 12 \\ \begin{bmatrix} 1.1^2 & 9.6 \\ 1.1^2 & 6.4 \end{bmatrix}^{-1} \begin{bmatrix} 0.4 \\ 3.6 \end{bmatrix} = \begin{bmatrix} 8.2645 \\ -1.0 \end{bmatrix} & \text{if } S(1) = 8 \end{cases}$$

The cost of ϑ at $t = 1$ is

$$V_{\vartheta}(1) = \begin{cases} 0.99174 \cdot 1.1 - 8.3333 \times 10^{-2} \cdot 12 = 0.09 & \text{if } S(1) = 12 \\ 8.2645 \cdot 1.1 - 1.0 \cdot 8 = 1.09 & \text{if } S(1) = 8 \end{cases}$$

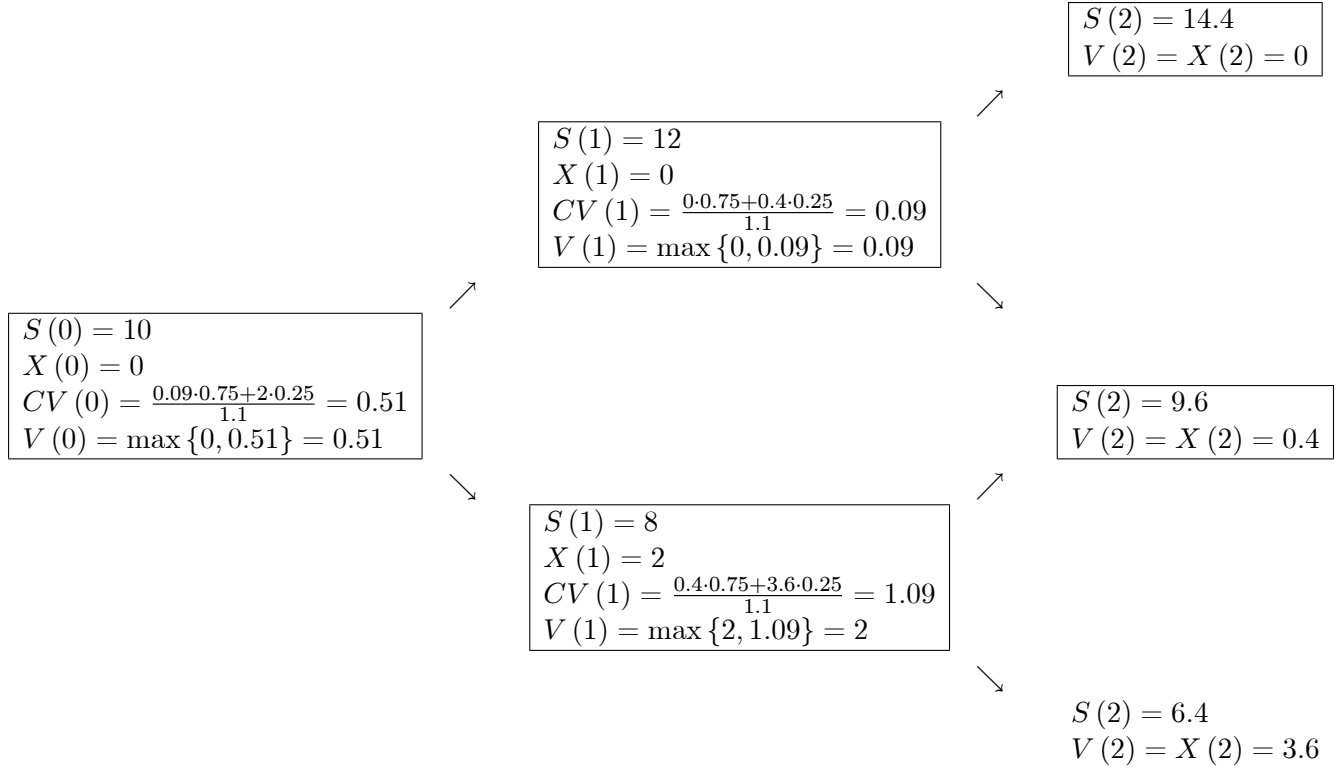
that coincides with the continuation value at $t = 1$, i.e. $V_{\vartheta}(1) = CV(1)$. Finally, at time $t = 0$, we select $\vartheta(0)$ such that the value of $\vartheta(0)$ at $t = 1$ equals $V(1)$. This gives

$$\begin{bmatrix} \vartheta_0(0) \\ \vartheta_1(0) \end{bmatrix} = \begin{bmatrix} 1.1 & 12 \\ 1.1 & 8 \end{bmatrix}^{-1} \begin{bmatrix} 0.09 \\ 2 \end{bmatrix} = \begin{bmatrix} 5.2909 \\ -0.4775 \end{bmatrix}$$

whose value at $t = 0$ is

$$V_{\vartheta}(0) = 5.2909 \cdot 1 - 0.4775 \cdot 10 = 0.51 = V(0).$$

*The put example:
Tree for S, X, CV and V*

 $t = 0$ $t = 1$ $T = 2$ 

When is the writer of the option allowed to consume? It depends on the exercise decision of the holder. The optimal exercise policy in this example is

$$\tau^* = \begin{cases} 2 & \text{if } S(1) = 12 \\ 1 & \text{if } S(1) = 8 \end{cases}$$

If this policy is adopted by the holder of the option, then the writer uses all the initial money to hedge his position. In fact, he enters in the strategy

$$\begin{bmatrix} \vartheta_0(0) \\ \vartheta_1(0) \end{bmatrix} = \begin{bmatrix} 5.2909 \\ -0.4775 \end{bmatrix}$$

at the initial date, at the cost $V_\vartheta(0) = 0.51$, which is equal to the initial option premium $V(0)$. At $t = 1$, if $S(1) = 12$ the $\vartheta(0)$ positions are worth 0.09 (by construction of $\vartheta(0)$ itself). The holder does not exercise, since

$\tau^* = 2$, and all this money is needed to enter in the new positions $\vartheta(1)$ to hedge the option at $T = 2$.

At $t = 1$, if $S(1) = 8$ the $\vartheta(0)$ positions are worth 2 (by construction of $\vartheta(0)$ itself). This amount is exactly equal to the immediate payoff $X(1) = 2$ that is optimally exercised by the holder $t = 1$, if $S(1) = 8$, since $\tau^* = 1$. Hence the contract is closed, and the writer does not have any future obligations, but no money withdrawal is possible from the original premium $V(0) = 0.51$.

On the contrary, if the option holder *does not optimally exercise* the option, and misses the early exercise opportunity at $t = 1$ when $S(1) = 8$, then the writer can consume, i.e. withdraw money from the replicating strategy. In fact, if the option is *not* exercised at $t = 1$ when $S(1) = 8$, then the writer needs only $V_{\vartheta}(1) = 1.09 = CV(1)$ to hedge the option at $T = 2$. But the writer was ready to pay at $t = 1$ the immediate payoff $X(1) = V(1) = 2$, that is the current value at $t = 1$ of $\vartheta(0)$. Therefore the writer withdraws $V(1) - CV(1) = 2 - 1.09 = 0.91$, liquidating the positions $\vartheta(0)$ at the value $V(1)$, and entering in the cheaper new positions $\vartheta(1)$, whose cost is $CV(1)$.

1.3 Path-dependent options and Markovianity

In order to be able to compute option prices and replicating strategies, we need to assume that security prices are Markovian.

A price process S is Markovian if for any bounded function φ

$$\mathbf{E}^Q[\varphi(S(t+1)) | \mathcal{P}_t] = \mathbf{E}^Q[\varphi(S(t+1)) | S(t)]$$

In words, the conditional expectation of $\varphi(S(t+1))$ given the information $\mathcal{P}_t$ coincides with the conditional expectation of $\varphi(S(t+1))$ given only the last value of S , $S(t)$: the past history of S entailed in $\mathcal{P}_t$ does not matter, and only $S(t)$ is relevant to determine $\mathbf{E}^Q[\varphi(S(t+1)) | \mathcal{P}_t]$.

Markovianity is an useful feature of a price process. It allows to price derivative securities, whose payoff depends only on the *current* underlying stock price, in a fast way. More precisely, suppose that the price process S_1 is Markovian (this is, for example, the case of a binomial model) and suppose you want to evaluate an option whose time t -payoff is

$$X(t) = \varphi(t, S_1(t))$$

This is, for instance, the case of call and put options (of American and European type).

Then, in the backward recursion (1.1) (or (1.7) if the option is American) every time you have to take conditional expectations or conditional covariances with respect to $\mathcal{P}_t$ you rely only on the value $S_1(t)$, disregarding the past. The recursion (1.7) can be restated, for instance, as follows:

$$\tilde{V}(T) = \tilde{X}(T)$$

and for $t = T - 1, \dots, 0$

$$\begin{aligned} \tilde{V}(t) &= \max \left\{ \tilde{X}(t); \mathbf{E}^Q \left[\tilde{V}(t+1) \middle| S_1(t) \right] \right\} \\ \vartheta_1^*(t) &= \frac{\text{Cov}^Q \left[\tilde{V}(t+1), \Delta \tilde{S}_1(t) \middle| S_1(t) \right]}{\text{Var}^Q \left[\Delta \tilde{S}_1(t) \middle| S_1(t) \right]} \\ \vartheta_0^*(t) &= \tilde{V}(t) - \vartheta_1^*(t) \tilde{S}_1(t) \\ \Delta \tilde{C}^*(t) &= -\mathbf{E}^Q \left[\Delta \tilde{V}(t) \middle| S_1(t) \right] \end{aligned} \quad (1.8)$$

where we simply replaced $\mathcal{P}_t$ with $S_1(t)$.

Hence, instead of computing the entire information structure for a price process S_1 , we can compute only the *tree* that describes the evolution of S_1 , and derive from this tree all the information we need to evaluate and to hedge the option on S_1 . In a binomial model, this is a great simplification: suppose that $T = 150$ (this is the usual number of periods you get when you discretize continuous-time price processes). If you have to compute the entire information structure, at time T you get 2^T cells: a tremendous number. If you just construct the recombining tree for S_1 , you have at maturity $T + 1$ different elements, that you can manage using an excel worksheet without any trouble.

What happens if the payoff process is *path-dependent*? Consider, for example, a lookback American put option, whose payoff at time t is

$$X(t) = \max_{0 \leq s \leq t} S_1(s) - S_1(t)$$

The strike of this put option is $\max_{0 \leq s \leq t} S_1(s)$, that is the running maximum of S_1 .

No difficulty arises in considering the entire information structure $\mathcal{P}$ and you can use the backward recursion (1.7) to compute everything you need. Notice that we cannot use the simplified recursion (1.8) on the tree of the price process S_1 , since the value of the running maximum of S_1 is missing in the simplified tree.

The solution in this case is to *Markovianize the problem*.

Introduce another state variable. Here we need the running maximum, so that we define for every t

$$M(t) = \max_{0 \leq s \leq t} S_1(s)$$

Construct now the *tree for S and M* . We observe that

$$\begin{aligned} M(t+1) &= \max_{0 \leq s \leq t+1} S_1(s) = \max \left\{ \max_{0 \leq s \leq t} S_1(s); S_1(t+1) \right\} = \\ &= \max \{M(t); S_1(t+1)\} \end{aligned}$$

The pair (S_1, M) is a Markovian process. In fact, for any bounded function (of two variables) φ we have that

$$\begin{aligned} \mathbf{E}^Q[\varphi(S_1(t+1), M(t+1)) | \mathcal{P}_t] &= \mathbf{E}^Q[\varphi(S_1(t+1), \max \{M(t); S_1(t+1)\}) | \mathcal{P}_t] = \\ &= \mathbf{E}^Q[\varphi(S_1(t+1), \max \{M(t); S_1(t+1)\}) | (S_1(t); M(t))] \end{aligned}$$

since $M(t+1)$ is known as soon as you remember $M(t)$ and $S_1(t+1)$, whose dependence on $\mathcal{P}_t$ can be reduced to the conditioning with respect to $S_1(t)$.

With this trick, the American lookback put option of the example can be evaluated by constructing *a tree in which you collect only the current values of S_1 and M* . The tree is not recombining, because of M . To avoid the exponential growth of the number of possible pairs (S_1, M) at time t , that is the number of elements of the tree at time t , you have to set

$$d = \frac{1}{u}$$

This assumption is common whenever you use binomial trees for asset pricing. In this case, it can be shown that the number of different nodes (i.e. the different pairs (S_1, M)) at time t grows as $\frac{t^2}{4}$.

Though the computational complexity implied by the quadratic number of nodes is not dramatic, there are various approximation methods that have been studied in order to squeeze the computational time, preserving numerical accuracy. For instance, you can attach at any node at time t of the binomial tree of $S = S_1$ an *auxiliary vector* of representative running maxima and compute the price of the option for any representative running maximum. A suitable interpolation of the components of the vector obtained in this way gives the approximated price of the lookback option. Such a technique, called *forward shooting grid approach*, can be applied also for

Asian options and is quite effective. The interested reader is addressed to the references quoted in the last section of this chapter. In the following section we explain the main features of the forward shooting grid method applied to the American lookback option we previously mentioned.

1.3.1 The forward shooting grid method

In this section we explain the main steps to price an American lookback option by means of the forward shooting grid approach. The immediate payoff at any date t is $X(t) = M(t) - S(t)$. Suppose you approximate S by means of a binomial tree. At time t you have $t + 1$ possible values for $S(t)$. At each node of $S(t)$ you associate a vector of *auxiliary representative* running maxima $\widehat{M}(t)$. Complete the tree from time $t = 0$ up to maturity T . This is the *forward-part* of the procedure. Then you start the *backward* procedure to price the option. At T for any possible $S(T)$ you have a vector of representative running maxima $\widehat{M}(T)$ and, consequently, a vector of representative terminal payoffs that are the prices at maturity of your option. Hence compute for any possible $S(T)$ the vector of representative prices of the option $\widehat{V}(T) = \widehat{M}(T) - S(T)$.

$$S(T) \rightarrow \underbrace{\widehat{M}(T) = [M_1 \quad \dots \quad M_K]}_{\text{Representative running maxima for } S(T)} \rightarrow \underbrace{\widehat{V}(T) = [M_1 - S(T) \quad \dots \quad M_K - S(T)]}_{\text{Terminal representative option prices for } S(T)}$$

Then for $t = T - 1, \dots, 2, 1, 0$ compute the vector of representative option prices $\widehat{V}$ as follows.

First, compute the vector of immediate payoffs associated to any node $S(t)$ for the tree at date t :

$$S(t) \rightarrow \underbrace{\widehat{M}(t) = [M_1 \quad \dots \quad M_K]}_{\text{Representative running maxima for } S(t)} \rightarrow \underbrace{\widehat{X}(t) = [M_1 - S(t) \quad \dots \quad M_K - S(t)]}_{\text{Representative immediate payoffs for } S(t)}$$

Then, compute the continuation value of the option at the node $S(t)$ as follows. Given the underlying price $S(t)$ at time t , at time $t + 1$ the underlying can be either $S(t + 1) = S(t) \cdot u$ with risk-neutral probability $q = \frac{1+r-d}{u-d}$ or $S(t + 1) = S(t) \cdot d$ with risk-neutral probability $1 - q$. As a consequence, the updated running maximum in case of an upwards movement is $M(t + 1) = \max\{M(t); S(t) \cdot u\}$. The updated running maximum in case of a downwards movement is $M(t + 1) = M(t)$. Pick the *corresponding* representative price for $\widehat{V}(t + 1)$. The corresponding price is the one you find for the representative running maximum $\widehat{M}(t + 1)$ given the underlying

price $S(t+1)$. If there is no perfect match between the updated $M(t+1)$ and any component of $\widehat{M}(t+1)$, the representative vector of running maxima associated to $S(t+1)$, then a *selection procedure* for $\widehat{V}(t+1)$ has to be defined. For instance, a selection procedure can:

1. select the representative running maximum $\widehat{M}(t+1)$ which is *closest* to the updated running maximum $M(t+1)$, and pick the option price $\widehat{V}(t+1)$ corresponding to $\widehat{M}(t+1)$.
2. Or find the two representative running maxima $\widehat{M}(t+1)$ that bound $M(t+1)$ by above and by below, and *interpolate* the corresponding option prices $\widehat{V}(t+1)$.

The discounted risk-neutral expectation of $\widehat{V}(t+1)$ conditional to $S(t), M(t)$ gives the continuation value of the option for $S(t), M(t)$. Take the maximum with the immediate payoff, and the *price of the option* is computed for $S(t)$ and any value of the representative $\widehat{M}(t)$.

Clearly this price depends on the choice of the values of the auxiliary variable $M(t)$ in the forward part of the algorithm, and on the selection procedure for $\widehat{V}$ in the backward part of the algorithm. The more exotic the payoff, the more difficult the choice of a proper algorithm both for auxiliary variable M in the forward part, and for the selection procedure for $\widehat{V}$ in the backward part of the algorithm. The papers quoted in the next section address these issues.

Table 1 visualizes the backward procedure just described, when the selection procedure for $\widehat{V}$ picks the *closest* value of M , as described in previous Point 1. More in detail, given $S(t)$ at date t , consider the component M_* of the representative running maximum $M(t)$. Compute the consequent running maximum at $t+1$. In case of an *upwards* movement $M(t+1) = \max\{M_*; S(t) \cdot u\}$. Denote with M'_* the component of $\widehat{M}(t+1)$, the representative running maxima at $t+1$ for $S(t+1) = S(t) \cdot u$, that is closest to $M(t+1) = \max\{M_*; S(t) \cdot u\}$. Let V'_* be the component of the representative option price $\widehat{V}(t+1)$ at $t+1$ for $S(t+1) = S(t) \cdot u$ corresponding to $\widehat{M}(t+1) = M'_*$.

In case of a *downwards* movement of S between t and $t+1$, given the time- t representative running maximum M_* , the consequent running maximum at $t+1$ is $M(t+1) = \max\{M_*; S(t) \cdot d\} = M_*$. Denote with M''_* the component of $\widehat{M}(t+1)$, the representative running maxima at $t+1$ for $S(t+1) = S(t) \cdot d$, that is closest to $M(t+1) = \max\{M_*; S(t) \cdot d\} = M_*$.

Let V''_* be the component of the representative option price $\widehat{V}(t+1)$ at $t+1$ for $S(t+1) = S(t) \cdot d$ corresponding to $\widehat{M}(t+1) = M''_*$.

Then at time t , for the node $S(t)$ and for the representative running maximum M_* , the continuation value of the option is

$$CV'_* = \frac{qV'_* + (1-q)V''_*}{1+r}$$

The value of the American option at time t , for the node $S(t)$ and for the representative running maximum M_* is therefore

$$V_* = \max \left\{ \underbrace{M_* - S(t)}_{\text{Immediate payoff for } S(t), M_*} ; \underbrace{\frac{qV'_* + (1-q)V''_*}{1+r}}_{\text{Continuation value for } S(t), M_*} \right\}$$

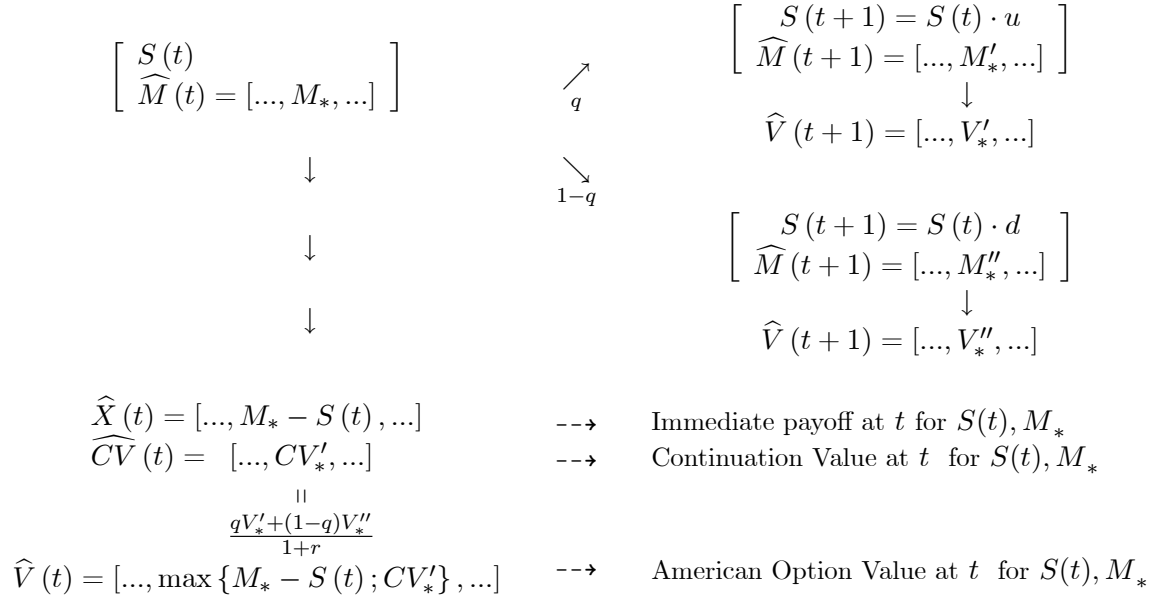
By letting M_* vary among all the possible components of the representative running maximum $\widehat{M}(t)$ associated to $S(t)$, we construct the vector of *representative* option prices $\widehat{V}(t)$ associated to $S(t)$

$$\begin{array}{c} S(t), \widehat{M}(t) = [M_1 \quad \dots \quad M_K] \\ \downarrow \\ \widehat{V}(t) = [V_1 \quad \dots \quad V_K] \end{array}$$

The backward procedure at time $t = 0$ delivers a unique price for the option because the entries of the vector $\widehat{M}(0)$ of representative running maxima at $t = 0$ coincide with the unique value $\widehat{M}(0) = S(0)$

$$S(0), \widehat{M}(0) = [S(0) \quad \dots \quad S(0)]$$

To reduce the computational cost of the implementation, one can employ vectors of representative running maximum whose dimension is increasing with respect to time, starting with single-entry vector for $\widehat{M}(0) = S(0)$.

Table 1: from $t + 1$ to t in the FSG pricing backward recursion

The general idea of the two-step procedure behind the forward shooting grid method to price American path-dependent options is now clear. What has still to be specified is what kind of vector of *auxiliary representative* running maxima $\widehat{M}(t)$ you associate to $S(t)$. The choice depends on the payoff of the derivative you want to price, and it is not unique at all. We provide here a very simple example, to illustrate one of the possibilities you have for the lookback option.

Suppose for sake of simplicity that $d = u^{-1}$, the maturity of the option is $T = 8$, and the maximum dimension for the vector of *auxiliary representative* running maxima $\widehat{M}(t)$ is 4 for any $t = 0, \dots, 8$. In the binomial model, the generic value of the underlying S at t is

$$S(t) = S_0 u^k d^{t-k}, \text{ for any } k = 0, \dots, t.$$

Given $S(t) = S_0 u^k d^{t-k} = S_0 u^{2k-t}$, the *highest possible value for the running maximum at t* is $S_0 u^k$, and the *lowest possible value for the running maximum at t* is $\max(S(t), S_0) = \max(S_0 u^k d^{t-k}, S_0)$. Consider the non-trivial case $k > 0$. If $2k - t > 0$, the lowest possible value for the running maximum at t is the current value of the underlying, i.e. $\max(S_0 u^k d^{t-k}, S_0) = S_0 u^k d^{t-k}$. If $2k - t \leq 0$, the lowest possible value for the running maximum

at t is the initial value of the underlying S_0 . We use the two extreme values for the running maximum to fill the extreme entries of the 4-dimensional vector $\widehat{M}(t)$, that is

$$S(t) = S_0 u^k d^{t-k} \mapsto \widehat{M}(t) = \begin{bmatrix} \max(S_0 u^k d^{t-k}, S_0) & \dots & \dots & S_0 u^k \end{bmatrix}.$$

How can we fill the two remaining central entries of $\widehat{M}(t)$? Consider the node $t = T = 8$, and $k = 5$. In this case $S(t) = S(8) = S_0 u^5 d^3 = S_0 u^2$, and the lowest (resp. highest) possible value for the running maximum is $S_0 u^2$ (resp. $S_0 u^5$). The (real) running maximum assumes therefore any value in $\{S_0 u^2, S_0 u^3, S_0 u^4, S_0 u^5\}$, depending on the paths followed by S to reach $S_0 u^5 d^3$ at $T = 8$. Hence we associate to $S(8) = S_0 u^5 d^3 = S_0 u^2$ the 4-dimensional vector of running maxima $\widehat{M}(8) = \begin{bmatrix} S_0 u^2 & S_0 u^3 & S_0 u^4 & S_0 u^5 \end{bmatrix}$.

Consider now the node $t = T = 8$, and $k = 4$. In this case $S(t) = S(8) = S_0 u^4 d^4 = S_0$, and the lowest (resp. highest) possible value for the running maximum is S_0 (resp. $S_0 u^4$). The (real) running maximum assumes therefore any value in $\{S_0, S_0 u^1, S_0 u^2, S_0 u^3, S_0 u^4\}$, depending on the paths followed by S to reach $S_0 u^4 d^4 = S_0$ at $T = 8$. To represent the 5 possible values of the running maximum with a vector of only 4 entries we can take for instance

$$S(8) = S_0 \mapsto \widehat{M}(8) = \begin{bmatrix} S_0 & S_0 u^{\frac{4}{3}} & S_0 u^{\frac{8}{3}} & S_0 u^4 \end{bmatrix}.$$

We illustrate now the backward pricing procedure in the one-period sub-market starting at the node $t = 7$, $S(7) = S_0 u^4 d^3 = S_0 u$, and finishing at $T = 8$ with the two terminal nodes just described, i.e. $S(8) = S_0 u^5 d^3$ and $S(8) = S_0 u^4 d^4$.

$$\begin{array}{c} \nearrow_q \left[\widehat{M}(8) = \begin{bmatrix} S(8) = S_0 u^5 d^3 \\ S_0 u^2 & S_0 u^3 & S_0 u^4 & S_0 u^5 \end{bmatrix} \right] \\ \left[\begin{array}{l} S(7) = S_0 u^4 d^3 = S_0 u \\ \widehat{M}(7) = [S_0 u, S_0 u^2, S_0 u^3, S_0 u^4] \end{array} \right] \\ \searrow_{1-q} \left[\widehat{M}(8) = \begin{bmatrix} S(8) = S_0 u^4 d^4 \\ S_0 & S_0 u^{\frac{4}{3}} & S_0 u^{\frac{8}{3}} & S_0 u^4 \end{bmatrix} \right] \end{array}$$

At the terminal date $T = 8$, the value of the American option is its terminal payoff. Hence we have the following vectors $\widehat{V}(8)$ of option prices

associated to the two terminal nodes:

$$S(8) = S_0 u^5 d^3 = S_0 u^2, \widehat{M}(8) = \begin{bmatrix} S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \\ S_0 u^5 \end{bmatrix} \mapsto \widehat{V}(8) = \begin{bmatrix} S_0 u^2 - S_0 u^2 \\ S_0 u^3 - S_0 u^2 \\ S_0 u^4 - S_0 u^2 \\ S_0 u^5 - S_0 u^2 \end{bmatrix}$$

and

$$S(8) = S_0 u^4 d^4 = S_0, \widehat{M}(8) = \begin{bmatrix} S_0 u \\ S_0 u^{\frac{4}{3}} \\ S_0 u^{\frac{8}{3}} \\ S_0 u^4 \end{bmatrix} \mapsto \widehat{V}(8) = \begin{bmatrix} S_0 u - S_0 \\ S_0 u^{\frac{4}{3}} - S_0 \\ S_0 u^{\frac{8}{3}} - S_0 \\ S_0 u^4 - S_0 \end{bmatrix}.$$

Now we focus on $S(7) = S_0 u^4 d^3 = S_0 u$ at $t = 7$. The immediate payoff of the American option is

$$X(7) = \widehat{M}(7) - S(7) = \begin{bmatrix} S_0 u - S_0 u \\ S_0 u^2 - S_0 u \\ S_0 u^3 - S_0 u \\ S_0 u^4 - S_0 u \end{bmatrix}.$$

We have to compare the immediate payoff with the continuation value. We first have to find which is the proper successor for each value of the pair $(S(7), \widehat{M}(7))$. For $S(7) = S_0 u^4 d^3 = S_0 u$, in case of an upwards movement, the updating rule applied to each component of $\widehat{M}(7)$ maps $\widehat{M}(7)$ into $M(8) = \max\{\widehat{M}(7); S(8)\}$

$$\widehat{M}(7) = \begin{bmatrix} S_0 u \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix} \xrightarrow{M(8)=\max\{\widehat{M}(7); S(8)\}} M(8) = \begin{bmatrix} S_0 u^2 \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix}.$$

All the entries of $M(8) = [S_0 u^2 \ S_0 u^2 \ S_0 u^3 \ S_0 u^4]$ can be found among the entries of $\widehat{M}(8) = [S_0 u^2 \ S_0 u^3 \ S_0 u^4 \ S_0 u^5]$, associated to $S(8) = S_0 u^5 d^3$, with a perfect match, because

$$M(8) = \begin{bmatrix} S_0 u^2 \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix} \begin{matrix} \rightarrow \\ \nearrow \\ \nearrow \\ \nearrow \end{matrix} \begin{bmatrix} S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \\ S_0 u^5 \end{bmatrix} = \widehat{M}(8)$$

Therefore, in case of an upwards movement starting from $S(7) = S_0 u^4 d^3$, $\widehat{M}(7) = [S_0 u \ S_0 u^2 \ S_0 u^3 \ S_0 u^4]$ the values for the American option at $T = 8$ are

$$M(7) = \begin{bmatrix} S_0 u \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix} \xrightarrow[\text{up}]{\text{with prob. } q} \begin{matrix} \nearrow \\ \nearrow \\ \nearrow \end{matrix} \begin{bmatrix} S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \\ S_0 u^5 \end{bmatrix} \mapsto \begin{bmatrix} S_0 u^2 - S_0 u^2 \\ S_0 u^3 - S_0 u^2 \\ S_0 u^4 - S_0 u^2 \end{bmatrix}$$

$\widehat{M}(8)$ $\widehat{V}(8)$

so that

$$M(7) = \begin{bmatrix} S_0 u \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix} \xrightarrow[\text{up}]{\text{with prob. } q} \begin{matrix} \rightarrow & S_0 u^2 & \rightarrow & S_0 u^2 - S_0 u^2 \\ \rightarrow & S_0 u^2 & \rightarrow & S_0 u^2 - S_0 u^2 \\ \rightarrow & S_0 u^3 & \rightarrow & S_0 u^3 - S_0 u^2 \\ \rightarrow & S_0 u^4 & \rightarrow & S_0 u^4 - S_0 u^2 \end{matrix}$$

entries of $\widehat{M}(8)$ entries of $\widehat{V}(8)$

For $S(7) = S_0 u^4 d^3 = S_0 u$, in case of an downwards movement, the updating rule $M(8) = \max \{M(7); S(8)\}$ maps $S(7) = S_0 u$, $\widehat{M}(7) = [S_0 u \ S_0 u^2 \ S_0 u^3 \ S_0 u^4]$ into $S(8) = S_0$, $M(8) = [S_0 u \ S_0 u^2 \ S_0 u^3 \ S_0 u^4]$. In this case only the entry $S_0 u^4$ of $M(8)$ can be exactly matched by one of the entries of $\widehat{M}(8) = [S_0 \ S_0 u^{\frac{4}{3}} \ S_0 u^{\frac{8}{3}} \ S_0 u^4]$. If we decide to approximate the entries of $M(8)$ with the closest (from above) entries of $\widehat{M}(8)$, then in this case the update $[S_0 u \ S_0 u^2 \ S_0 u^3 \ S_0 u^4]$ of the representative vector $\widehat{M}(7) = [S_0 u \ S_0 u^2 \ S_0 u^3 \ S_0 u^4]$ at $t = 8$ is approximated via $[S_0 u^{\frac{4}{3}} \ S_0 u^{\frac{8}{3}} \ S_0 u^4 \ S_0 u^4]$, by picking the closest (from above) entries of $\widehat{M}(8)$ as follows:

$$M(8) = \begin{bmatrix} S_0 u \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix} \begin{matrix} \searrow \\ \searrow \\ \searrow \\ \rightarrow \end{matrix} \begin{bmatrix} S_0 \\ S_0 u^{\frac{4}{3}} \\ S_0 u^{\frac{8}{3}} \\ S_0 u^4 \end{bmatrix} = \widehat{M}(8)$$

Therefore, in case of a downwards movement starting from $S(7) = S_0 u^4 d^3$, $\widehat{M}(7) = [S_0 u \ S_0 u^2 \ S_0 u^3 \ S_0 u^4]$ the associated values for the American

option at $T = 8$ are

$$M(7) = \begin{bmatrix} S_0 u \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix} \xrightarrow[\text{down}]{\text{with prob. } 1-q} \begin{matrix} \searrow \\ \searrow \\ \searrow \\ \rightarrow \end{matrix} \begin{bmatrix} S_0 \\ S_0 u^{\frac{4}{3}} \\ S_0 u^{\frac{8}{3}} \\ S_0 u^4 \end{bmatrix} \xrightarrow{\quad} \begin{bmatrix} S_0 u^{\frac{4}{3}} - S_0 \\ S_0 u^{\frac{8}{3}} - S_0 \\ S_0 u^4 - S_0 \end{bmatrix}$$

$\widehat{M}(8)$ $\widehat{V}(8)$

so that

$$M(7) = \begin{bmatrix} S_0 u \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix} \xrightarrow[\text{down}]{\text{with prob. } 1-q} \begin{matrix} \rightarrow \\ \rightarrow \\ \rightarrow \\ \rightarrow \end{matrix} \begin{matrix} S_0 u^{\frac{4}{3}} \\ S_0 u^{\frac{8}{3}} \\ S_0 u^4 \\ S_0 u^4 \end{matrix} \xrightarrow{\quad} \begin{matrix} S_0 u^{\frac{4}{3}} - S_0 \\ S_0 u^{\frac{8}{3}} - S_0 \\ S_0 u^4 - S_0 \\ S_0 u^4 - S_0 \end{matrix}$$

entries of $\widehat{M}(8)$ entries of $\widehat{V}(8)$

This gives the following entries for the continuation value vector $CV(7)$ at $t = 7$, $S(7) = S_0 u^4 d^3$:

$$M(7) = \begin{bmatrix} S_0 u \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix} \mapsto CV(7) = \begin{bmatrix} \frac{q(S_0 u^2 - S_0 u^2) + (1-q)(S_0 u^{\frac{4}{3}} - S_0)}{1+r} \\ \frac{q(S_0 u^2 - S_0 u^2) + (1-q)(S_0 u^{\frac{8}{3}} - S_0)}{1+r} \\ \frac{q(S_0 u^3 - S_0 u^2) + (1-q)(S_0 u^4 - S_0)}{1+r} \\ \frac{q(S_0 u^4 - S_0 u^2) + (1-q)(S_0 u^4 - S_0)}{1+r} \end{bmatrix}$$

The vector of values $V(7)$ of the American option at $t = 7$, $S(7) = S_0 u^4 d^3$, is provided by $\max(X(7), CV(7))$:

$$M(7) = \begin{bmatrix} S_0 u \\ S_0 u^2 \\ S_0 u^3 \\ S_0 u^4 \end{bmatrix} \mapsto V(7) = \begin{bmatrix} \max \left(S_0 u - S_0 u, \frac{q(S_0 u^2 - S_0 u^2) + (1-q)(S_0 u^{\frac{4}{3}} - S_0)}{1+r} \right) \\ \max \left(S_0 u^2 - S_0 u, \frac{q(S_0 u^2 - S_0 u^2) + (1-q)(S_0 u^{\frac{8}{3}} - S_0)}{1+r} \right) \\ \max \left(S_0 u^3 - S_0 u, \frac{q(S_0 u^3 - S_0 u^2) + (1-q)(S_0 u^4 - S_0)}{1+r} \right) \\ \max \left(S_0 u^4 - S_0 u, \frac{q(S_0 u^4 - S_0 u^2) + (1-q)(S_0 u^4 - S_0)}{1+r} \right) \end{bmatrix}$$

This step concludes the pricing procedure in the node $S(7) = S_0 u^4 d^3$ at $t = 7$.

1.3.2 References on lattice algorithms for exotic options

We conclude this chapter quoting some important references on lattice methods (i.e. trees) for the valuation of plain vanilla and exotic options.

We start with the seminal paper:

“Option pricing: A simplified approach” by Cox, J. Ross, S.A. and Rubinstein, M. on *Journal of Financial Economics*; 1979 Vol. 7, p229-263.

This paper is considered the historical reference for the valuation of options via lattice methods.

As we already observed, when options are path-dependent (e.g. lookback options, Asian options) the pure tree-representation of the underlying does not carry the necessary information to price the derivative. Hence such information is added to each node of the tree by means of an auxiliary state vector. Several papers adapt lattice methods to exotic options in such directions. We recall here a few.

- “Efficient procedures for valuing European and American path dependent options” by Hull, J. and White, A. on *Journal of Derivatives*; 1993 Vol. 1, p21-31.
- Pricing of American path-dependent contingent claims; Barraquand, J.; Pudet, T. *Math. Finance*; 1996 Vol. 6, p17-51
- “Pricing Algorithms for Options with Exotic Path-Dependence” by Kwok, Y.K. and Lau, K.W. on the *Journal of Derivatives*, 2001 Vol. 9, p28-38.

The idea in these papers is to exploit the forward shooting grid approach to price exotic path-dependent options. As the authors of the last paper explain in the abstract, the forward shooting grid approach is characterized by the augmentation of an auxiliary state vector at each grid node on a lattice tree that simulates the discrete asset price process. The state vector is used to capture the specific path-dependent feature of the option contract. Such method is very flexible. For instance, in the last paper the authors apply this method to various types of Parisian options², with reset features, and alpha quantile options.

²The Parisian Option is a barrier option for which the barrier feature (knock in or knock out) is only triggered after the underlying asset has spent a certain prescribed time beyond the barrier.

To enhance the accuracy of the lattice methods without burdening the computational cost it is also possible to *refine* the tree representation of the underlying in *option-specific regions*. A classical reference for this technique is

“The Adaptive Mesh Model: A New Approach to Efficient Option Pricing”, by Figlewski, S. and Gao, B on Journal of Financial Economics, 53 (1999) 313-351.

As the authors explain, the Adaptive Mesh Model (AMM) sharply reduces the nonlinearity error. The nonlinearity error refers to the fact that when the option value is highly nonlinear with respect to the underlying asset (for instance around the strike at expiration), a uniform refinement of the step size does not efficiently increase accuracy, because much of the computational effort is wasted on unimportant regions. The idea of the AMM is to graft one or more small sections of the fine high-resolution lattice onto a tree with coarser time and price steps to increase the computational accuracy only on those regions where needed. Three different AMM structures are presented, one for pricing ordinary options, one for barrier options, and one for computing delta and gamma efficiently. The AMM approach can be adapted to a wide variety of contingent claims. For some common problems, accuracy increases by several orders of magnitude with no increase in execution time.

1.3.3 Barrier options

Barrier options deserve some specific comment as they are the most popular options with path-dependent payoffs. These options can be written on a specified asset price, index, or rate. They are activated or extinguished when the specified underlying reaches the barrier. The simplest barrier options are otherwise standard calls and puts that are knocked in or knocked out by the underlying asset itself. Some variants tie the barrier crossing to one variable and the payoff to another; others specify binary payoffs in place of the usual payoffs for calls and puts. The knock-in and knock-out features lower the price of an option, and may provide a payoff distribution that better matches a hedger’s risk or a speculator’s view. European discrete knock-out barrier options can be easily priced via lattice methods. With the usual backward recursion, at any node of the tree we check whether the underlying asset has reached the barrier or not, and update consistently the *value* of the option. This procedure returns at the *final* step the *initial*

no-arbitrage price of the option. In particular, consider the final payoff $X = \varphi(S(T))$, where φ is a (sufficiently smooth) deterministic function of the terminal value of the underlying. Examples include digital options by setting $\varphi(S(T)) = \mathbb{I}_{S(T) > K}$, call options with $\varphi(S(T)) = (S(T) - K)^+$, and put options by setting $\varphi(S(T)) = (K - S(T))^+$. The *European knock-out option on S* delivers the final payoff $X = \varphi(S(T))$ if the barrier is not triggered before maturity, i.e.

$$X_{KO}(T) = \begin{cases} X & \text{if } \max_{0 \leq t \leq T} S(t) < b \\ 0 & \text{otherwise} \end{cases}$$

where the barrier $b > S(0)$. Let

$$M(t) = \max_{0 \leq s \leq t} S(s)$$

the running maximum at t of the underlying S . The final payoff of the knock-out option can be rewritten

$$\begin{aligned} X_{KO}(T) &= \begin{cases} X & \text{if } M(T) < b \\ 0 & \text{otherwise} \end{cases} \\ &= X \cdot \mathbb{I}_{M(T) < b} \end{aligned}$$

where $\mathbb{I}_{M(T) < b}$ denotes the indicator function of $M(T) < b$. Thus the no-arbitrage price of our knock-out option is

$$V_{KO}(0) = \mathbf{E}^Q \left[\frac{X \cdot \mathbb{I}_{M(T) < b}}{B(T)} \right]. \quad (1.9)$$

The price $V_{KO}(0)$ can be easily computed via the following backward recursion, where we check node by node on the underlying's tree whether or not the barrier is knocked out.

Proposition 6 (The backward recursion for knock-out barrier options) *The initial no-arbitrage price of the knock-out barrier option defined in Equation (1.9) is $V_{KO}(0) = \hat{V}_{KO}(0)$, where $\hat{V}_{KO}$ is defined via the following backward recursion*

$$\hat{V}_{KO}(T) = \begin{cases} X & \text{if } S(T) < b \\ 0 & \text{otherwise} \end{cases} \quad (1.10)$$

and for $t = T - 1, \dots, 1, 0$

$$\hat{V}_{KO}(t) = \begin{cases} \mathbb{E}_t \left[\frac{\hat{V}_{KO}(t+1)}{1+r(t)} \right] & \text{if } S(t) < b \\ 0 & \text{if } S(t) \geq b \end{cases} \quad (1.11)$$

Proof. We show now that the recursion in Equations (1.10) and (1.11) provides the fair price of the knock-out option at $t = 0$, $V_{KO}(0)$. We see that

$$\begin{aligned}\mathbb{I}_{M(T)<b} &= \mathbb{I}_{S(T)<b} \cdot \mathbb{I}_{M(T-1)<b} \\ &= \mathbb{I}_{S(T)<b} \cdot \mathbb{I}_{S(T-1)<b} \cdot \mathbb{I}_{M(T-2)<b} \\ &= \mathbb{I}_{S(T)<b} \cdot \mathbb{I}_{S(T-1)<b} \cdot \dots \cdot \mathbb{I}_{S(0)<b}\end{aligned}$$

Thus

$$\begin{aligned}V_{KO}(0) &= \mathbf{E}^Q \left[\frac{X \cdot \mathbb{I}_{M(T)<b}}{B(T)} \right] \\ &= \mathbf{E}^Q \left[\frac{X \cdot \mathbb{I}_{S(T)<b} \cdot \mathbb{I}_{S(T-1)<b} \cdot \dots \cdot \mathbb{I}_{S(0)<b}}{(1+r(T-1)) B(T-1)} \right] \\ &= \mathbf{E}^Q \left[\frac{X \cdot \mathbb{I}_{S(T)<b} \cdot \mathbb{I}_{S(T-1)<b} \cdot \dots \cdot \mathbb{I}_{S(0)<b}}{(1+r(T-1)) B(T-1)} \right] \\ &\stackrel{(I)}{=} \mathbf{E}^Q \left[\mathbf{E}_{T-1}^Q \left[\frac{X \cdot \mathbb{I}_{S(T)<b}}{(1+r(T-1))} \right] \cdot \mathbb{I}_{S(T-1)<b} \cdot \frac{\mathbb{I}_{S(T-2)<b} \cdot \dots \cdot \mathbb{I}_{S(0)<b}}{B(T-1)} \right] \\ &\stackrel{(II)}{=} \mathbf{E}^Q \left[\widehat{V}_{KO}(T-1) \frac{\mathbb{I}_{S(T-2)<b} \cdot \dots \cdot \mathbb{I}_{S(0)<b}}{B(T-1)} \right]\end{aligned}$$

where equality (I) follows by the the law of iterated conditional expectations and the $\mathcal{F}_{T-1}$ -measurability of the factor $\frac{\mathbb{I}_{S(T-1)<b} \cdot \dots \cdot \mathbb{I}_{S(0)<b}}{B(T-1)}$, equality (II) follows from Equation (1.11) for $t = T-1$. With similar steps

$$\begin{aligned}V_{KO}(0) &= \mathbf{E}^Q \left[\widehat{V}_{KO}(T-1) \cdot \frac{\mathbb{I}_{S(T-2)<b} \cdot \dots \cdot \mathbb{I}_{S(0)<b}}{B(T-1)} \right] \\ &= \mathbf{E}^Q \left[\mathbf{E}_{T-2}^Q \left[\frac{\widehat{V}_{KO}(T-1) \mathbb{I}_{S(T-2)<b}}{(1+r(T-2))} \right] \cdot \frac{\mathbb{I}_{S(T-3)<b} \cdot \dots \cdot \mathbb{I}_{S(0)<b}}{B(T-2)} \right] \\ &= \mathbf{E}^Q \left[\widehat{V}_{KO}(T-2) \cdot \frac{\mathbb{I}_{S(T-3)<b} \cdot \dots \cdot \mathbb{I}_{S(0)<b}}{B(T-2)} \right],\end{aligned}$$

and thus, by continuing the induction, one can prove that

$$V_{KO}(0) = \widehat{V}_{KO}(0). \quad \blacksquare$$

The initial price of the *knock-in option with payoff* X , $V_{KI}(0)$, whose final payoff is

$$V_{KI}(0)(T) = \begin{cases} X & \text{if } \max_{0 \leq t \leq T} S(t) \geq b \\ 0 & \text{otherwise} \end{cases}$$

can then be computed by exploiting the parity relation

$$V(0) = V_{KI}(0) + V_{KO}(0),$$

where

$$V(0) = \mathbf{E}^Q \left[\frac{X}{B(T)} \right]$$

is the initial no-arbitrage price of the European option whose payoff at T is X .

In general, however, the high nonlinearity of the option value around the barrier produces a nonlinearity error. This problem can be solved by exploiting an AMM, as in the following specific paper:

“Pricing Discrete Barrier Options with an Adaptive Mesh Model” by Dong-Hyun Ahn, Figlewski, Stephen and Bin Gao on *Journal of Derivatives*, Summer 99, Vol. 6 Issue 4, p33-43.

As the authors explain, the binomial option pricing model and the trinomial model, its more versatile relative, are invaluable tools for pricing complex derivatives, especially those with American exercise. But while these models converge to the correct option values as the time and price step sizes go to zero, for certain kinds of problems getting close enough may require a very large amount of calculation. The cause is often non-linearity or discontinuity in the option payoff that occurs only in a small region. One example is a barrier option with a barrier that is only monitored at discrete intervals. This article describes how to solve the discrete barrier problem with an adaptive mesh model, a general approach to lattice building that constructs small sections of fine high-resolution mesh in the critical areas and then grafts them onto a base lattice with coarser time and price steps elsewhere. The technique achieves an increase in accuracy that is quite remarkable.

Discrete barrier options are often approximated with continuous barrier options (i.e. options where the barrier is monitored continuously in time), by using the closed formula that can be derived in the continuous-time framework. Such approximation overprices systematically the knock-in discrete option and underprices the knock-out discrete options. To reduce this error one can apply a continuity correction for barrier options suggested in the paper: “Continuity correction for barrier options”, by: Broadie, Mark, Glasserman, Paul and Kou, Steven on *Mathematical Finance*, Oct.97, Vol. 7 Issue 4, p325-350.

In this paper the authors suggest to use the continuous formula with a modified barrier. In particular, if the barrier is monitored at M dates

during the life of the option $[0, T]$, and the initial value of the underlying is *under* the barrier, i.e. $S(0) < b$, they suggest to use the continuous explicit formula with a suitable *higher* barrier

$$\bar{b} = b \cdot e^{\beta\sigma\sqrt{\Delta t}} > b,$$

where $\beta \approx 0.5826$, σ is the underlying's volatility and $\sqrt{\Delta t} = T/M$ is the time between two successive monitoring dates. Viceversa, if the initial value of the underlying is *above* the barrier, they suggest to use the continuous explicit formula with a suitable *lower* barrier

$$\bar{b} = b \cdot e^{-\beta\sigma\sqrt{\Delta t}} < b$$

Such correction is based on limit results for random walks in sequential analysis, and reduces the error estimate as

$$\left(\begin{array}{c} \text{Value of barrier option} \\ M \text{ monitoring dates, barrier } b \end{array} \right) = \left(\begin{array}{c} \text{Value of barrier option} \\ \textit{continuously} \text{ monitored, barrier } \bar{b} \end{array} \right) + o\left(\frac{1}{\sqrt{M}}\right).$$

The correction can be applied to further enhance the accuracy of the lattice estimation of discrete barrier options, as explained in:

“Connecting discrete and continuous path-dependent options”, by Broadie, Mark, Glasserman, Paul and Kou, S.G. on Finance & Stochastics, 1999, Vol. 3 Issue 1, p55-83.

In this paper the authors develop a trinomial model to price barrier options. First, they correct the barrier as previously explained. Then they build up the trinomial tree with a *stretch* parameter that allows to match layers of nodes with the modified barrier. They also adapt this method to discrete lookback options.

Part II

**Continuos Time Financial
Markets**

Chapter 2

Diffusive models

When trading takes place continuously in time, the diffusive models for the asset prices are the most common ones. The simple one-dimensional Black-Scholes model, as well as stochastic volatility models (like the Heston one) employ diffusions (i.e. Brownian motions) to describe the (random) dynamics of the assets. All these methods produce asset prices with *continuous* trajectories¹. Therefore, they cannot describe the presence of jumps or spikes in the prices. Nevertheless, diffusive models can be tailored to produce fat tails, mean-reverting behaviors, and many other important features of asset prices quite efficiently. For these reasons, the diffusive models are extremely important in quantitative finance. In this chapter we study the basic ones.

2.1 The one-dimensional Black-Scholes model

In this section we review briefly the one-dimensional Black-Scholes model, and then we deal with the multi-dimensional extension. The Black-Scholes model, as well as its extensions, assumes that trading takes place continuously in time. Investors share a common probability law on Ω , the *historical probability measure* P . The information structure of discrete-time models is substituted by the *filtration* $\mathbb{F}$ (for more details see [2] or [1]). Two securities are traded on the market:

1. the riskless security $B(t) = e^{rt} = e^{\int_0^t r(s)ds}$ where r is the constant or deterministic riskless interest rate

¹The simplest way to produce jumps in the trajectories is to consider the *jump-diffusion model*, presented in the final chapter of these notes.

2. the risky security $S = \{S(t)\}_{t \in [0, T]}$ described by

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dW^P(t) \quad (2.1)$$

where W^P is a standard Brownian motion under P with respect to $\mathbb{F}$.

We recall that a W^P is a standard Brownian motion under P with respect to $\mathbb{F}$ iff

1. $W^P(0) = 0$;
2. $W^P(\cdot)(\omega) : [0; T] \rightarrow \mathbb{R}$ is a continuous function *for almost every* $\omega \in \Omega$ i.e. the process has continuous trajectories;
3. for all $0 \leq s < t \leq T$, the increment $W^P(t) - W^P(s)$ is independent of $\mathcal{F}_s$ and its distribution is $W^P(t) - W^P(s) \stackrel{P}{\sim} N(0, t - s)$ (where $N(m, v)$ denotes the distribution of a normally distributed random variable under P , with mean m and variance v).

For all $s, t \geq 0$, the increment of the process W^P from t to $t + s$ is distributed under P as a normal random variable with zero mean and variance equal to the length of the time interval considered, i.e. $W^P(t + s) - W^P(t) \stackrel{P}{\sim} N(0, s)$.

The parameters μ and σ in (2.1) are called *drift* and *volatility*. The drift μ is the *mean of the instantaneous rate of return* given by security S . The volatility σ represents the sensitivity of the security S to the *risk factor* W^P .

The stochastic differential equation (2.1) is a short notation for

$$S(t) = S(0) + \int_0^t S(s) \mu ds + \int_0^t S(s) \sigma dW^P(s)$$

The second integral, $\int_0^t S(s) \sigma dW^P(s)$ is the stochastic integral, whose definition differs from the *traditional* one (see [2] for a discussion on this argument). The presence of an integral with respect to $dW^P(s)$ prevents from applying the usual rules of differential calculus. This fact has important consequences. Suppose we are interested in the value of a derivative, whose payoff at time t depends only on t , and on the current stock price, $S(t)$. It can be proved that the unique no-arbitrage price of the derivative is also a deterministic function of time t and the current stock price $S(t)$. Denote this function with $f(t, S(t))$. The quantity $f(t, S(t))$ is the fair price at time t

of the derivative written on S . If we want to compute the *differential* of the derivative value in $[t; t + dt[$, we cannot use the standard chain rule, because of the presence of the stochastic integral. Instead, we have to use the

Ito formula: *Suppose that X satisfies*

$$\begin{cases} dX(s) = a(s, X(s)) ds + b(s, X(s)) dW(s) \\ X(0) = X_0 \end{cases}$$

with a and b such that X is well defined. Let $f : [0; T] \times \mathfrak{R} \rightarrow \mathfrak{R}$ be continuously differentiable, once with respect to the first variable, denoted by t , twice with respect to the second, denoted by x . Let

$$Y(t) = f(t; X(t))$$

Then $Y(t)$ is itself an Ito process and its integral decomposition is given by:

$$\begin{aligned} Y(t) = & Y(0) + \int_0^t \frac{\partial f(s; x)}{\partial s} \Big|_{(s; X(s))} ds + \int_0^t \frac{\partial f(s; x)}{\partial x} \Big|_{(s; X(s))} dX(s) \\ & + \int_0^t \frac{1}{2} \frac{\partial^2 f(s; x)}{\partial x^2} \Big|_{(s; X(s))} \cdot b^2(s, x) \Big|_{(s; X(s))} ds \end{aligned}$$

which can be written for short as

$$dY(t) = \frac{\partial f(t; X(t))}{\partial t} dt + \frac{\partial f(t; X(t))}{\partial x} \cdot dX(t) + \frac{1}{2} \frac{\partial^2 f(t; X(t))}{\partial x^2} b^2(t, X(t)) dt \quad (2.2)$$

By using Ito Formula to differentiate $Y(t) = f(t, S(t))$ we get

$$dY(t) = \frac{\partial f(t; S(t))}{\partial t} dt + \frac{\partial f(t; S(t))}{\partial S} dS(t) + \underbrace{\frac{1}{2} \frac{\partial^2 f(t; S(t))}{\partial S^2} (\sigma S(t))^2 dt}_{\text{additional term}}$$

Ito Formula is a powerful tool when working with stochastic processes. It also allows us to *solve* the SDE (2.1). In fact, let $f(t, S) = \ln S$ then

$$\frac{\partial f}{\partial t} = 0, \frac{\partial f}{\partial S} = \frac{1}{S}, \frac{\partial^2 f}{\partial S^2} = -\frac{1}{S^2},$$

and these derivatives must be computed in $(t, S(t))$. By applying Ito's Formula (2.2), we get

$$\begin{aligned} Y(t) &= Y(0) + \int_0^t \left[\frac{1}{S(s)} \mu S(s) + \frac{1}{2} \left(-\frac{1}{S^2(s)} \right) \sigma^2 S^2(s) \right] ds + \int_0^t \frac{1}{S(s)} \sigma S(s) dW^P(s) \\ &= Y(0) + \int_0^t \left[\mu - \frac{1}{2} \sigma^2 \right] ds + \int_0^t \sigma dW^P(s) \\ &= Y(0) + \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W^P(t) \end{aligned}$$

Now, recalling that $Y(t) = \ln S(t)$, we have $Y(0) = \ln S_0$ and hence

$$Y(t) = \ln S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W^P(t)$$

We thus have an explicit expression for $Y(t) = \ln S(t)$. To obtain $S(t)$, we just need to note that

$$\ln S(t) = \ln S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W^P(t)$$

and hence we can take exponential on both sides of the equality

$$\ln \frac{S(t)}{S_0} = \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W^P(t)$$

to get

$$\frac{S(t)}{S_0} = e^{(\mu - \frac{1}{2} \sigma^2)t + \sigma W^P(t)}$$

so that we finally obtain that the solution $S(t)$ of (2.1) is

$$S(t) = S_0 \cdot e^{(\mu - \frac{1}{2} \sigma^2)t + \sigma W^P(t)} \quad (2.3)$$

Hence, Ito Formula allows us to solve the SDE (2.1) and to find the distribution under P of S , by exploiting the properties of W^P .

2.2 Dividends in the Black-Scholes market

2.2.1 Discrete Dividends

Dividends constitute an important feature of traded financial assets. By Gordon's formula, the value of an asset is given by the present value of the

sum of the (risk-adjusted) cash-flow provided by the asset, i.e. its dividends. Usually dividends are paid out at discrete dates, and their amount is announced in advance. At the dividend payment date the asset value drops by the dividend amount². In the continuous-time setting discrete dividends can be embedded in the standard Black Scholes model, by resetting the dynamics of the asset at the dividend payment dates $T_1, \dots, T_M$. More precisely, we consider the standard Black-Scholes dynamics until the first dividend payment, that is

$$\begin{cases} dS(t) = \mu S(t) dt + \sigma S(t) dW(t) & \text{for } 0 \leq t < T_1 \\ S(0) = S_0 \end{cases} \quad (2.4)$$

where W is a standard Brownian motion on $(\Omega, \mathbb{F}, P)$, with μ and σ real positive constants. At the dividend payment date T_1 , the *cum-dividend price* is therefore

$$\begin{aligned} S(T_1^-) &= S_0 e^{(\mu - \frac{\sigma^2}{2})T_1^- + \sigma W(T_1^-)} \\ &= S_0 e^{(\mu - \frac{\sigma^2}{2})T_1 + \sigma W(T_1)} \quad \text{by the continuity of } W. \end{aligned}$$

At T_1 the dividend D_1 is paid out, and the *ex-dividend stock price* is

$$S(T_1) = S(T_1^-) - D_1.$$

Then S evolves following equation (2.4) until the next dividend payment date T_2 , starting from the *ex-dividend stock price* $S(T_1)$:

$$\begin{cases} dS(t) = \mu S(t) dt + \sigma S(t) dW(t) & \text{for } T_1 \leq t < T_2 \\ S(T_1) = S(T_1^-) - D_1 \end{cases}$$

Hence for $T_1 \leq t < T_2$ the stock price is given by

$$S(t) = \underbrace{(S(T_1^-) - D_1)}_{S(T_1)} e^{(\mu - \frac{\sigma^2}{2})(t - T_1) + \sigma(W(t) - W(T_1))} \quad (2.5)$$

If dividends are *proportional to cum-dividend stock price*, i.e. if

$$D_i = \delta_i \cdot S(T_i^-) \quad i = 1, \dots, M,$$

²Due to different taxation or disclosure rules, the jump of the asset price is often smaller than the dividend amount. By no-arbitrage, however, the jump size equals on risk-neutral average the dividend amount.

with $\delta_i \in (0; 1)$, equation (2.5) simplifies to

$$\begin{aligned}
 S(t) &= \underbrace{(S(T_1^-) - \delta_1 \cdot S(T_1^-))}_{S(T_1)} \cdot e^{\left(\mu - \frac{\sigma^2}{2}\right)(t-T_1) + \sigma(W(t) - W(T_1))} \\
 &= (1 - \delta_1) \cdot S(T_1^-) \cdot e^{\left(\mu - \frac{\sigma^2}{2}\right)(t-T_1) + \sigma(W(t) - W(T_1))} \\
 &= (1 - \delta_1) \cdot \underbrace{S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)T_1 + \sigma W(T_1)}}_{S(T_1^-)} \cdot e^{\left(\mu - \frac{\sigma^2}{2}\right)(t-T_1) + \sigma(W(t) - W(T_1))} \\
 &= (1 - \delta_1) \cdot S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W(t)}
 \end{aligned}$$

for all $T_1 \leq t < T_2$. After the payment of $D_1 = \delta_1 \cdot S(T_1^-)$, ..., $D_n = \delta_n \cdot S(T_n^-)$, proportional discrete dividends, the stock price is

$$S(t) = (1 - \delta_1) \cdot (1 - \delta_2) \cdot \dots \cdot (1 - \delta_n) \cdot S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W(t)}$$

for all $T_n \leq t < T_{n+1}$ (see Bjork's book, chapter 16, for a comprehensive analysis of this model).

2.2.2 Continuous dividends.

A perhaps more common way to model dividends in the continuous-time setting is to consider dividends as a *continuous* cash-flow, proportional to the stock price:

$$\underbrace{dD(t)}_{\text{dividend paid in } [t; t+dt[} = qS(t) dt$$

with $q \geq 0$. The dividend yield q can be thought as an average dividend yield, estimated by equating the total amount of the dividend paid in $[0; T]$:

$$D_1 + \dots + D_M = \int_0^T \underbrace{qS(t) dt}_{dD(t)}$$

The SDE of the gain process G of the holder of one unit of the stock is given by

$$\begin{aligned}
 dG(t) &= dS(t) + dD(t) \\
 &= dS(t) + qS(t) dt \\
 &= S(t) ((\mu + q) dt + \sigma dW(t))
 \end{aligned}$$

In discounted terms

$$\begin{aligned}
d\tilde{G}(t) &= d\tilde{S}(t) + d\tilde{D}(t) \\
&= d\tilde{S}(t) + \underbrace{q\tilde{S}(t)dt}_{d\tilde{D}(t)} \\
&= \tilde{S}(t)((\mu - r)dt + \sigma dW(t)) + q\tilde{S}(t)dt \\
&= \tilde{S}(t)((\mu + q - r)dt + \sigma dW(t)) \\
&= \tilde{S}(t)\sigma \left(\frac{(\mu + q - r)}{\sigma}dt + dW(t) \right)
\end{aligned}$$

By the 1st FTAP the market is free of arbitrage opportunities if there exists a *risk neutral measure* $\mathbb{Q}$ that makes the *discounted gain process* a martingale.

Hence, thanks to Girsanov theorem, $\mathbb{Q}$ is the probability measure such that

$$dW^{\mathbb{Q}}(t) = \frac{(\mu + q - r)}{\sigma}dt + dW(t)$$

is a standard Brownian motion with respect to $\mathbb{Q}$. In fact by

$$\begin{aligned}
d\tilde{G}(t) &= \tilde{S}(t)\sigma \left(\frac{(\mu + q - r)}{\sigma}dt + dW(t) \right) \\
&= \tilde{S}(t)\sigma dW^{\mathbb{Q}}(t),
\end{aligned}$$

it follows that $\tilde{G}$ is a pure stochastic integral with respect to $W^{\mathbb{Q}}$, and therefore a $\mathbb{Q}$ -martingale.

As a consequence, the risk-neutral dynamics of the stock price can be derived as follows:

$$\begin{aligned}
\frac{dS(t)}{S(t)} &= \mu dt + \sigma dW(t) \\
&= \mu dt + \sigma \left(\underbrace{dW^{\mathbb{Q}}(t) - \frac{(\mu + q - r)}{\sigma}dt}_{dW(t)} \right) \\
&= (r - q)dt + \sigma dW^{\mathbb{Q}}(t)
\end{aligned} \tag{2.6}$$

2.2.3 European options on dividend-paying assets.

European call prices can be computed by taking the risk-neutral expectation of their discounted terminal payoff. The result is the following:

Proposition 7 (Call Option Formula) *Let us consider a European call option with maturity T and strike price K . Its time- t no-arbitrage price is then given by*

$$c(t) = S(t) e^{-q(T-t)} N(d_1) - K e^{-r(T-t)} N(d_2), \quad (2.7a)$$

where $N(z)$ is the distribution function of a standard normal random variable, i.e.

$$N(y) = \int_{-\infty}^y \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz,$$

while

$$d_1 = \frac{1}{\sigma \sqrt{(T-t)}} \left(\ln \left(\frac{S(t)}{K} \right) + \left(r - q + \frac{1}{2} \sigma^2 \right) (T-t) \right)$$

and

$$d_2 = d_1 - \sigma \sqrt{(T-t)}.$$

Once the call price has been computed, the put price with same strike and same maturity can be derived via the put-call parity.

The **put-call parity** is a law-of-one-price relation, that can be deduced by the equality of the following payoffs at maturity:

$$\begin{array}{ccc} p(T) & = & K - S(T) + c(T) \\ \text{put payoff} & & \text{call payoff} \end{array} \quad (2.8)$$

When no dividends are paid, the right-hand side term in equation (2.8) is the terminal value of the buy-and-hold strategy settled at $t = 0$ as follows

$$\begin{array}{ll} \vartheta_0 & = K e^{-rT} \text{ units of } B \\ \vartheta_1 & = -1 \text{ units of } S \\ \vartheta_c & = +1 \text{ units of the call} \end{array}$$

This buy-and-hold strategy is self-financing and generates the same terminal payoff of the put option. Therefore at any date t the cost of this strategy and the price of the put option must coincide:

$$\begin{array}{ccc} p(t) & = & K e^{-rT} \cdot e^{rt} - 1 \cdot S(t) + 1 \cdot c(t) \\ \text{put price} & & \text{call price} \end{array}$$

This equation delivers the put price for any $t \in [0; T]$, when no dividends are paid.

When the asset S pays a dividend stream, the *buy-and-hold strategy of 1 unit of the asset S* is **not** self-financing. To obtain a self-financing strategy, the dividend paid in $[t; t + dt]$ can be instantaneously reinvested in the stock S itself. Let $\vartheta_1(t)$ denote the number of units of S hold at t . The holder of $\vartheta_1(t)$ units of the stock at t receives in $[t; t + dt]$ the dividend amount

$$\vartheta_1(t) \cdot dD(t) = \vartheta_1(t) qS(t) dt$$

With this money the stock holder buys instantaneously at t the *additional* number of units of asset S

$$\begin{aligned} d\vartheta_1(t) &= \frac{\vartheta_1(t) \cdot dD(t)}{S(t)} && \leftarrow \text{dividend amount} \\ & && \leftarrow \text{price of } S \text{ at } t \\ &= \frac{\vartheta_1(t) qS(t) dt}{S(t)} \\ &= \vartheta_1(t) qdt \end{aligned}$$

The equality

$$d\vartheta_1(t) = \vartheta_1(t) qdt$$

is an ordinary differential equation that can be immediately solved for any interval $[t_1; t_2] \subset [0; T]$ by integration:

$$\begin{aligned} \int_{t_1}^{t_2} \frac{d\vartheta_1(t)}{\vartheta_1(t)} &= \int_{t_1}^{t_2} qdt \\ \int_{t_1}^{t_2} d \ln(\vartheta_1(t)) &= \int_{t_1}^{t_2} qdt \end{aligned}$$

leading to

$$\vartheta_1(t_2) = \vartheta_1(t_1) e^{q(t_2 - t_1)}. \quad (2.9)$$

In particular, if $t_1 = 0$ and $t_2 = t$, the total number of units of S hold at t by investing the dividends continuously in S itself is

$$\vartheta_1(t) = \vartheta_1(0) e^{qt}.$$

To generalize the put-call parity to the case of dividends, we compute the number of units of S that we must hold at time t to have $\vartheta_1(T) = -1$ units of S at maturity, by continuously reinvesting the dividends in S itself. Equation (2.9) with $t_1 = t$ and $t_2 = T$ gives

$$\begin{aligned} \vartheta_1(T) &= \vartheta_1(t) e^{q(T-t)} \\ -1 &= \vartheta_1(t) e^{q(T-t)} \\ \vartheta_1(t) &= -e^{-q(T-t)} \end{aligned}$$

The right-hand side term in equation (2.8) is now the terminal value of the *self-financing strategy* defined as follows

$$\begin{aligned}\vartheta_0(t) &= \vartheta_0 = K e^{-rT} \text{ units of } B \text{ at } t \\ \vartheta_1(t) &= -e^{-q(T-t)} \text{ units of } S \text{ at } t \\ \vartheta_c(t) &= \vartheta_c = +1 \text{ units of the call at } t\end{aligned}$$

This self-financing strategy generates the same terminal payoff of the put option. Therefore at any date t the cost of this strategy and the price of the put option must coincide:

$$\underset{\text{put price}}{p(t)} = K e^{-rT} \cdot e^{rt} - e^{-q(T-t)} \cdot S(t) + \underset{\text{call price}}{1 \cdot c(t)}$$

This equation is the **put-call parity** with dividends, that delivers the put price for any $t \in [0; T]$, when dividends are paid:

Proposition 8 (Put Option Price Formula) *Let us consider a European put option written on the underlying S , with maturity T and strike price K . Then, its time- t no-arbitrage price is given by*

$$p(t) = K e^{-r(T-t)} N(-d_2) - S(t) e^{-q(T-t)} N(-d_1), \quad (2.10)$$

with $N(z)$ denoting the distribution function of a standard normal random variable,

$$N(y) = \int_{-\infty}^y \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz,$$

while

$$d_1 = \frac{1}{\sigma \sqrt{(T-t)}} \left(\ln \left(\frac{S(t)}{K} \right) + \left(r - q + \frac{1}{2} \sigma^2 \right) (T-t) \right)$$

and

$$d_2 = d_1 - \sigma \sqrt{(T-t)}.$$

The Black-Scholes Partial Differential Equation can also be extended to the case of a dividend-paying underlying asset.

Proposition 9 (Black-Scholes PDE with dividends) *Let us consider a European derivative security whose payoff at maturity T is a function of $S(T)$, i.e.*

$$V_X(T)(\omega) = X(T)(\omega) = f(S(T)(\omega))$$

for each $\omega \in \Omega$, with $f : \mathbb{R}_+ \rightarrow \mathbb{R}$. Suppose S pays a continuous dividend yield q , as in Equation (2.6). Let assume that $E^{\mathbb{Q}} \left[(X(T))^2 \right] < \infty$ denote by F the function giving the time- t no-arbitrage price of the derivative as a function of t and $S(t)$, i.e.

$$V_X(t) = F(t, S(t)).$$

Suppose that F is continuously differentiable once with respect to time, t , and twice with respect to space, S . Then, F solves the following PDE for all $t \in [0; T[$ and $S \in \mathbb{R}_+$ with boundary condition at time T :

$$\begin{cases} \frac{\partial}{\partial t} F(t, S) + \frac{\partial}{\partial S} F(t, S) \cdot (r - q) S + \frac{1}{2} \frac{\partial^2}{\partial S^2} F(t, S) \cdot \sigma^2 S^2 = r F(t, S) \\ F(T, S) = f(S) \end{cases} \quad (2.11)$$

Proof. Equation(2.11) can be retrieved via risk-neutral derivation or by constructing the hedging portfolio. We perform here the risk-neutral derivation. We first compute the SDE describing the dynamics of the discounted price of the European derivative

$$\begin{aligned} d(F(t, S(t))e^{-rt}) &= -re^{-rt}F(t, S(t))dt + e^{-rt}dF(t, S(t)) \\ &= e^{-rt} \left(-rF(t, S)dt + \frac{\partial}{\partial t} F(t, S)dt + \frac{\partial}{\partial S} F(t, S)dS + \frac{1}{2} \frac{\partial^2}{\partial S^2} F(t, S) \cdot \sigma^2 S^2 dt \right) \end{aligned}$$

(for simplicity, we omit the dependence on t of S). By no arbitrage, the discounted price of the European derivative is a risk neutral martingale. We therefore write the differential of $F(t, S(t))e^{-rt}$ with respect to the risk neutral measure according to Equation (2.6) for $dS(t)$

$$\begin{aligned} d(F(t, S(t))e^{-rt}) &= \\ &= e^{-rt} \left(-rF(t, S)dt + \frac{\partial}{\partial t} F(t, S)dt + \frac{\partial}{\partial S} F(t, S) \underbrace{S \left((r - q) dt + \sigma dW^{\mathbb{Q}}(t) \right)}_{dS} + \frac{1}{2} \frac{\partial^2}{\partial S^2} F(t, S) \cdot \sigma^2 S^2 dt \right) \\ &= e^{-rt} \left(\left(-rF(t, S) + \frac{\partial}{\partial t} F(t, S) + \frac{\partial}{\partial S} F(t, S) S (r - q) + \frac{1}{2} \frac{\partial^2}{\partial S^2} F(t, S) \cdot \sigma^2 S^2 \right) dt + \right. \\ &\quad \left. + \frac{\partial}{\partial S} F(t, S) S(t) \sigma dW^{\mathbb{Q}}(t) \right) \end{aligned}$$

In order $F(t, S(t))e^{-rt}$ to be a risk-neutral martingale, the risk neutral drift term of its SDE must be zero, i.e.

$$\left(-rF(t, S) + \frac{\partial}{\partial t} F(t, S) + \frac{\partial}{\partial S} F(t, S) S (r - q) + \frac{1}{2} \frac{\partial^2}{\partial S^2} F(t, S) \cdot \sigma^2 S^2 \right) = 0$$

which is Equation (2.11).

2.3 The Black-Scholes model with many underlying assets

In this section we provide a first extension of the Black-Scholes model, by considering a market with one riskless security and *many* risky securities³. In order to describe a given correlation structure among the log-returns of the risky securities, we are going to employ many risk factors. In particular, a k -dimensional Brownian motion on the filtered probability space $(\Omega, \mathbb{F}, P)$ is used to represent the riskiness of the market.

As in the previous sections, we denote by B the riskless security, and denote with $S_1, \dots, S_d$ the d risky securities ($d \geq 1$). The riskless security B provides a constant or deterministic interest rate, i.e. $B(t) = e^{rt}$ where r is the constant or $B(t) = e^{\int_0^t r(s)ds}$ if r is a deterministic riskless interest rate (note that in some market models the interest rate r varies stochastically, i.e. the spot rate $r(t)$ is a stochastic process).

We assume that each risky security S_i satisfies

$$\frac{dS_i(t)}{S_i(t)} = \mu_i(S(t), t)dt + \sigma_i^T(S(t), t)dW^P(t) \quad (2.12)$$

where

- $S(t) = (S_1(t), \dots, S_d(t))^T$
- W^P is a k -dimensional standard Brownian motion:

$$W^P = \begin{bmatrix} W_1^P \\ \dots \\ W_k^P \end{bmatrix}$$

where $W_1^P, \dots, W_k^P$ are *independent* (real valued) standard Brownian motions under P .

- $\mu_i : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ are deterministic functions of the current prices $S = (S_1, \dots, S_d)^T$ and of time⁴.

In the simplest case, $\mu_i(S, t) = \mu_i$ is constant for every $i = 1, \dots, d$.

³We use the notations of [4], chapter 1.2.1.

⁴The μ_i 's have to satisfy some requirement in order S to be well defined from equations (2.12)(see [[4]: Appendix B]).

- $\sigma_i^T : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^k$ are deterministic functions of the current prices and of time⁵ with values in $\mathbb{R}^k$.

In the simplest case, $\sigma_i^T(S, t) = [\sigma_{i,1}, \dots, \sigma_{i,k}]$ is a k -vector of constants. The component $\sigma_{i,j}$ represents the sensitivity of the instantaneous return of security S_i to risk factor j , represented by the standard Brownian motion W_j^P . For instance, $\sigma_{1,3}$ is the sensitivity of the instantaneous return of security S_1 to the *third* risk factor, represented by the standard Brownian motion W_3^P .

We also denote by

$$\Sigma_{i,j} = \sigma_i^T \sigma_j = \sum_{l=1}^k \sigma_{il} \sigma_{jl}$$

$\Sigma_{i,j}$ is the covariance between the instantaneous return of S_i and the instantaneous return of S_j .

The extension to many risk factors is necessary as soon as we have to describe a non-trivial variance-covariance structure among risky securities. In fact, with a unique risk factor W_1^P , the risky securities can only be perfectly correlated. We can check this sentence immediately: for instance, if

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i dW_1^P(t)$$

for $i = 1, 2$, with $\sigma_i \in \mathbb{R}$, then

$$Corr \left(\frac{dS_1(t)}{S_1(t)}, \frac{dS_2(t)}{S_2(t)} \right) = \frac{\sigma_1 \sigma_2 dt}{\sigma_1 \sqrt{dt} \sigma_2 \sqrt{dt}} = 1$$

On the contrary, if we employ two independent Brownian motions W_1^P and W_2^P to describe the dynamics of S_1 and S_2 we get

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_{i1} dW_1^P(t) + \sigma_{i2} dW_2^P(t)$$

for $i = 1, 2$, with $\sigma_{i,j} \in \mathbb{R}$, then

$$\begin{aligned} Cov \left(\frac{dS_1(t)}{S_1(t)}, \frac{dS_2(t)}{S_2(t)} \right) &= (\sigma_{11} \sigma_{21} + \sigma_{12} \sigma_{22}) dt \\ Var \left(\frac{dS_1(t)}{S_1(t)} \right) &= (\sigma_{11}^2 + \sigma_{12}^2) dt \\ Var \left(\frac{dS_2(t)}{S_2(t)} \right) &= (\sigma_{21}^2 + \sigma_{22}^2) dt \end{aligned}$$

⁵ As the μ_i 's, also the σ_i 's have to satisfy some requirement in order S to be well defined from equations (2.12) (see [[4]: Appendix B]).

Suppose that we want to assign variance v_i to each log-return:

$$\text{Var} \left(\frac{dS_i(t)}{S_i(t)} \right) = v_i dt$$

for $i = 1, 2$, and correlation

$$\text{Corr} \left(\frac{dS_1(t)}{S_1(t)}, \frac{dS_2(t)}{S_2(t)} \right) = \rho.$$

Then we can set

$$\begin{aligned} \sigma_{11} &= \sqrt{v_1} \text{ and } \sigma_{12} = 0 \\ \sigma_{21} &= \rho\sqrt{v_2} \text{ and } \sigma_{22} = \sqrt{v_2(1-\rho^2)} \end{aligned}$$

and our goal is reached. The same result can be obtained by employing two *correlated one-dimensional* Brownian motions B_1 and B_2 , with

$$\text{Corr}(dB_1(t), dB_2(t)) = \rho$$

and setting

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sqrt{v_i} dB_i(t)$$

for $i = 1, 2$. In general, the two approaches (the one with independent Brownian motions and the one with correlated Brownian motions) are equivalent (see [2] for a discussion with more than two securities). For sake of simplicity, we will consider here follows independent Brownian motions.

2.3.1 The No-Arbitrage analysis

The First Fundamental Theorem of Asset Pricing guarantees that No-Arbitrage holds if and only if there exists an Equivalent Martingale Measure Q (see [2] or [1] for more details). In this section we show how to work out an equivalent martingale measure (EMM) in this multidimensional framework.

By definition, an EMM Q is an equivalent probability under which *all* the discounted risky securities $\tilde{S}_i$

$$\tilde{S}_i(t) = \frac{S_i(t)}{B(t)}$$

are martingales.

Suppose now that the filtration is generated by the P -Brownian motion $W^P : \mathbb{F} = \mathbb{F}(W^P)$.

2.3. THE BLACK-SCHOLES MODEL WITH MANY UNDERLYING ASSETS⁵¹

In this case, a representation theorem says that a process M is an $\mathbb{F}$ -martingale under P if and only if⁶ it can be written as

$$dM = (...) \cdot dW^P$$

where W^P is a P -Brownian motion.

Hence, up to technical requirements, *martingales are stochastic integrals*.

From equation (2.12) we get for each $i = 1, \dots, d$

$$\frac{d\tilde{S}_i(t)}{\tilde{S}_i(t)} = (\mu_i(S(t), t) - r(t)) dt + \sigma_i^T(S(t), t) dW^P(t) \quad (2.13)$$

How can we find a measure Q under which *all* discounted risky securities $\tilde{S}_i$ for $i = 1, \dots, d$ are martingales? As we are going to see, this is possible if *there exists a k -valued process $\nu = \{\nu(t)\}_{t \in [0; T]}$ such that for all $i = 1, \dots, d$ we have*

$$\mu_i(S(t), t) - r(t) = \sigma_i^T(S(t), t) \nu(t) \quad (2.14)$$

In this case, in fact, equation (2.13) can be written as

$$\frac{d\tilde{S}_i(t)}{\tilde{S}_i(t)} = \sigma_i^T(S(t), t) (\nu(t) dt + dW^P(t))$$

The term in the parenthesis $\nu(t) dt + dW^P(t)$ is not the differential of a standard Brownian motion under P . But we can now apply the (multi-dimensional) Girsanov Theorem to find a probability Q under which

$$\int_0^t \nu(s) ds + \int_0^t dW^P(s) = \int_0^t \nu(s) ds + W^P(t)$$

is a standard Brownian motion.

In fact, by Girsanov Theorem (see [[4]: Appendix B, Theorem B.4.1] the process

$$W^Q(t) = W^P(t) + \int_0^t \nu(s) ds$$

is a standard Brownian motion under Q where

$$L(T) = \frac{dQ}{dP}$$

⁶ M has to satisfy some integrability requirement

is given by

$$L(t) = \exp \left(-\frac{1}{2} \int_0^t \|\nu(s)\|^2 ds - \int_0^t \nu(s) dW^P(s) \right)$$

(up to some technical assumptions on L : see Theorem B.4.1 in [4]).

Hence, under Q

$$\frac{d\tilde{S}_i(t)}{\tilde{S}_i(t)} = \sigma_i^T(S(t), t) dW^Q(t)$$

the processes $\tilde{S}_i$ are martingales for all $i = 1, \dots, d^7$!

Our key assumption is the existence of the process ν as in (2.14).

The existence of such a ν is *equivalent* by Girsanov Theorem to the existence of an EMM Q . By the first fundamental theorem of asset pricing, the existence of an EMM is equivalent to No Arbitrage in the market.

But what is the meaning of ν ? ν is the **market price of risk process**!

Suppose first that all coefficients are constant and that $k = d = 1$ (as in the standard Black-Scholes model).

In this case, ν has to satisfy (2.14) that becomes $\mu - r = \sigma\nu$, that is

$$\nu = \frac{\mu - r}{\sigma}$$

the ratio of

- $\mu - r$, the excess instantaneous return of S on the risk-free interest rate r
- and the volatility σ which is a measure of the riskiness of security S .

If all coefficients are *constant*, but $k, d \neq 1$, then ν is a vector of $\mathbb{R}^k$ that satisfies (2.14)

$$\mu_i - r = \sigma_i^T \nu \text{ for } i = 1, \dots, d$$

Equivalently

$$\begin{bmatrix} \sigma_{1,1} & \dots & \sigma_{1,k} \\ \dots & \dots & \dots \\ \dots & \dots & \dots \\ \sigma_{d,1} & \dots & \sigma_{d,k} \end{bmatrix} \begin{bmatrix} \nu_1 \\ \dots \\ \nu_k \end{bmatrix} = \begin{bmatrix} \mu_1 - r \\ \dots \\ \mu_d - r \end{bmatrix} \quad (2.15)$$

Summing up:

⁷We should verify that the filtrations generated by W^P and W^Q coincide. This fact follows trivially if the coefficients μ_i and σ_i are constants.

1. No Arbitrage holds if and only if there exists a market price of risk vector ν satisfying (2.15) (first fundamental theorem of asset pricing). In particular, if such vector ν exists, then the (multi-dimensional) Girsanov theorem delivers an equivalent probability Q that makes the discounted value of *any* traded security a martingale.
2. The market is complete and No Arbitrage holds if and only if the market price of risk vector ν satisfying (2.15) is unique (second fundamental theorem of asset pricing).

If we denote the volatility matrix with

$$[\sigma_{i,j}] = \begin{bmatrix} \sigma_{1,1} & \cdots & \sigma_{1,k} \\ \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots \\ \sigma_{d,1} & \cdots & \sigma_{d,k} \end{bmatrix}$$

and assume that $[\sigma_{i,j}]$ has full rank (securities are not redundant) we have the following rule of thumb:

1. If $rk[\sigma_{i,j}] = d < k$ then (2.15) admits infinitely many solutions ν and No Arbitrage holds, but the market is incomplete.
2. If $rk[\sigma_{i,j}] = k = d$ then the market is complete and No Arbitrage holds, since (2.15) admits a unique solution ν

Hence, the market satisfies No Arbitrage and completeness *if and only if the number of risk factors k = the number of risky (non redundant) securities d .*

2.3.2 The distribution of the risky securities under the historical and the risk neutral measure

We want to solve (2.12) in case

$$\begin{aligned} \mu_i(S(t), t) &= \mu_i \\ \sigma_i^T(S(t), t) &= \sigma_i^T \end{aligned}$$

where, with a small abuse of notations, μ_i and σ_i^T on the right-hand side are constants.

To this aim, we have to use the multidimensional Ito formula (Theorem B.1.1. in [4]), because W^P is a k -dimensional standard Brownian motion. Ito formula will also be useful to derive the hedging strategy for an European derivative with many underlying assets.

Proposition 10 (Multi-dimensional) Ito formula: Suppose that X satisfies

$$\begin{cases} dX_i(s) = a_i(s, X(s)) ds + b_i^T(s, X(s)) dW(s) \\ X(0) = X_0 \end{cases}$$

where W^P is a k -dimensional standard Brownian motion and $a_i : [0; T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ and $b_i : [0; T] \times \mathbb{R}^d \rightarrow \mathbb{R}^k$ are such that X is well defined. Let $f : [0; T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ be continuously differentiable, once with respect to the first variable, denoted by t , twice with respect to the second vector variable, denoted by $x \in \mathbb{R}^d$. Let

$$Y(t) = f(t; X(t))$$

Then $Y(t)$ is itself an Ito process is given by:

$$\begin{aligned} dY(t) &= \frac{\partial f(t; X(t))}{\partial t} dt + \sum_{i=1}^d \frac{\partial f(t; X(t))}{\partial x_i} \cdot dX_i(t) + .. \\ &.. + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 f(t; X(t))}{\partial x_i \partial x_j} b_i^T(t, X(t)) b_j(t, X(t)) dt \end{aligned} \quad (2.16)$$

To solve the SDE (2.12) for S_i we set as in the one-dimensional case $f(t, S) = \ln S_i$ then

$$\frac{\partial f}{\partial t} = 0, \frac{\partial f}{\partial S_i} = \frac{1}{S_i}, \frac{\partial^2 f}{\partial S_i^2} = -\frac{1}{S_i^2},$$

and

$$\frac{\partial f}{\partial S_j} = 0, \frac{\partial^2 f}{\partial S_i \partial S_j} = 0 \text{ if } j \neq i$$

By applying Ito's Formula (2.16), we get

$$Y(t) = Y(0) + \left(\mu_i - \frac{1}{2} \sigma_i^T \sigma_i \right) t + \sigma_i^T W^P(t)$$

Now, recalling that $Y(t) = \ln S_i(t)$, we get

$$S_i(t) = S_i(0) \cdot e^{(\mu_i - \frac{1}{2} \sigma_i^T \sigma_i)t + \sigma_i^T W^P(t)} \quad (2.17)$$

Hence, Ito Formula allows us to solve the stochastic differential equation (2.12) and to find the distribution under P of S , by exploiting the properties of W^P .

To find the distribution under the (or an) EMM Q we recall that $\mu_i - r = \sigma_i^T \nu$ and that

$$W^Q(t) = W^P(t) + \nu t$$

is a standard Brownian motion under Q

Hence

$$\begin{aligned} \left(\mu_i - \frac{1}{2} \sigma_i^T \sigma_i \right) t + \sigma_i^T W^P(t) &= \left(\mu_i \pm r - \frac{1}{2} \sigma_i^T \sigma_i \right) t + \sigma_i^T W^P(t) = \\ &= \left(\sigma_i^T \nu + r - \frac{1}{2} \sigma_i^T \sigma_i \right) t + \sigma_i^T W^P(t) = \\ &= \left(r - \frac{1}{2} \sigma_i^T \sigma_i \right) t + \sigma_i^T (W^P(t) + \nu t) = \\ &= \left(r - \frac{1}{2} \sigma_i^T \sigma_i \right) t + \sigma_i^T W^Q(t) \end{aligned}$$

so that

$$S_i(t) = S_i(0) \cdot e^{(r - \frac{1}{2} \sigma_i^T \sigma_i)t + \sigma_i^T W^Q(t)}$$

2.3.3 No Arbitrage Valuation of derivatives on several underlying assets

We are ready to price any European derivative with payoff

$$X(T) = f(S_1(T), \dots, S_d(T))$$

at maturity T by computing

$$V(0) = \mathbb{E}^Q [e^{-rT} X(T)]$$

In fact, we can deduce the *joint* probability density with respect to Q of the risky assets at the terminal date $S_1(T), \dots, S_d(T)$. However in some cases it is possible to compute the risk neutral expectation of the discounted terminal payoff by using *ad hoc* arguments.

Example

$$X(T) = (S_1(T)S_2(T) - K)^+$$

Suppose that $\sigma_i^T = [\sigma_{i,1}, \sigma_{i,2}, 0, \dots, 0]$ for $i = 1, 2$ (only 2 risk factors affect S_1 and S_2)

$$\begin{aligned} V(0) &= \mathbb{E}^Q [e^{-rT} X(T)] = \\ &= \mathbb{E}^Q [e^{-rT} (S_1(T)S_2(T) - K)^+] = \\ &= S_1(0)S_2(0)e^{(-r + a + 0.5\sigma_Y^2)T} Q[Z > z^* - \sigma_Y \sqrt{T}] - Ke^{-rT} Q[Z > z^*] \end{aligned}$$

where Z is a standard (one-dimensional) normal random variable under Q

$$a = 2r - \frac{\sigma_1^T \sigma_1 + \sigma_2^T \sigma_2}{2}$$

$$\sigma_Y^2 = (\sigma_{11} + \sigma_{21})^2 + (\sigma_{12} + \sigma_{22})^2$$

$$z^* = \frac{1}{\sigma_Y \sqrt{T}} \left(\ln \frac{K}{S_1(0)S_2(0)} - aT \right)$$

If the coefficients (drift and volatility) of S_1 and S_2 are not constant, and we are not able to find the distribution under Q of the S_i 's, then Ito formula can help us in finding at least the SDE solved by the product

$$Y(t) = S_1(t) \cdot S_2(t)$$

In this case, in fact, we have that

$$dY(t) = Y(t) \left((2r + \sigma_1^T \sigma_2) dt + (\sigma_1^T + \sigma_2^T) dW^Q(t) \right)$$

that leads to the same solution we found just above.

If S_1 is a stock denominated in dollars and S_2 is the *dollars to euros* exchange ratio (0.78 on 28th September 2006), then $S_1(T)S_2(T)$ gives the value in euros of the stock at maturity date. Some care must be taken when choosing the drift under the risk-neutral measure Q (see the chapter on currency derivatives).

In case S_1 is a domestic stock denominated in euros, but the strike price K and the call option is denominated in dollars, then the dollar-payoff at time T becomes

$$X(T) = \left(\frac{S_1(T)}{S_2(T)} - K \right)^+$$

where $\frac{1}{S_2}$ is the *euro to dollar* exchange ratio (1.2719 at the same date).

To find the distribution of

$$Y(t) = \frac{S_1(t)}{S_2(t)}$$

we differentiate $Y(t)$ by using Ito Formula.

This leads in our case to

$$dY(t) = Y(t) \left((\mu_1 - \mu_2 + (-\sigma_1^T + \sigma_2^T) \sigma_2) dt + (\sigma_1^T - \sigma_2^T) dW^P(t) \right)$$

(check the computations as an exercise).

2.3.4 The Black-Scholes PDE and the hedging ratios

Suppose X is an European derivative whose payoff depends at maturity only at the underlying stock prices at T .

$$X(T) = f(S_1(T), \dots, S_d(T))$$

It can be shown that

$$V(t) = \mathbb{E}^Q \left[e^{-r(T-t)} X(T) \middle| \mathcal{F}_t \right]$$

is also a deterministic function of time t and current stock prices $S(t) = S_1(t), \dots, S_d(t)$.

Suppose that

$$\begin{aligned} V(t) &= F(t, S(t)) = \\ &= F(t, S_1(t), \dots, S_d(t)) \end{aligned}$$

is once continuously differentiable with respect to t and twice with respect to S .

Then it is possible to construct a self-financing portfolio with F and S_i 's that is locally riskless. In particular, let π be constituted at time t by

- a *short* position on the derivative F
- h_i units of each risky security S_i for $i = 1, \dots, d$

Our aim is to find h_i such that

$$d\pi(t) = \pi(t) ((\dots) dt + 0 \cdot dW^P(t))$$

so that No Arbitrage will force

$$d\pi(t) = \pi(t) r dt$$

By applying Ito Formula, we see that

$$h_i = \frac{\partial F}{\partial S_i}$$

makes the diffusive coefficient of π equal to 0, so that the drift of π must be equal to the riskless interest rate r . In this way we get the following Partial Differential Equation that F must satisfy

$$\frac{\partial F(t; S)}{\partial t} + \sum_{i=1}^d \frac{\partial F(t; S)}{\partial S_i} \cdot S_i r + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 F(t; S)}{\partial S_i \partial S_j} \sigma_i^T \sigma_j S_i S_j = F(t, S) r$$

that together with the boundary condition

$$F(T, S(T)) = f(S(T))$$

determines the function F .

The number of shares invested in the risky security

$$h_i = \frac{\partial F}{\partial S_i}$$

for $i = 1, \dots, d$ are the *hedge ratios* of F . Each h_i represents the *sensitivity* of the value of the European option to variations of the i^{th} underlying risky asset S_i . We remark that the hedge ratios have to be continuously rebalanced in order π to remain locally riskless.

It is easy to find the hedging strategy for the European option, since the most difficult part of the task has been already accomplished. In fact, the self-financing strategy that replicates F is given by

$$\begin{aligned} \vartheta_0(t) &= \left(F(t, S(t)) - \underbrace{\sum_{i=1}^d \frac{\partial F}{\partial S_i}(t, S(t)) S_i(t)}_{h_i(t)} \right) \frac{1}{B(t)} \\ \vartheta_i(t) &= \underbrace{\frac{\partial F}{\partial S_i}(t, S(t))}_{h_i(t)} \text{ for } i = 1, \dots, d \end{aligned}$$

2.4 Exercises

In the following exercises, we denote with $N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{u^2}{2}} du$ the cumulative distribution function of a standard normal random variable.

2.4.1 The one dimensional Black-Scholes model

In a Black-Scholes model, the riskless security provides the instantaneous riskless interest rate $r = 8\%$ and the risky security S follows the SDE

$$dS(t) = S(t)(\mu dt + \sigma dW^P(t))$$

with $\mu = 10\%$ and $\sigma = 20\%$, initial value $S(0) = 10$ and where W^P is a standard Brownian motion under the historical probability measure P .

1. Determine the expected value of $S(T)$ for $T = 4$ under the risk-neutral measure Q and under the historical probability measure P .

If at time $t = 3$ the price of S is $S(3) = 12$, what is the expected value of $S(T)$ for $T = 4$ under the risk-neutral measure Q and under the historical probability measure P conditional to the fact that $S(3) = 12$?

2. Find the historical probability and the risk-neutral probability that an European call option written on S with strike $K = 8$ and maturity $T = 4$ closes at maturity *in the money*.
3. At time $t = 3$ the stock S is worth $S(3) = 12$. What is at time $t = 3$ the *historical* probability that the European call option written on S with strike $K = 8$ and maturity $T = 4$ closes at maturity *in the money* given the fact that $S(3) = 12$? What is at time $t = 3$ the *risk-neutral* probability that the European call option written on S with strike $K = 8$ and maturity $T = 4$ closes at maturity *in the money* given the fact that $S(3) = 12$?

Answer.

1. The risky security S is given by $S(t) = S(0)e^{\left(\mu - \frac{\sigma^2}{2}\right) \cdot t + \sigma W^P(t)} = S(0)e^{\left(r - \frac{\sigma^2}{2}\right) \cdot t + \sigma W^Q(t)}$ where W^P is a standard Brownian motion under P and W^Q is a standard Brownian motion under Q .

Hence:

$$\begin{aligned}\mathbb{E}^P[S(T)] &= S(0)e^{\mu T} = 14.91 \\ \mathbb{E}^Q[S(T)] &= S(0)e^{rT} = 13.77\end{aligned}$$

If at time $t = 3$ the price of S is $S(3) = 12$, the expected value of $S(T)$ for $T = 4$ under the historical probability measure P conditional to the fact

that $S(3) = 12$ is

$$\begin{aligned}
\mathbb{E}^P [S(T) | \mathcal{F}_t] &= \mathbb{E}^P [S(T) | \mathcal{F}_t] = \\
&= \mathbb{E}^P \left[S(0) e^{\left(\mu - \frac{\sigma^2}{2}\right) \cdot (T \pm t) + \sigma(W^P(T) \pm W^P(t))} \middle| \mathcal{F}_t \right] = \\
&= \mathbb{E}^P \left[\underbrace{S(t)}_{\mathcal{F}_t\text{-measurable}} e^{\left(\mu - \frac{\sigma^2}{2}\right) \cdot (T-t) + \sigma(W^P(T) - W^P(t))} \middle| \mathcal{F}_t \right] = \\
&= S(t) \cdot \mathbb{E}^P \left[e^{\left(\mu - \frac{\sigma^2}{2}\right) \cdot (T-t) + \sigma(W^P(T) - W^P(t))} \middle| \mathcal{F}_t \right] = \\
&= S(t) \cdot \mathbb{E}^P \left[e^{\left(\mu - \frac{\sigma^2}{2}\right) \cdot (T-t) + \sigma(W^P(T) - W^P(t))} \right]
\end{aligned}$$

since $W^P(T) - W^P(t)$ is independent under P from $\mathcal{F}_t$. This means that the conditional expectation of $S(T)$ with respect to $\mathcal{F}_t$ relies only on the time t value of S , $S(t)$. Moreover

$$W^P(T) - W^P(t) \stackrel{P}{\sim} W^P(T - t)$$

so that

$$\begin{aligned}
\mathbb{E}^P [S(T) | \mathcal{F}_t] &= \mathbb{E}^P [S(T) | S(t)] = \\
&= S(t) \mathbb{E}^P \left[e^{\left(\mu - \frac{\sigma^2}{2}\right) \cdot (T-t) + \sigma W^P(T-t)} \right] = \\
&= S(t) \cdot e^{\mu \cdot (T-t)}
\end{aligned}$$

Hence in our case

$$\mathbb{E}^P [S(T) | S(3) = 12] = 12 \cdot e^{\mu \cdot (T-3)} = 13.26$$

With similar computations under Q we have that

$$\mathbb{E}^Q [S(T) | S(3) = 12] = 12 \cdot e^{r \cdot (T-3)} = 12.99$$

2. The historical probability that the call option closes in the money is

$$\begin{aligned}
\mathbf{P} [S(T) > K] &= \mathbf{P} \left[e^{\left(\mu - \frac{\sigma^2}{2}\right) \cdot T + \sigma W_T^P} > \frac{K}{S(0)} \right] = \\
&= \mathbf{P} \left[\left(\mu - \frac{\sigma^2}{2} \right) \cdot T + \sigma W_T^P > \ln \frac{K}{S(0)} \right] = \\
&= \mathbf{P} \left[Y > \frac{1}{\sigma \sqrt{T}} \left(\ln \frac{K}{S(0)} - \left(\mu - \frac{\sigma^2}{2} \right) T \right) \right] = \\
&= 1 - N(-1.357) = 0.912
\end{aligned}$$

where Y denotes a standard normal random variable under $\mathbf{P}$. The risk-neutral probability that the call option closes *in the money* is

$$\begin{aligned}
 \mathbf{Q}[S(T) > K] &= \mathbf{Q}\left[e^{\left(r - \frac{\sigma^2}{2}\right) \cdot T + \sigma W_T^Q} > \frac{K}{S(0)}\right] = \\
 &= \mathbf{Q}\left[\left(r - \frac{\sigma^2}{2}\right) \cdot T + \sigma W_T^Q > \ln \frac{K}{S(0)}\right] = \\
 &= \mathbf{Q}\left[Z > \frac{1}{\sigma\sqrt{T}} \left(\ln \frac{K}{S(0)} - \left(r - \frac{\sigma^2}{2}\right) T\right)\right] = \\
 &= 1 - N(-1.157) = 0.876
 \end{aligned}$$

where Z denotes a standard normal random variable under $\mathbf{Q}$.

3. The historical probability that the call option closes at maturity *in the money*, given the fact that $S(3) = 12$ at $t = 3$ is

$$\begin{aligned}
 \mathbf{P}[S(T) > K | S(3) = 12] &= \mathbf{P}\left[S(3) \cdot e^{\left(\mu - \frac{\sigma^2}{2}\right) \cdot (T-3) + \sigma(W^P(T) - W^P(3))} > K \mid S(3) = 12\right] = \\
 &= \mathbf{P}\left[12 \cdot e^{\left(\mu - \frac{\sigma^2}{2}\right) \cdot (T-3) + \sigma(W^P(T) - W^P(3))} > K\right] \\
 &= \mathbf{P}\left[\left(\mu - \frac{\sigma^2}{2}\right) \cdot (T-3) + \sigma\sqrt{T-3}Y > \ln \frac{K}{12}\right]
 \end{aligned}$$

where Y denotes a standard normal random variable under P . Hence

$$\begin{aligned}
 \mathbf{P}[S(T) > K | S(3) = 12] &= \mathbf{P}\left[Y > \frac{1}{\sigma\sqrt{T-3}} \left(\ln \frac{K}{12} - \left(\mu - \frac{\sigma^2}{2}\right) (T-3)\right)\right] = \\
 &= 1 - N(-2.427) = 0.9923
 \end{aligned}$$

The risk-neutral probability that the call option closes at maturity *in the money*, given the fact that $S(3) = 12$ at $t = 3$ is

$$\begin{aligned}
 \mathbf{Q}[S(T) > K | S(3) = 12] &= \mathbf{Q}\left[S(3) \cdot e^{\left(r - \frac{\sigma^2}{2}\right) \cdot (T-3) + \sigma(W^P(T) - W^P(3))} > K \mid S(3) = 12\right] = \\
 &= \mathbf{Q}\left[12 \cdot e^{\left(r - \frac{\sigma^2}{2}\right) \cdot (T-3) + \sigma(W^P(T) - W^P(3))} > K\right] \\
 &= \mathbf{Q}\left[\left(r - \frac{\sigma^2}{2}\right) \cdot (T-3) + \sigma\sqrt{T-3}Z > \ln \frac{K}{12}\right]
 \end{aligned}$$

where Z denotes a standard normal random variable under $\mathbf{Q}$. Hence

$$\begin{aligned}
 \mathbf{Q}[S(T) > K | S(3) = 12] &= \mathbf{Q}\left[Z > \frac{1}{\sigma\sqrt{T-3}} \left(\ln \frac{K}{12} - \left(r - \frac{\sigma^2}{2}\right) (T-3)\right)\right] = \\
 &= 1 - N(-2.327) = 0.9902
 \end{aligned}$$

In the previous exercise we computed conditional probabilities and conditional expectations. In general, a useful tool to compute conditional expectation is given in the following

Proposition *Let X and Y two random variables on Ω with values in $\mathfrak{R}$. Assume that X is measurable with respect to $\mathcal{F}_s$ and Y is independent of $\mathcal{F}_s$ under Q . Then, if⁸ $g : \mathfrak{R} \times \mathfrak{R} \rightarrow \mathfrak{R}$, we have that*

$$E^{\mathbf{Q}}[g(X, Y) | \mathcal{F}_s] = G(X)$$

where

$$G(x) := E^{\mathbf{Q}}[g(x, Y)]$$

Hence, to compute conditional expectations we may exploit the previous proposition in the following way.

- Fix two random variables: the first one, X , must be *measurable* with respect to $\mathcal{F}_s$; the second random variable, Y , has to be *independent* under $\mathbf{Q}$ from the sigma-algebra $\mathcal{F}_s$.
- Write the object inside the conditional expectation with respect to $\mathcal{F}_s$ under $\mathbf{Q}$ as a deterministic function of X and Y , that is write the object whose conditional expectation you want to compute as $g(X, Y)$ (where g is a real function of two real random variables).
- Fix $x \in \mathfrak{R}$ and compute $G(x) := E^{\mathbf{Q}}[g(x, Y)]$. Since Y is independent of $\mathcal{F}_s$ under $\mathbf{Q}$, the conditional expectation reduces to the unconditional expectation.
- Finally, the conditional expectation we want to compute, $E^{\mathbf{Q}}[g(X, Y) | \mathcal{F}_s]$, is the function $G(x)$ evaluated at $x = X$ *path by path*, that is for all ω you set $x = X(\omega)$ and

$$E^{\mathbf{Q}}[g(X, Y) | \mathcal{F}_s](\omega) = G(X(\omega))$$

⁸Assume g is bounded, to avoid integrability problems.

2.4.2 The Black-Scholes model with many underlying assets

Consider a market model with two risky securities S_1 and S_2 and a riskless asset that provides the interest rate $r = 7\%$. Each risky security is driven by

$$dS_i(t) = S_i(t)(\mu_i dt + \sigma_i^T dW^P(t))$$

where W^P is an $\mathfrak{R}^2$ -Brownian motion under P and the coefficients are

$$\begin{array}{ll} \mu_1 = 12\% & \mu_2 = 10\% \\ \sigma_1^T = [10\% \ 8\%] & \sigma_2^T = [4\% \ 6\%] \end{array}$$

At time $t = 0$ the risky security prices are $S_1(0) = 15$ and $S_2(0) = 10$

1. Find the vector of the market price of risk. Does the market satisfy No Arbitrage? Is it complete?
2. Write the SDE for each S_i with respect to Q . What is the Q -covariance between the instantaneous return of S_1 and S_2 . What is the covariance computed under P ?
3. Find the Q -variance and the P -variance of $S_1(t)$ and $S_2(t)$ for $t = 4$.
4. Find the P -mean and the P -variance of the *instantaneous return*⁹ of the process $Y(t) = S_1(t)(S_2(t))^2$.

Answer.

1. To find the market price of risk ν we have to solve

$$\mu_i - r = \sigma_i^T \nu \text{ for } i = 1, 2$$

that in our case leads to

$$\begin{array}{l} \sigma_{11}\nu_1 + \sigma_{12}\nu_2 = \mu_1 - r \\ \sigma_{21}\nu_1 + \sigma_{22}\nu_2 = \mu_2 - r \end{array}$$

that is

$$\begin{array}{l} 0.1\nu_1 + 0.08\nu_2 = 0.12 - 0.07 \\ 0.04\nu_1 + 0.06\nu_2 = 0.10 - 0.07 \end{array}$$

⁹The mean instantaneous return refers to the drift of the process Y , as in the Glasserman's book.

By solving the system we obtain

$$\begin{aligned}\nu_1 &= 0.214 \\ \nu_2 &= 0.357\end{aligned}$$

Since the market price of risk is unique, the market is complete and arbitrage-free.

2. The SDE for S_i under Q is given by

$$\begin{aligned}dS_1(t) &= S_1(t)(0.07 dt + 0.1 dW_1^Q(t) + 0.08 dW_2^Q(t)) \\ dS_2(t) &= S_2(t)(0.07 dt + 0.04 dW_1^Q(t) + 0.06 dW_2^Q(t))\end{aligned}$$

The Q -covariance between the instantaneous return of S_1 and S_2 is

$$\begin{aligned}\sigma_1^T \sigma_2 &= (\sigma_{11}\sigma_{21} + \sigma_{12}\sigma_{22}) = \\ &= (0.1 \cdot 0.04 + 0.08 \cdot 0.06) = \\ &= 0.0088\end{aligned}$$

The covariance computed under P does not change because the volatility vectors σ_i^T , i.e. the *diffusive coefficients*, are the same under P and Q .

3. We know that each $S_i(t) = S_i(0)e^{(r-0.5\sigma_i^T\sigma_i)\cdot t + \sigma_i^T W^Q(t)}$. Since

$$\sigma_i^T W^Q(t) \stackrel{Q}{\sim} \sqrt{t}\mathcal{N}(0, \sigma_{i1}^2 + \sigma_{i2}^2)$$

where $\mathcal{N}(m, v)$ denotes a normal variable with mean m and variance v , setting

$$\sigma_{i*} = \sqrt{\sigma_{i1}^2 + \sigma_{i2}^2}$$

we have that under Q

$$S_i(t) = S_i(0)e^{(r-0.5\sigma_i^T\sigma_i)\cdot t + \sigma_{i*}\sqrt{t}Z}$$

where $Z \stackrel{Q}{\sim} \mathcal{N}(0, 1)$. By using the formulas of the one-dimensional Black-Scholes

$$\mathbb{E}^Q [S_i^2(t)] = S_i(0)^2 \cdot e^{2rt + \sigma_{i*}^2 t}$$

and hence that the variance of $S_i(t)$ is

$$Var^Q [S_i(t)] = S_i(0)^2 \cdot e^{2rt} (e^{\sigma_{i*}^2 t} - 1).$$

Hence for $i = 1$ we get

$$\begin{aligned} Var^Q [S_1(t)] &= S_1(0)^2 \cdot e^{2rt} \left(e^{\sigma_{1*}^2 t} - 1 \right) = \\ &= 15^2 e^{2 \cdot 0.07 \cdot 4} \left(e^{(0.0164)^4} - 1 \right) = 26.70 \end{aligned}$$

and for

$$Var^Q [S_2(t)] = S_2(0)^2 \cdot e^{2rt} \left(e^{\sigma_{2*}^2 t} - 1 \right) = 3.679$$

Under the historical probability measure, each $S_i(t) = S_i(0)e^{(\mu_i - 0.5\sigma_i^T \sigma_i) \cdot t + \sigma_i^T W^P(t)}$. Since

$$\sigma_i^T W^P(t) \stackrel{P}{\sim} \sqrt{t} \mathcal{N}(0, \sigma_{i1}^2 + \sigma_{i2}^2)$$

where $\mathcal{N}(m, v)$ denotes a normal variable with mean m and variance v , setting

$$\sigma_{i*} = \sqrt{\sigma_{i1}^2 + \sigma_{i2}^2}$$

we have that under P

$$S_i(t) = S_i(0)e^{(\mu_i - 0.5\sigma_i^T \sigma_i) \cdot t + \sigma_{i*} \sqrt{t} Y}$$

where $Y \stackrel{P}{\sim} \mathcal{N}(0, 1)$. Hence,

$$\begin{aligned} Var^P [S_1(t)] &= S_1(0)^2 \cdot e^{2\mu_1 t} \left(e^{\sigma_{1*}^2 t} - 1 \right) = \\ &= 15^2 e^{2 \cdot 0.12 \cdot 4} \left(e^{(0.0164)^4} - 1 \right) = 39.84 \end{aligned}$$

and for

$$Var^P [S_2(t)] = S_2(0)^2 \cdot e^{2\mu_2 t} \left(e^{\sigma_{2*}^2 t} - 1 \right) = 4.677$$

4. We apply Ito formula to derive the instantaneous return of Y . To this aim, set

$$f(S_1, S_2) = S_1 (S_2)^2$$

so that

$$\frac{\partial f}{\partial t} = 0, \quad \frac{\partial f}{\partial S_1} = S_2^2, \quad \frac{\partial^2 f}{\partial S_1^2} = 0,$$

$$\frac{\partial f}{\partial S_2} = 2S_1 S_2, \quad \frac{\partial^2 f}{\partial S_2^2} = 2S_1$$

and

$$\frac{\partial^2 f}{\partial S_1 \partial S_2} = 2S_2$$

By applying Ito multidimensional formula we get

$$\begin{aligned} dY(t) = & S_2^2 dS_1 + 2S_1 S_2 dS_2 + \frac{1}{2} 2S_1 S_2^2 \sigma_2^T \sigma_2 dt + \\ & + 2S_2 S_1 S_2 \sigma_2^T \sigma_1 dt \end{aligned}$$

that leads to

$$dY(t) = Y(t) [(\mu_1 + 2\mu_2 + \sigma_2^T \sigma_2 + 2\sigma_2^T \sigma_1) dt + (\sigma_1^T + 2\sigma_2^T) dW^P(t)]$$

Hence the P -mean of the instantaneous return is

$$\mu_1 + 2\mu_2 + \sigma_2^T \sigma_2 + 2\sigma_2^T \sigma_1 = 0.3428$$

whereas the variance is given by

$$\begin{aligned} (\sigma_1^T + 2\sigma_2^T) (\sigma_1 + 2\sigma_2) &= (\sigma_{11} + 2\sigma_{21})^2 + (\sigma_{12} + 2\sigma_{22})^2 = \\ &= 0.0724 \end{aligned}$$

Chapter 3

Valuation of American options in continuous time

In this section we show the most common analytic approaches to state and solve the American option problem in the continuous time framework, i.e. the variational inequality and the free boundary problem. We use some properties highlighted in the discrete-time case, and give intuitive and financial interpretation to the required technical tools. For a rigorous analysis of the subject we address the interested reader to [5] (and the references therein).

In the continuous-time setting an **American** option with immediate payoff X and maturity T can be exercised at *any* date $\tau \in [0; T]$. As we said in the discrete-time setting, the time- t No-Arbitrage value of the discounted payoff exercised at τ is

$$\mathbf{E}^Q \left[\frac{X(\tau)}{B(\tau)} \middle| \mathcal{F}_t \right].$$

If the holder of the option is rational, he chooses the exercise policy that maximizes the expected discounted payoff. Therefore the value of the American option at time t is (by definition)

$$\tilde{V}(t) = \text{ess sup}_{t \leq \tau \leq T} \mathbf{E}^Q \left[\frac{X(\tau)}{B(\tau)} \middle| \mathcal{F}_t \right] \quad (3.1)$$

where τ runs among all possible exercise policies.

The *ess* that you find before the sup in (3.1) stays for *essential supremum*. It allows to define the supremum of a (uncountable) family of

random variables $\left(\mathbf{E}^Q \left[\frac{X(\tau)}{B(\tau)} \middle| \mathcal{F}_t \right], \text{ as } \tau \text{ varies} \right)$ that are defined up to zero-probability sets (see for instance [3]). The result of such operation is the one you expect: the supremum of the expected discounted payoffs. In the continuous-time case the formal definition of the exercise policy i.e. the *stopping time* is similar to the one we gave in the discrete-time case: $\tau : \Omega \rightarrow T$ is a *stopping time* (with respect to the filtration $\mathcal{F}$) if

$$\{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t$$

for every $t \in [0; T]$.

As in the discrete-time case, this property simply means that the holder decides to exercise the option at or before time t using the information he has at t , $\mathcal{F}_t$.

$\tilde{V}$ defined in (3.1) is called *Snell envelope* of the discounted payoff process $\tilde{X}$. The process $\tilde{V}$ enjoys the same properties listed in the discrete-time case. In particular, $\tilde{V}$ is the *smallest* super-martingale under Q that dominates the immediate discounted payoff $\tilde{X}$. Moreover, the optimal exercise policy is τ^* the first instant in which $\tilde{V}$ touches $\tilde{X}$.

In the discrete-time case we stated the *backward recursive formula* to compute $\tilde{V}$. The recursive formula cannot be used in the continuous-time case, but still can help us in the characterization of the value of the American option. Suppose that we discretize the life of the option in many subintervals with length Δt . The *discretized* discounted value of the American option $\tilde{V}$ at any intermediate date follows the backward recursion:

$$\tilde{V}(t) = \max \left\{ \tilde{X}(t); \mathbf{E}^Q \left[\tilde{V}(t + \Delta t) \middle| \mathcal{F}_t \right] \right\}$$

for $t \in (0, T)$. Then

1. either $\tilde{X}(t) < \mathbf{E}^Q \left[\tilde{V}(t + \Delta t) \middle| \mathcal{F}_t \right]$ and $\tilde{V}(t) = \mathbf{E}^Q \left[\tilde{V}(t + \Delta t) \middle| \mathcal{F}_t \right]$
2. or $\tilde{X}(t) \geq \mathbf{E}^Q \left[\tilde{V}(t + \Delta t) \middle| \mathcal{F}_t \right]$ and $\tilde{V}(t) = \tilde{X}(t) \geq \mathbf{E}^Q \left[\tilde{V}(t + \Delta t) \middle| \mathcal{F}_t \right]$.

In the first case the early exercise is not optimal, and we are in the *continuation region* (the region where it is optimal to wait until $t + \Delta t$). In such region $\tilde{V}$ exhibits the *martingale property* under Q .

In the second case the early exercise is optimal, and we are in the *early exercise region*. In such region $\tilde{V}$ is not a martingale, but a supermartingale under Q .

The two properties have a clear financial interpretation: when early exercise is not profitable, the American option coincides (locally) with the European option. Therefore, its discounted price is a Q -martingale, as the price of any traded (European) security. On the contrary, when the American option is optimally early exercised, the value of the American option is greater than the continuation value, and equals the immediate discounted payoffs.

When the time mesh $\Delta t \rightarrow 0$, such properties translate into the *variational inequality problem for the American option*. We first observe that, as in the European case, if the immediate payoff $X(t)$ is a deterministic function of the current underlying risky security¹, i.e. if there exists $f : \mathfrak{R} \rightarrow \mathfrak{R}$ such that $X(t) = f(S(t))$, then the no-arbitrage price of the American derivative is a deterministic function of time and the current price of the underlying security. We denote such function by $F : [0; T] \times \mathfrak{R} \rightarrow \mathfrak{R}$. This is so because

$$\begin{aligned} V(t) &= \operatorname{ess\,sup}_{t \leq \tau \leq T} \mathbf{E}^Q \left[f(S(\tau)) e^{-r(\tau-t)} \middle| \mathcal{F}_t \right] \\ &= F(t, S(t)) \end{aligned}$$

where

$$F(t, x) = \sup_{0 \leq \theta \leq T-t} \mathbf{E}^Q \left[f(x e^{(r-0.5\sigma^2)\theta + \sigma W^*(\theta)}) e^{-r\theta} \right].$$

The properties we discussed above for the value of the American option lead to the following characterization for the function F :

Proposition 11 (*The variational inequality problem*) *With respect to previous notations and definitions, the function F satisfies the following four statements*

1. $\frac{\partial F(t;S)}{\partial t} + \frac{\partial F(t;S)}{\partial S} \cdot Sr + \frac{1}{2} \frac{\partial^2 F(t;S)}{\partial^2 S} \sigma^2 S^2 - F(t, S)r \leq 0$ for any $(t, S) \in [0; T] \times \mathfrak{R}^+$
2. $\left(\frac{\partial F(t;S)}{\partial t} + \frac{\partial F(t;S)}{\partial S} \cdot Sr + \frac{1}{2} \frac{\partial^2 F(t;S)}{\partial^2 S} \sigma^2 S^2 - F(t, S)r \right) (F(t, S) - f(S)) = 0$ for any $(t, S) \in [0; T] \times \mathfrak{R}^+$
3. $F(t, S) \geq f(S)$ for any $(t, S) \in [0; T] \times \mathfrak{R}^+$
4. $F(T, S) = f(S)$ for any $S \in \mathfrak{R}^+$

¹If there are many underlying assets, i.e. when $X(t) = f(S_1(t), \dots, S_d(t))$, then set $f : \mathfrak{R}^d \rightarrow \mathfrak{R}$.

We comment briefly the statement just given. The first property describes the supermartingale property of $\tilde{V}(t) = F(t, S(t)) e^{-rt}$. In fact, $\tilde{V}$ is a Q -supermartingale if and only if its *drift* under the risk neutral measure Q is *negative*. If we apply Ito formula to differentiate $\tilde{V}(t) = F(t, S(t)) e^{-rt}$ we have that

$$\begin{aligned} d\tilde{V}(t) &= e^{-rt} \left(\frac{\partial F(t; S)}{\partial t} + \frac{\partial F(t; S)}{\partial S} S r + \frac{1}{2} \frac{\partial^2 F(t; S)}{\partial^2 S} \sigma^2 S^2 - F(t, S) r \right) dt + \\ &\quad + e^{-rt} \frac{\partial F(t; S)}{\partial S} S \sigma dW^*(t) \end{aligned}$$

In order $\tilde{V}$ to be a *supermartingale*, its conditional expectation must *decrease*. And it is so if and only if the *drift* is *negative*, that is if and only if the first inequality in the variational problem is satisfied.

The second property specifies that only two alternatives can occur. Either we are in the *continuation region* and $\tilde{V}$ is a Q -martingale, i.e. its *drift* under the risk neutral measure Q is *zero*, that is F satisfies the Black-Scholes PDE:

$$\frac{\partial F(t; S)}{\partial t} + \frac{\partial F(t; S)}{\partial S} \cdot S r + \frac{1}{2} \frac{\partial^2 F(t; S)}{\partial^2 S} \sigma^2 S^2 - F(t, S) r = 0.$$

Or we are in the *early exercise* region, but in this case the value of the American option equals the immediate payoff $F(t, S) = f(S)$.

The third property says that in any case the value of the American option dominates the immediate payoff.

The fourth and last property highlights the boundary condition at maturity, when the option has to be exercised.

In the variational inequality problem, as we see, the description of the continuation region and the early exercise region is *implicit*. The variational inequality problem can be tackled with numerical techniques such as finite difference schemes or finite elements techniques (see [5] and the references therein).

Another way to address the American option problem is to first describe the continuation region and the early exercise region and then impose the Black-Scholes PDE only on the continuation region. This approach leads to the free boundary problem. The free boundary is the boundary between the continuation region and the early exercise region. Its features depends on the payoff you consider, and on the parameters of the model. We focus on the case of the put option, where the immediate payoff is $f(S) = (K - S)^+$. It can be proved that the American put option price $F(t, S)$ inherits the

convexity with respect to the underlying S and the decreasing monotonicity property with respect to S from the payoff function f . Moreover F is decreasing with respect to t . Hence, since for any fixed $t \in [0; T]$,

$$\begin{aligned} F(t, 0) &= \sup_{0 \leq \theta \leq T-t} \mathbf{E}^Q \left[(K - 0e^{(r-0.5\sigma^2)\theta + \sigma W^*(\theta)})^+ e^{-r\theta} \right] \\ &= K \end{aligned}$$

either $F(t, S)$ is above $(K - S)^+$ for any $S > 0$ (i.e. early exercise is never optimal at t) **or** there exists a critical value $S^*(t) \in (0; K)$ such that $F(t, S) = (K - S)^+$ for $S \in [0; S^*(t)]$ and then $F(t, S) > (K - S)^+$ for all $S > S^*(t)$.

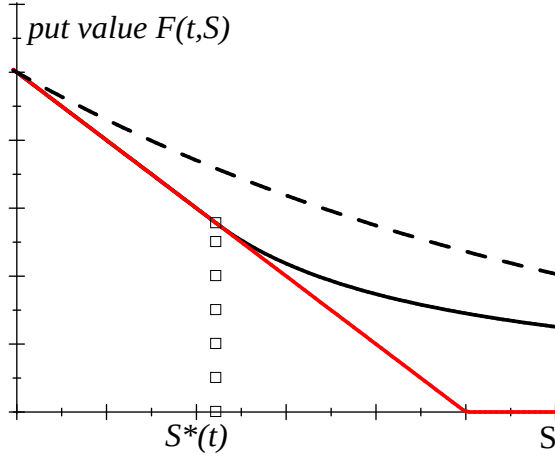


Fig.1: Finite-maturity put value

The first case is depicted in Figure 1 with the dashed black curve. As you see, in this case $F(t, S) > (K - S)^+$ for all S . The second case is represented in Figure 1 with the solid black curve. In this case the put value $F(t, S) > (K - S)^+$ only if S is sufficiently high. In other words, the American option is optimally exercised at date t if S is sufficiently *in the money*, i.e. $S \leq S^*(t)$.

The value $S^*(t)$ is called the *critical price* of S at t . It is the threshold under which it is optimal to exercise the option at t . Unfortunately, no analytical formula is available to compute S^* as a function of time (and the other parameters of the American option problem). This is why no closed formulae are available in the finite maturity case for plain vanilla put options. The graph of the unknown function $t \in [0; T] \rightarrow S^*(t)$ in the plane

(t, S) is called the *free boundary*. Some features of the free boundary can be deduced from the properties of the American option.

For instance, the value of the American put option F is decreasing in time. For this reason S^* *increases* as time goes by, as you can see from Figure 2.

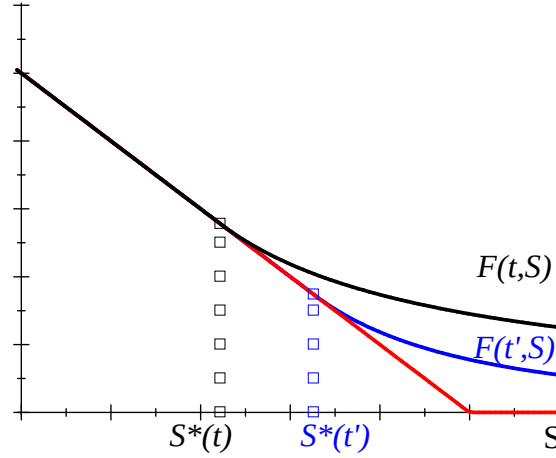


Fig.2: For $t < t'$ we have $S^*(t) \leq S^*(t')$

The graph of S^* as a function of time in the plane (t, S) is the (free) boundary that separates the early exercise region (below) from the continuation region (above). If the underlying asset² pays a dividend yield $q \leq r$ (or does not pay any dividend yield at all, i.e. $q = 0$), then the free boundary S^* is continuous on the whole interval $[0; T]$ and $S^*(T) = K$. On the contrary, if the underlying asset pays a *high* continuous dividend yield $q > r$, then the free boundary S^* displays a discontinuity at maturity, because $S^*(T^-) = \frac{r}{q}K < K = S^*(T)$. Figure 3 illustrates the qualitative behavior³ of the free boundary for the American put option when no high dividends are paid out by the underlying asset (i.e. $0 \leq q \leq r$).

²If the underlying asset $S(t)$ pays the dividend yield q , its *risk-neutral* dynamics is given by $\frac{dS(t)}{S(t)} = (r - q) dt + \sigma dW^*(t)$.

³An investigation of the asymptotic behavior of the free boundary near maturity can be found in the paper by J.D. Evans, R. Kuske and J.B. Keller (2002): American options on asset with dividends near expiry, *Mathematical Finance*, 12 (3), 219-237.

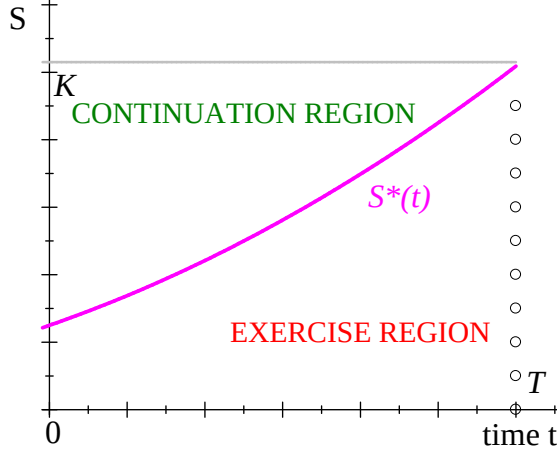


Fig.3: Free boundary, Continuation Region and Exercise Region.

We are now ready to state the free boundary problem for the American put option:

Proposition 12 *(The free boundary problem for the put option) With respect to previous notations and definitions, the function F that gives the price of the American put option satisfies the following statements*

1. $\frac{\partial F(t;S)}{\partial t} + \frac{\partial F(t;S)}{\partial S} \cdot Sr + \frac{1}{2} \frac{\partial^2 F(t;S)}{\partial S^2} \sigma^2 S^2 - F(t,S)r = 0$ for any $t \in [0; T], S > S^*(t)$
2. $F(t, S) = (K - S)^+$ for any $t \in [0; T], S = S^*(t)$ (value matching)
3. $\frac{\partial F(t;S)}{\partial S} = -1$ for any $t \in [0; T], S = S^*(t)$ (smooth pasting).
4. $F(T, S) = f(S)$ for any $S \in \mathbb{R}^+$

We comment briefly the statement just given. The first property describes the fact that in the continuation region the American option coincides with the European one, and therefore F satisfies the Black-Scholes PDE.

Property 2 and 3 describe the behavior of the American price where it begins to coincide with the immediate payoff. In particular, both the value

of the American option and the delta must coincide at the free boundary with the value of immediate payoff and its delta.

The free-boundary statement of the American put-option problem has a more *friendly* appearance than the variational inequality one. Are the two statements equivalent?

Properties 1. and 2. of the free-boundary statement follow by the variational inequality formulation. This is also the case for the smooth-pasting property, because F is a smooth function of the underlying.

Hence if F is a solution to the variational inequality problem, then F is also a solution of the free-boundary problem. Does the converse implication hold true? Fortunately, the answer is positive. Since the function F is convex with respect to the underlying S , properties 2. and 3. also imply that $F(t, S) \geq (K - S)^+$ for any $S \geq 0$. The only property that remains to be checked is that

$$\frac{\partial F(t; S)}{\partial t} + \frac{\partial F(t; S)}{\partial S} \cdot Sr + \frac{1}{2} \frac{\partial^2 F(t; S)}{\partial^2 S} \sigma^2 S^2 - F(t, S)r \leq 0$$

for any $t \in [0; T]$, $S \leq S^*(t)$, i.e. in the early exercise region.

Because in this region $F(t; S) = (K - S)^+$, it can be verified that also this (variational) inequality is satisfied.

Hence the two formulations are perfectly equivalent for put options.

When the maturity is infinite, the free boundary problem leads to an explicit solution. The key point is that in the perpetual case, the option value does not depend explicitly on t because

$$F(t, S) = \sup_{0 \leq \theta \leq \infty} \mathbf{E}^Q \left[f(S e^{(r - 0.5\sigma^2)\theta + \sigma W^*(\theta)}) e^{-r\theta} \right] = F(S).$$

Hence there is also a constant critical price S_∞^* . By imposing an educated guess on F of exponential form

$$F(S) = AS^\alpha \text{ for } S > S_\infty^*$$

the Black-Scholes PDE in the continuation region delivers the unknown parameter α . Value matching and smooth pasting deliver A and S_∞^* .

We derive the explicit formula in the following

Proposition 13 (*The American Perpetual Put Option*) *The no-arbitrage price of the perpetual American put option is given by the function*

$$F_\infty(S) = \begin{cases} AS^\alpha & \text{for } S > S_\infty^* \\ K - S & \text{for } S \leq S_\infty^* \end{cases}$$

where

$$\begin{aligned}\alpha &= \frac{-\left(r - \frac{\sigma^2}{2}\right) - \sqrt{\left(r - \frac{\sigma^2}{2}\right)^2 + 2r\sigma^2}}{\sigma^2} \\ &= \frac{-\left(r - \frac{\sigma^2}{2}\right) - \sqrt{r^2 + \sigma^2 r + \frac{\sigma^4}{4}}}{\sigma^2} < 0\end{aligned}$$

The critical asset price is

$$S_\infty^* = -\frac{\alpha}{1-\alpha}K < K \quad \text{for } i = 1, 2,$$

and the constant A is given by

$$A = \frac{(S_\infty^*)^{1-\alpha}}{-\alpha} > 0$$

Proof. By imposing condition 1. of the free boundary formulation on the function $F(S) = AS^\alpha$ we get

$$\underbrace{0}_{\frac{\partial F(t;S)}{\partial t}} + \underbrace{A\alpha S^{\alpha-1}}_{\frac{\partial F(t;S)}{\partial S}} \cdot Sr + \frac{1}{2} \underbrace{A\alpha(\alpha-1)S^{\alpha-2}\sigma^2 S^2}_{\frac{\partial^2 F(t;S)}{\partial^2 S}} - AS^\alpha r = 0$$

that leads to

$$\frac{\sigma^2}{2}\alpha^2 + \left(r - \frac{\sigma^2}{2}\right)\alpha - r = 0$$

Such equation has two solutions, because the determinant is always positive:

$$\alpha_{1,2} = \frac{-\left(r - \frac{\sigma^2}{2}\right) \pm \sqrt{\left(r - \frac{\sigma^2}{2}\right)^2 + 2r\sigma^2}}{\sigma^2}$$

These solutions have opposite sign. Only

$$\alpha = \frac{-\left(r - \frac{\sigma^2}{2}\right) - \sqrt{\left(r - \frac{\sigma^2}{2}\right)^2 + 2r\sigma^2}}{\sigma^2}$$

is acceptable, because it is negative (and makes the function F decreasing with respect to S).

By imposing value matching and smooth pasting we get the value of the other parameters:

$$\begin{aligned} F(t; S)|_{S=S_\infty^*} &= A (S_\infty^*)^\alpha = K - S_\infty^* \\ \frac{\partial F(t; S)}{\partial S} \Big|_{S=S_\infty^*} &= A\alpha (S_\infty^*)^{\alpha-1} = -1 \end{aligned}$$

We conclude this section by observing that the value of the perpetual American put option dominates the value of finite maturity American put

$$F(t, S) \leq F_\infty(S)$$

for any possible value of S and t . Therefore $S^*(t)$, the finite-maturity critical price at date t , is always above the perpetual critical price S_∞^* , i.e.

$$S^*(t) > S_\infty^* \text{ for all } t \in [0; T].$$

Chapter 4

Foreign markets and Quanto Options

In this section we analyze a simple foreign exchange market, using notations and arguments of [2]. This market model is quite simple: the exchange rate is assumed to be lognormal as well as the traded domestic and foreign risky securities. The lognormal assumption for the exchange rate is often criticized, because it is not realistic. However, this model allows us to perform the no arbitrage analysis of the market without too many technicalities. In particular, we are able to deduce the dynamics of the exchange rate and of the foreign risky security with respect to the domestic risk neutral measure. We also derive the covered interest rate parity that relates the forward exchange rate to the spot exchange rate.

4.1 The domestic and the foreign market

In the *domestic* market there are two securities:

1. the riskless security $B_d(t) = e^{r_d t}$ where r_d is the constant *domestic* riskless interest rate
2. the risky security $S_d = \{S_d(t)\}_{t \in [0, T]}$ described by

$$\frac{dS_d(t)}{S_d(t)} = \mu_d dt + \sigma_d^T dW^P(t)$$

where W^P is a $\mathfrak{R}^3$ -Brownian motion under the historical probability measure P with respect to the filtration $\mathbb{F}$ and σ_d is the $\mathfrak{R}^3$ -vector of volatilities.

In the *foreign* market there are two securities as well:

1. the riskless security $B_f(t) = e^{r_f t}$ where r_f is the constant *foreign* riskless interest rate
2. the risky security $S_f = \{S_f(t)\}_{t \in [0, T]}$ described by

$$\frac{dS_f(t)}{S_f(t)} = \mu_f dt + \sigma_f^T dW^P(t)$$

where W^P is the $\mathfrak{R}^3$ -Brownian motion of above and σ_f is the $\mathfrak{R}^3$ -vector of volatilities of the foreign security.

The two markets are connected via the *foreign to domestic exchange rate* X . For instance you may think at the US market as the foreign one. In this case, X is the *dollar to euros* exchange rate. Assume that X is lognormal and driven by

$$\frac{dX(t)}{X(t)} = \mu_X dt + \sigma_X^T dW^P(t)$$

under the historical probability measure P (σ_X is an $\mathfrak{R}^3$ -vector).

4.2 The domestic risk-neutral dynamics of the domestic and foreign securities

In order to price any derivative sold in the domestic market, we need to know the probability distribution of its payoff under the *domestic risk neutral measure* Q . As we know, the *domestic* security S_d is driven under Q by

$$\frac{dS_d(t)}{S_d(t)} = r_d dt + \sigma_d^T dW^Q(t)$$

where W^Q is a $\mathfrak{R}^3$ -Brownian motion. But what about the *foreign* security S_f and the exchange rate X ?

Since the change of measure does not affect the volatilities, S_f and X are driven under the *domestic* risk neutral measure Q by

$$\frac{dS_f(t)}{S_f(t)} = \mu_f^Q dt + \sigma_f^T dW^Q(t)$$

4.2. THE DOMESTIC RISK-NEUTRAL DYNAMICS OF THE DOMESTIC AND FOREIGN SECURITIES

and

$$\frac{dX(t)}{X(t)} = \mu_X^Q dt + \sigma_X^T dW^Q(t)$$

We want to determine μ_f^Q and μ_X^Q . The task can be accomplished by observing that the *foreign bond converted in the domestic currency*

$$B_f^*(t) = B_f(t)X(t)$$

and the *foreign security converted in the domestic currency*

$$S_f^*(t) = S_f(t)X(t)$$

can be treated as (*risky*) *domestic securities*.

Hence, since

$$\begin{aligned} dB_f^*(t) &= d(B_f(t)X(t)) = (dB_f(t))X(t) + B_f(t)dX(t) + 0 = \\ &= B_f^*(t) \left[(r_f + \mu_X^Q) dt + \sigma_X^T dW^Q(t) \right] \end{aligned}$$

then

$$r_f + \mu_X^Q = r_d$$

that is

$$\mu_X^Q = r_d - r_f.$$

Hence, the instantaneous rate of growth of the exchange rate under the domestic risk-neutral measure is given by the difference of the *riskless domestic interest rate* and the *foreign one*. This equality delivers also the *covered interest rate parity*. In fact, the *forward exchange rate* F_0 , i.e. the number of units of domestic currency settled at $t = 0$ to receive one unit of the foreign currency at the maturity T , is determined by

$$E^Q [e^{-r_d T} (1 \cdot X(T) - F_0)] = 0$$

that implies the following relation between the forward and the spot exchange rate in terms of the domestic and foreign interest rates:

$$F_0 = E^Q [X(T)] = X(0) e^{\mu_X^Q T} = X(0) e^{(r_d - r_f)T}.$$

that is known as the covered interest rate parity.

Moreover, by forcing S_f^* to have drift r_d under the domestic risk neutral measure Q we get (by multidimensional Ito formula)

$$\begin{aligned} dS_f^*(t) &= d(S_f(t)X(t)) = (dS_f(t))X(t) + S_f(t)dX(t) + \sigma_f^T S_f(t)\sigma_X X(t)dt = \\ &= S_f^*(t) \left[(\mu_f^Q + \mu_X^Q + \sigma_f^T \sigma_X) dt + (\sigma_X^T + \sigma_f^T) dW^Q(t) \right] \end{aligned}$$

and therefore

$$\mu_f^Q + \mu_X^Q + \sigma_f^T \sigma_X = r_d$$

that is

$$\begin{aligned} \mu_f^Q &= r_d - (r_d - r_f) - \sigma_f^T \sigma_X = \\ &= r_f - \sigma_f^T \sigma_X \end{aligned}$$

4.3 Pricing a quanto option

We want to price an *European put option* on the *foreign* security S_f with maturity T *denominated in the domestic currency*. The terminal payoff is therefore

$$V(T) = (K - S_f(T) \cdot X(T))^+$$

so that the fair price of the option at $t = 0$ is

$$\begin{aligned} V(0) &= E^Q [e^{-r_d T} (K - S_f(T) \cdot X(T))^+] = \\ &= E^Q [e^{-r_d T} (K - S_f^*(T))^+] \end{aligned}$$

Since

$$\begin{aligned} dS_f^*(t) &= S_f^*(t) [r_d dt + (\sigma_X^T + \sigma_f^T) dW^Q(t)] = \\ &= S_f^*(t) [r_d dt + \sigma^* dB^Q(t)], \end{aligned}$$

where

$$\begin{aligned} (\sigma^*)^2 &= (\sigma_X^T + \sigma_f^T) (\sigma_X + \sigma_f) = \\ &= \sum_{i=1}^3 (\sigma_{X,i} + \sigma_{f,i})^2 \end{aligned}$$

and B^Q is one-dimensional Q -Brownian motion, $V(0)$ is the value of an European put option on a lognormal asset (whose drift under Q is the interest rate r_d), with initial value $S_f^*(0) = S_f(0)X(0)$ and volatility σ^* . We therefore apply the standard Black-Scholes formula for a put option with these parameters and get

$$\begin{aligned} V(0) &= e^{-r_d T} K \cdot N \left[\frac{1}{\sigma^* \sqrt{T}} \left(-\ln \frac{S_0 X_0}{K} - \left(r_d - \frac{(\sigma^*)^2}{2} \right) T \right) \right] - \\ &\quad - S_0 X_0 N \left[\frac{1}{\sigma^* \sqrt{T}} \left(-\ln \frac{S_0 X_0}{K} - \left(r_d + \frac{(\sigma^*)^2}{2} \right) T \right) \right] \end{aligned}$$

where $N()$ denotes as usual the cumulative distribution function of a standard normal random variable.

Chapter 5

The jump diffusion model

In this chapter we study a market model where the risky security has discontinuous trajectories. A Brownian motion drives the continuous evolution of the trajectories of the risky security, that jumps at random times with stochastic amplitude. This model was introduced by Merton¹ in 1976, and it is nowadays the basic model for all those securities that display jumps in the trajectories. The presence of the jumps in the trajectories of an asset increases its variance; hence a jump diffusive asset has fatter tails in its distribution than a lognormal one. A generalization of the jump diffusion model leads to the *marketed point processes* (see [6] for details) and to the Levy processes. Since these more general processes are also more complicated for a technical point of view, we restrict ourselves to Merton's model, that we study in details. We perform the no arbitrage analysis of the market, and discuss the market price of both the diffusive and the jump risk. Ito formula for jump diffusions is also presented at the end of this chapter to enable us to differentiate a smooth function of a jump diffusion security.

We use the same notations of [4], Chapter 3.5. These pages are in fact an integration to the description of the jump diffusion model you find in [4].

5.1 The Jump Diffusion model

The market is constituted by the riskless bond $B(t) = e^{rt}$ and the risky security driven by

$$dS(t) = S(t^-)(\mu dt + \sigma dW(t) + dJ(t)) \quad (5.1)$$

¹“Option pricing when underlying stock returns are discontinuous”, Journal of Financial Economics, 1976.

where J is a *jump process* given by

$$J(t) = \sum_{j=1}^{N(t)} (Y_j - 1) \quad (5.2)$$

with Y_j i.i.d. *positive* random variables and N is a *counting process*. W , N and Y_j 's are assumed to be independent.

$N(t)$ counts the number of jumps until t (t *included*). Jumps occur at the random times $\tau_1, \tau_2, \dots, \tau_j, \tau_{j+1}, \dots$ and $N(t)$, the number of jumps occurred before t is therefore

$$N(t) = \sup \{n : \tau_n \leq t\} \quad (5.3)$$

The process N is *right-continuous*:

$$N(t^+) = \lim_{s \rightarrow t^+} N(s) = N(t),$$

describing the fact that if a jump *occurs at* t , the value $N(t)$ includes this jump. Hence the arrival of a jump *at time* t produces a discontinuity in the trajectories of N from the left:

$$N(t^-) = \lim_{s \rightarrow t^-} N(s) \begin{cases} \neq N(t) & \text{if a jump occurs at } t \\ = N(t) & \text{if no jumps occur at } t \end{cases}.$$

In other words, there is no chance at time $t^- = t - dt$ to unveil the uncertainty about the occurrence of a jump at t . In fact, jumps describe the arrival of sudden shocks that produce discontinuities in the price process of S . On the contrary, the Brownian motion W affects continuously the price of S , and produces a random but continuous variation in its trajectories.

In the simplest case, the process N is a Poisson process.

Definition N defined in (5.3) is a **Poisson process with intensity** $\lambda > 0$ if the **waiting times between jumps** $\tau_1, \tau_2 - \tau_1, \dots, \tau_{j+1} - \tau_j, \dots$ are i.i.d. random variables, exponentially distributed with parameter² λ . It can be shown that

$$P[N(t) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$$

for any³ $n \in \mathbb{N}$.

²This means that $\mathbf{P}[\tau_{j+1} - \tau_j \leq t] = 1 - e^{-\lambda t}$ for $t > 0$ and $\mathbf{P}[\tau_{j+1} - \tau_j \leq t] = 0$ for $t \leq 0$.

³With the convention $N(0) = 0$.

In particular

$$\begin{aligned} E[N(t)] &= \lambda t \\ \text{Var}[N(t)] &= \lambda t \end{aligned}$$

and the probability of having a jump between $t^- = t - dt$ and t is

$$P[\text{a jump occurs between } t^- = t - dt \text{ and } t] = \lambda \cdot dt$$

that explains why λ is called the *intensity* of the Poisson process.

It can be also shown that a process with independent and stationary increments, right-continuous, non-decreasing, and whose jump amplitude is equal to 1 is a Poisson process. Hence, Poisson processes are special but important counting processes⁴.

The process J defined in (5.2) is a *compound Poisson process*. In a compound Poisson process, the number of jumps until t is still given by $N(t)$, but the *size* of the jumps of J is random and determined by a sequence of i.i.d. random variables. In our case, the size of the j^{th} jump of J is given by $Y_j - 1$.

In the next section we solve equation (5.1) by applying Ito formula for jump diffusions. Nevertheless, we provide here an intuitive argument that allows us to guess the solution of equation (5.1). To this aim, we observe that if S satisfies equation (5.1) then

- after the $(j-1)^{\text{th}}$ jump has occurred and before the j^{th} jump arrives, that is on the (random) time interval $[\tau_{j-1}; \tau_j[$

$$dS(t) = S(t)(\mu dt + \sigma dW(t))$$

- at τ_j the j^{th} jump arrives and produces the following variation of S

$$\Delta S(\tau_j) = S(\tau_j) - S(\tau_j^-) = S(\tau_j^-)(Y_j - 1)$$

Hence

$$S(\tau_j) = S(\tau_j^-) + \Delta S(\tau_j) = S(\tau_j^-) + S(\tau_j^-)(Y_j - 1) = S(\tau_j^-)Y_j.$$

⁴It is possible to obtain more general counting processes by taking a positive *deterministic time-varying* intensity $\lambda = \lambda(t)$: such processes are called *inhomogeneous Poisson processes*. And further if $\lambda = \lambda(t)$ is a positive *stochastic process*, we obtain the *Cox processes*.

Therefore, between jumps, S follows the standard lognormal dynamics. When the j^{th} jump occurs, S is multiplied by Y_j . Hence:

$$S(t) = S(0)e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W(t)} \left(\prod_{j=1}^{N(t)} Y_j \right)$$

with the convention $\prod_{j=1}^0 = 1$.

5.2 The Ito formula for jump diffusions

Equation (5.1) has been just solved following an intuitive approach. We derive here rigorously the solution of equation (5.1), exploiting the Ito formula for jump diffusions, that with respect to previous notations and definitions *sounds* like this:

Ito formula *Let $\varphi : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ continuously differentiable with respect to the first argument and twice continuously differentiable with respect to the second argument. Then*

$$\begin{aligned} d\varphi(t, S(t)) &= \frac{\partial \varphi(t; S(t^-))}{\partial t} dt + \frac{\partial \varphi(t; S(t^-))}{\partial S} \cdot S(t^-) (\mu dt + \sigma dW(t)) + \\ &\quad + \frac{1}{2} \frac{\partial^2 \varphi(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 dt + (\varphi(t, S(t)) - \varphi(t, S(t^-))) \Delta N(t) \end{aligned}$$

or in integral form

$$\begin{aligned} \varphi(T, S(T)) - \varphi(0, S(0)) &= \int_0^T \frac{\partial \varphi(t; S(t^-))}{\partial t} dt + \dots \\ &\quad \dots + \int_0^T \frac{\partial \varphi(t; S(t^-))}{\partial S} \cdot S(t^-) (\mu dt + \sigma dW(t)) + \dots \\ &\quad \dots + \int_0^T \frac{1}{2} \frac{\partial^2 \varphi(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 dt + \dots \\ &\quad \dots + \sum_{t \leq T} (\varphi(t, S(t)) - \varphi(t, S(t^-))) \end{aligned}$$

Remark *The last sum in Ito formula*

$$\sum_{t \leq T} (\varphi(t, S(t)) - \varphi(t, S(t^-)))$$

has only a finite number of terms. In fact, S jumps $N(T)$ times in the interval $[0; T]$ at the random times τ_j . Hence

$$\sum_{t \leq T} (\varphi(t, S(t)) - \varphi(t, S(t^-))) = \sum_{j=1}^{N(T)} \left(\varphi(\tau_j, S(\tau_j^-) Y_j) - \varphi(\tau_j, S(\tau_j^-)) \right)$$

We are now ready to solve equation (5.1).

If we set $\varphi(t, S(t)) = \log(S(t))$, Ito formula gives

$$\begin{aligned} \log(S(T)) - \log(S(0)) &= \int_0^T \frac{1}{S(t^-)} \cdot S(t^-) (\mu dt + \sigma dW(t)) + .. \\ .. + \int_0^T \frac{1}{2} \frac{(-1)}{(S(t^-))^2} (S(t^-))^2 \sigma^2 dt &+ \sum_{j=1}^{N(T)} \left(\log(S(\tau_j^-) Y_j) - \log(S(\tau_j^-)) \right) = \\ &= \mu T + \sigma W(T) - \frac{1}{2} \sigma^2 T + \sum_{j=1}^{N(T)} \log Y_j \end{aligned}$$

and by taking the exponential we get

$$\begin{aligned} S(T) &= S(0) e^{\mu T + \sigma W(T) - \frac{1}{2} \sigma^2 T + \sum_{j=1}^{N(T)} \log Y_j} = \\ &= S(0) e^{\mu T + \sigma W(T) - \frac{1}{2} \sigma^2 T} \prod_{j=1}^{N(T)} Y_j \end{aligned}$$

that was the solution derived intuitively before.

Thanks to the independence of W , N and Y_j we can easily compute *the mean and the variance* of $S(t)$ under P :

$$\mathbb{E}^P[S(t)] = S(0) e^{\mu t} \mathbb{E}^P \left[\prod_{j=1}^{N(t)} Y_j \right] = S(0) e^{\mu t} e^{\lambda t (\mathbb{E}[Y_1] - 1)}$$

$$\mathbb{E}^P[S^2(t)] = S^2(0) e^{(2\mu + \sigma^2)t} e^{\lambda t (\mathbb{E}[Y_1^2] - 1)}$$

If $\hat{S}$ is the price process of a lognormal security with drift $\hat{\mu}$ and volatility $\hat{\sigma}$, we have to set $\hat{\mu} = \mu + \lambda (\mathbb{E}[Y_1] - 1)$ to match the first moment of S .

The variance of the jump-diffusion S is

$$\text{Var}^P[S(t)] = S^2(0) \left(e^{(2\mu+\sigma^2)t} e^{\lambda t(\mathbb{E}[Y_j^2]-1)} - e^{2\mu t} e^{2\lambda t(\mathbb{E}[Y_1]-1)} \right)$$

It can be shown that, if we assign to the pure lognormal security $\hat{S}$ the volatility $\hat{\sigma} = \sigma$, then

$$\begin{aligned} \text{Var}^P[S(t)] &> \text{Var}^P[\hat{S}(t)] = \\ &= S^2(0) e^{2(\mu+\lambda(\mathbb{E}[Y_1]-1))t} (e^{\sigma^2 t} - 1) \end{aligned}$$

whenever $P[Y_j \neq 1] > 0$. This condition is obviously met, because if $P[Y_j = 1] = 1$ then $S(\tau_j) = S(\tau_j^-)Y_j = S(\tau_j^-)$, i.e. the stock price is unaffected by the arrival of the j^{th} jump.

A log-jump-diffusion process has therefore more variance than the lognormal process with the same mean instantaneous rate of return. This helps to describe *fat tails in the returns of S* , a stylized fact that is missed by the lognormal model.

By exploiting the properties of the conditional expectation it is also possible to prove that the discounted asset price $\tilde{S}(t) = S(t)e^{-rt}$ satisfies for any $s < t$

$$\mathbb{E}^P \left[\tilde{S}(t) \middle| \mathcal{F}_s \right] = \tilde{S}(s) e^{(\mu-r)(t-s)} e^{\lambda(t-s)\mathbb{E}[Y_1-1]}$$

5.3 The No Arbitrage analysis

We show in this section that the market is free of arbitrage but incomplete. We deduce this result by exploiting the second fundamental theorem of asset pricing, proving that there are many equivalent martingale measures for the discounted asset price $\tilde{S}(t) = S(t)e^{-rt}$.

To this aim, we need to know the impact of a change of measure on a jump diffusion process. Such an impact is described in the

Girsanov Theorem for jump diffusions (*very “light” version*⁵)

*Given θ and $\phi > 0$, there exists $Q \sim P$ whose density*⁶

$$L(T) = \frac{dQ}{dP}$$

⁵A rigorous statement of Girsanov theorem in the marketed point process framework can be found in “*Credit derivatives pricing models: models, pricing and implementation*” by P.J.Schoenbucher, Wiley, 2003.

⁶The $()^P$ emphasizes the reference to the measure P .

is given by

$$\frac{dL(t)}{L(t^-)} = -\theta dW^P(t) + (\phi - 1) (dN(t) - \lambda^P dt)$$

such that under Q

$$W_t^Q = W_t^P + \theta t$$

is a Q -Brownian motion and N_t is a Q -Poisson process with intensity

$$\lambda^Q = \phi \lambda^P$$

All processes are independent under Q . The conditional (to the arrival of a jump) distribution of the jump size of J , i.e. the distribution of $Y_j - 1$, is unchanged.

We are now ready to find the EMMs for $\tilde{S}$. We first compute the stochastic differential of the discounted asset price $\tilde{S}$ by applying Ito formula to the function $\varphi(t, S) = S \cdot e^{-rt}$. We get

$$\begin{aligned} d\varphi(t, S(t)) &= \frac{\partial \varphi(t; S(t^-))}{\partial t} dt + \frac{\partial \varphi(t; S(t^-))}{\partial S} \cdot S(t^-) (\mu dt + \sigma dW(t)) + \\ &\quad + \frac{1}{2} \frac{\partial^2 \varphi(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 dt + (\varphi(t, S(t)) - \varphi(t, S(t^-))) \Delta N(t) \\ &= -re^{-rt} S(t^-) dt + e^{-rt} S(t^-) (\mu dt + \sigma dW(t)) + 0 + (S(t)e^{-rt} - S(t^-)e^{-rt}) \Delta N(t) \\ &= -re^{-rt} S(t^-) dt + e^{-rt} S(t^-) (\mu dt + \sigma dW(t)) + 0 + e^{-rt} S(t^-) \Delta J(t). \end{aligned}$$

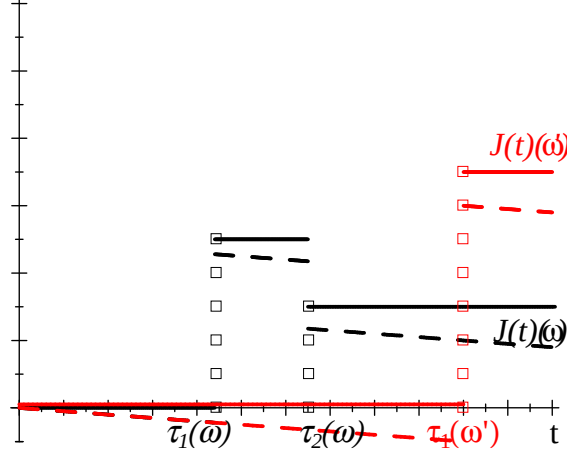
Hence the discounted risky security $\tilde{S}$ satisfies the SDE

$$\begin{aligned} \frac{d\tilde{S}(t)}{\tilde{S}(t^-)} &= (\mu - r) dt + \sigma dW^P(t) + \Delta J(t) = \\ &= (\mu - r) dt + \sigma (dW_t^Q - \theta dt) + \Delta J(t) \pm (\mu_Y - 1) \lambda^Q dt = \\ &= (\mu - r - \sigma \theta + (\mu_Y - 1) \lambda^Q) dt + \sigma dW_t^Q + (\Delta J(t) - (\mu_Y - 1) \lambda^Q dt) \end{aligned}$$

where $\mu_Y = \mathbb{E}^P[Y_j]$ and $\Delta J(t)$ emphasizes the *discontinuous nature* of the differential of J . The last term in the SDE

$$\Delta J(t) - (\mu_Y - 1) \lambda^Q dt$$

is the differential of a *pure jump martingale under Q* . The process $CJ(t) = J(t) - (\mu_Y - 1) \lambda^Q t$ is called the *compensated* compound Poisson process, and it is a pure-jump martingale.



Trajectories for J (solid) and CJ (dashed)
with $\mu_Y = 1.1$ and $\lambda^Q = 1$.

In fact,

$$\begin{aligned}
 \mathbb{E}^Q [\Delta J(t) | \mathcal{F}_{t-}] &= \mathbb{E}^Q [\Delta J(t) | \text{a jump occurs between } t^- \text{ and } t] \cdot \\
 &\quad \cdot Q [\text{a jump occurs between } t^- \text{ and } t] \\
 &= \mathbb{E}^Q [(Y_j - 1)] \cdot Q [\text{a jump occurs between } t^- \text{ and } t] = \\
 &= \mathbb{E}^Q [(Y_j - 1)] \cdot \lambda^Q dt = (\mu_Y - 1) \cdot \lambda^Q dt.
 \end{aligned}$$

Therefore, by recalling that $\lambda^Q = \phi \lambda^P$, we have that the *drift under Q* of $\tilde{S}$, or more precisely the term with non zero Q -conditional expectation in the differential $d\tilde{S}$, is

$$\mu - r - \sigma\theta + (\mu_Y - 1)\phi\lambda^P.$$

In fact

$$\begin{aligned}
 \mathbb{E}^Q \left[\frac{d\tilde{S}(t)}{\tilde{S}(t-)} \middle| \mathcal{F}_{t-} \right] &= (\mu - r - \sigma\theta + (\mu_Y - 1)\lambda^Q) dt + \\
 &\quad + \mathbb{E}^Q [\sigma dW_t^Q | \mathcal{F}_{t-}] + \mathbb{E}^Q [(\Delta J(t) - (\mu_Y - 1)\lambda^Q dt) | \mathcal{F}_{t-}] \\
 &= (\mu - r - \sigma\theta + (\mu_Y - 1)\lambda^Q) dt + 0 + 0.
 \end{aligned}$$

Hence $\tilde{S}$ is a Q -martingale if the parameters θ and ϕ kill the *drift under Q* of $\tilde{S}$:

$$\mu - r - \sigma\theta + (\mu_Y - 1)\phi\lambda^P = 0$$

This shows that there are infinitely many EMMs Q for $\tilde{S}$. By the second fundamental theorem of asset pricing, the market is free of arbitrage and incomplete.

The parameter θ is the analogous of the *risk premium* for the *diffusive risk* W in the pure diffusive framework. For any ϕ , the parameter θ is computed as follows:

$$\theta = \frac{\mu + (\mu_Y - 1)\phi\lambda^P - r}{\sigma}$$

The parameter ϕ describes the risk aversion of the investors to the *jump risk* represented by J . In his famous paper, Merton claims that the jump risk may be diversified: hence no risk premium is required for jumps. This means that $\phi = 1$ and the jump intensity of N is the same under P and Q . Merton's assumption

$$\phi = 1$$

and therefore

$$\theta = \frac{\mu + (\mu_Y - 1)\lambda^P - r}{\sigma}$$

is quite popular in some finance literature. By adopting this assumption, we select *a particular* EMM among the infinitely many we derived by imposing No Arbitrage. The chosen EMM corresponds to the assumption that investors do not require any risk premium for the jump risk. This is not always a proper assumption: for instance, in the evaluation of defaultable securities, where default is described as the arrival of the first jump of N , investors do require a risk premium for the default risk. When $\phi = 1$, the parameter θ is the ratio of the mean excess rate of return of the risky security S and its volatility. The mean excess rate of return of the risky security S is computed under the historical probability measure $\mathbf{P}$, and coincides with the mean excess rate of return of the lognormal security $\hat{S}$ (that is, jumps are considered only for their contribution to the mean rate of return of the security, by adding $(\mu_Y - 1)\lambda^P$ to the drift μ).

In general, if ϕ is free to vary in $\mathbb{R}^+$ and

$$\theta = \frac{\mu - r + (\mu_Y - 1)\phi\lambda^P}{\sigma}$$

the pair (θ, ϕ) determines an EMM Q under which

$$\begin{aligned} \frac{dS(t)}{S(t^-)} &= r dt + \sigma dW_t^Q + (\Delta J(t) - (\mu_Y - 1)\lambda^Q dt) = \\ &= (r - (\mu_Y - 1)\phi\lambda^P) dt + \sigma dW_t^Q + \Delta J(t) \end{aligned}$$

Repeating the same arguments that led to the solution of (5.1) we get

$$S(t) = S(0)e^{\left(r - (\mu_Y - 1)\phi\lambda^P - \frac{\sigma^2}{2}\right)t + \sigma W^Q(t)} \left(\prod_{j=1}^{N(t)} Y_j \right)$$

We observe that under the EMM Q the parameter $\phi\lambda^P$ replaces λ^P under the historical probability measure P , and $r - (\mu_Y - 1)\phi\lambda^P$ replaces μ , whereas the volatility σ is the same both under Q and P . We can therefore exploit the computations performed under P to conclude that

$$\mathbb{E}^Q[S(t)] = S(0)e^{(r - (\mu_Y - 1)\phi\lambda^P)t} \mathbb{E}^Q \left[\prod_{j=1}^{N(t)} Y_j \right] = S(0)e^{rt}.$$

Note that this result is not surprising: each traded security grows under *any* EMM Q at the mean rate of return of the riskless security.

Moreover the second moment of the risky security under any EMM

$$\mathbb{E}^Q[S^2(t)] = S^2(0)e^{(2(r - (\mu_Y - 1)\phi\lambda^P) + \sigma^2)t} e^{\lambda\phi t(\mathbb{E}[Y_j^2] - 1)}$$

Finally, we remark that the “light” version of Girsanov theorem stated at the beginning of this section leads to price the *time component* associated to the jump risk. In fact, the expected number of jumps under Q in the period $[0; t]$ is $\mathbb{E}^Q[N(t)] = \phi\lambda^P t$, hence ϕ times the number of jumps expected under P . The jump size Y_j is not affected by the change of measure. More general versions of Girsanov theorem provide probability measures Q under which both the intensity and the jump size have a modified distribution. Such versions of Girsanov theorem are a proper tool to price also the *size component* associated to the jump risk⁷.

5.4 The price of an European call option

Under Merton’s assumptions we derive in this section the pricing formula for an European call option on S with maturity T and strike price K .

In particular assume that:

$$\begin{aligned} \phi &= 1 \\ \theta &= \frac{\mu - r + (\mu_Y - 1)\lambda^P}{\sigma} \end{aligned}$$

⁷See [6] for more general version of Girsanov theorem.

and

$$Y_j \sim \log normal(a, b^2)$$

Denote with $m = \mu_Y - 1$, $\lambda = \lambda^P$ and with X_j a sequence of i.i.d. random variables $X_j \stackrel{Q}{\sim} N(a, b^2)$ such that $Y_j \stackrel{Q}{\sim} e^{X_j}$.

Since

$$S(t) = S(0)e^{\left(r - m\lambda - \frac{\sigma^2}{2}\right)t + \sigma W^Q(t)} \left(\prod_{j=1}^{N(t)} Y_j \right)$$

then

$$\begin{aligned} S(T)|_{N(T)=n} &= S(0) \left(\prod_{j=1}^n Y_j \right) e^{\left(r - m\lambda - \frac{\sigma^2}{2}\right)T + \sigma W^Q(T)} = \\ &= S(0) \cdot \exp \left(\left(r - m\lambda - \frac{\sigma^2}{2} \right) T + \sum_{j=1}^n X_j + \sigma W^Q(T) \right) = \\ &= S(0) \cdot \exp \left(\left(r - m\lambda - \frac{\sigma^2}{2} \right) T + na + \sigma_n \sqrt{T} Z' \right) \end{aligned}$$

where $Z' \stackrel{Q}{\sim} N(0, 1)$ and $\sigma_n^2 T = \sigma^2 T + nb^2$.

Observing that $m + 1 = \mathbb{E}[Y] = \mathbb{E}[e^X] = e^{a+0.5b^2}$ and setting

$$r_n = r - m\lambda + \frac{n \log(1 + m)}{T}$$

we get

$$S(T)|_{N(T)=n} = S(0) \cdot \exp \left(r_n T - \frac{\sigma_n^2}{2} T + \sigma_n \sqrt{T} Z' \right)$$

Hence the price at time $t = 0$ of the European call option on S is

$$\begin{aligned} c_0 &= \mathbb{E}^Q \left[(S(T) - K)^+ e^{-rT} \right] = \\ &= \sum_{n \in \mathbb{N}} e^{-\lambda T} \frac{(\lambda T)^n}{n!} \mathbb{E}^Q \left[(S(T) - K)^+ e^{-rT} \mid N(T) = n \right] = \\ &= \sum_{n \in \mathbb{N}} e^{-\lambda T} \frac{(\lambda T)^n}{n!} \mathbb{E}^Q \left[\left(S(0) \cdot \exp \left(r_n T - \frac{\sigma_n^2}{2} T + \sigma_n \sqrt{T} Z' \right) - K \right)^+ e^{-rT} \right] = \\ &= \sum_{n \in \mathbb{N}} e^{-\lambda' T} \frac{(\lambda' T)^n}{n!} \mathbb{E}^Q \left[\left(S(0) \cdot \exp \left(r_n T - \frac{\sigma_n^2}{2} T + \sigma_n \sqrt{T} Z' \right) - K \right)^+ e^{-r_n T} \right] \end{aligned}$$

where $\lambda' = \lambda(1 + m)$. Moreover

$$\mathbb{E}^Q \left[\left(S(0) \cdot \exp \left(r_n T - \frac{\sigma_n^2}{2} T + \sigma_n \sqrt{T} Z' \right) - K \right)^+ e^{-r_n T} \right]$$

is the price at time 0 of an European call option on a standard Black-Scholes risky asset with riskless interest rate r_n and volatility σ_n .

$$bsc(S_0, r_n, \sigma_n) = S_0 N(d_1) - K e^{-r_n T} N(d_2)$$

where $N(z)$ is the cumulative distribution function of a standard normal random variable and

$$d_1 = \frac{1}{\sigma_n \sqrt{T}} \left(\ln \left(\frac{S_0}{K} \right) + \left(r_n + \frac{1}{2} \sigma_n^2 \right) T \right)$$

and

$$d_2 = d_1 - \sigma_n \sqrt{T}.$$

Hence

$$c_0 = \sum_{n \in \mathbb{N}} e^{-\lambda' T} \frac{(\lambda' T)^n}{n!} bsc(S_0, r_n, \sigma_n).$$

5.5 A note on market incompleteness

As we have seen in the previous section, the price of a call option in the jump diffusion model depends on the specific choice of the risk-neutral probabilities, that in turn depends on the specific choice of the risk-neutral intensity of the Poisson process. The incompleteness of the jump-diffusion model is more severe than the incompleteness of a multivariate Black-Scholes model with more risk factors than risky securities. In the latter case, any European derivative whose final payoff depends only on the final values of the traded assets $S_1, \dots, S_d$

$$X(T) = f(S_1(T), \dots, S_d(T))$$

can be replicated with a self-financing strategy. Indeed, if its no-arbitrage price is a smooth deterministic function of the current values of the underlying assets

$$V(t) = F(t, S(t)) = F(t, S_1(t), \dots, S_d(t)),$$

then it is possible to construct a locally risk-less and self-financing hedging portfolio π with a short position on F and $h_i(t)$ units of each underlying asset such that

$$d\pi(t) = \pi(t) \left((\dots) dt + 0 \cdot dW^P(t) \right).$$

In fact, Ito's lemma implies

$$\begin{aligned}
d\pi(t) &= -dF(t, S(t)) + \sum_{i=1}^d h_i (dS_i(t) + q_i S_i(t) dt) \\
&= \left(-\frac{\partial F(t; S)}{\partial t} + \sum_{i=1}^d \left(-\frac{\partial F(t; S)}{\partial S_i} \mu_i + h_i \mu_i + h_i q_i \right) S_i(t) - \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 F(t; S)}{\partial S_i \partial S_j} \sigma_i^T \sigma_j S_i S_j \right) dt + \\
&\quad + \sum_{i=1}^d S_i(t) \left(h_i - \frac{\partial F(t; S)}{\partial S_i} \right) \sigma_i \cdot dW^P(t)
\end{aligned}$$

And setting

$$h_i = h_i(t) = \frac{\partial F}{\partial S_i}(t, S(t))$$

we immediately get

$$\begin{aligned}
d\pi(t) &= \left(-\frac{\partial F(t; S)}{\partial t} + \sum_{i=1}^d \left(-\frac{\partial F(t; S)}{\partial S_i} \mu_i + h_i \mu_i + h_i q_i \right) S_i(t) - \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 F(t; S)}{\partial S_i \partial S_j} \sigma_i^T \sigma_j S_i S_j \right) dt + 0 \cdot dW \\
&= \left(-\frac{\partial F(t; S)}{\partial t} + \sum_{i=1}^d \frac{\partial F(t; S)}{\partial S_i} q_i S_i(t) - \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 F(t; S)}{\partial S_i \partial S_j} \sigma_i^T \sigma_j S_i S_j \right) dt
\end{aligned}$$

Since it is possible to construct an hedging portfolio for F , then F can be replicated and its price is unique. Finally, no-arbitrage implies that

$$d\pi(t) = \pi(t) r dt$$

In this way we get the following Partial Differential Equation that F must satisfy

$$\frac{\partial F(t; S)}{\partial t} + \sum_{i=1}^d \frac{\partial F(t; S)}{\partial S_i} \cdot S_i (r - q_i) + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 F(t; S)}{\partial S_i \partial S_j} \sigma_i^T \sigma_j S_i S_j = F(t, S) r$$

that together with the boundary condition

$$F(T, S(T)) = f(S(T))$$

On the contrary, in the jump diffusion model even a plain vanilla derivative as the European call option on S cannot be perfectly replicated, and it is impossible to find an hedging portfolio. Let's try to replicate the steps of

the diffusive markets, for a European derivative whose final payoff depends only on the jump-diffusive asset at maturity T ,

$$X(T) = f(S(T)).$$

As in the standard Black-Scholes model we look for a locally risk-less and self-financing hedging portfolio π with a short position on F and $h(t)$ units of the underlying asset such that

$$d\pi(t) = \pi(t) ((...) dt + 0 \cdot dW^P(t)) + 0 \cdot dJ(t)$$

We start imposing the self-financing constraint and then use Ito's lemma to compute dF :

$$\begin{aligned} d\pi(t) &= -dF(t, S(t)) + h(dS(t) + qS(t^-)dt) \\ &= -\left(\frac{\partial F(t; S(t^-))}{\partial t} dt + \frac{\partial F(t; S(t^-))}{\partial S} \cdot S(t^-) (\mu dt + \sigma dW(t)) + \frac{1}{2} \frac{\partial^2 F(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 dt + \right. \\ &\quad \left. + hS(t^-) (\mu dt + \sigma dW(t) + dJ(t) + qdt) \right) \end{aligned}$$

This time, setting

$$h = h(t^-) = \frac{\partial F}{\partial S}(t; S(t^-))$$

we get

$$\begin{aligned} d\pi(t) &= -dF(t, S(t)) + h(dS(t) + qS(t^-)dt) \\ &= -\left(\frac{\partial F(t; S(t^-))}{\partial t} dt + \frac{1}{2} \frac{\partial^2 F(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 dt + (F(t, S(t)) - F(t, S(t^-))) \Delta N(t) \right) \\ &\quad + \frac{\partial F}{\partial S}(t; S(t^-)) S(t^-) (dJ(t) + qdt) \\ &= \left(-\frac{\partial F(t; S(t^-))}{\partial t} + \frac{\partial F}{\partial S}(t; S(t^-)) S(t^-) q - \frac{1}{2} \frac{\partial^2 F(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 \right) dt - \\ &\quad - (F(t, S(t)) - F(t, S(t^-))) \Delta N(t) + \frac{\partial F}{\partial S}(t; S(t^-)) S(t^-) dJ(t) \end{aligned}$$

As we can see, π here is no longer locally riskless. In fact, π is locally riskless only if there are no jumps at t (from t^- to t), since then

$$- (F(t, S(t)) - F(t, S(t^-))) \Delta N(t) + \frac{\partial F}{\partial S}(t; S(t^-)) S(t^-) dJ(t) = 0.$$

But if on the contrary a jump arrives at $t = \tau_j$ with size $Y_j - 1$ for the process J then

$$- (F(\tau_j, S(\tau_j)) - F(\tau_j^-, S(\tau_j^-))) + \frac{\partial F}{\partial S}(\tau_j^-; S(\tau_j^-)) S(\tau_j^-) ((Y_j - 1)) \neq 0$$

The discrete jump of the derivative price at τ_j cannot be hedged by the jump of the underlying asset, as the “delta” of our derivative at τ_j^- captures only a portion of this discrete jump. Indeed, the “delta” of our derivative at τ_j^- is

$$\frac{\partial F}{\partial S}(\tau_j^-; S(\tau_j^-)) = \lim_{\varepsilon \rightarrow 1} \frac{F(\tau_j, S(\tau_j^-)\varepsilon) - F(\tau_j^-, S(\tau_j^-))}{S(\tau_j^-)(\varepsilon - 1)}$$

Since the jump size Y_j can be far from 1, the discrete jump in the derivative price cannot be fully hedged with our usual predictable strategy $h(t^-) = \frac{\partial F}{\partial S}(t; S(t^-))$.

Consequently, we cannot proceed any further with the constructive approach. Instead, we rely on the First Fundamental Theorem of Asset Pricing, that requires the existence of a risk-neutral probability measure for the market to be arbitrage free. If the market includes the derivative F , then NA implies that there exists a risk-neutral probability for the original market that makes the discounted price of the derivative, $F(t, S(t))e^{-rt}$, a martingale. This implies that when computing

$$\begin{aligned} d(F(t, S(t))e^{-rt}) &= e^{-rt}(-rF(t; S(t^-))dt + dF(t; S(t^-))) \\ &= \underbrace{(\dots)dt}_0 + \underbrace{(\dots)dW_t^Q}_{\text{Differential of Brownian Martingale}} + (\dots) \underbrace{(\Delta N(t) - \lambda^Q dt)}_{\text{Differential of Jump Martingale}} \end{aligned}$$

the coefficient of dt must be equal to zero. In case of dividends, with respect to any risk neutral measure S satisfies the equation

$$\frac{dS(t)}{S(t^-)} = (r - q)dt + \sigma dW_t^Q + (\Delta J(t) - (\mu_Y - 1)\lambda^Q dt)$$

$$\begin{aligned} d(F(t, S(t))e^{-rt}) &= \\ &= e^{-rt} \left(-rF(t; S(t^-))dt + \frac{\partial F(t; S(t^-))}{\partial t}dt + \frac{\partial F(t; S(t^-))}{\partial S} \cdot S(t^-) \left((r - q - (\mu_Y - 1)\lambda^Q)dt + \sigma dW(t) \right) \right. \\ &\quad \left. + \frac{1}{2} \frac{\partial^2 F(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 dt + (F(t, S(t)) - F(t, S(t^-))) \Delta N(t) \right) \end{aligned}$$

For sake of space we omit the arguments of $F(t; S(t^-))$ and its derivatives

$\frac{\partial F(t; S(t^-))}{\partial t}$, $\frac{\partial F(t; S(t^-))}{\partial S}$ and $\frac{\partial^2 F(t; S(t^-))}{\partial S^2}$, rewriting

$$\begin{aligned}
& d(F(t, S(t))e^{-rt}) = \\
& = e^{-rt} \left(-rFdt + \frac{\partial F}{\partial t}dt + \frac{\partial F}{\partial S} \cdot S(t^-) \left((r - q - (\mu_Y - 1)\lambda^{\mathbb{Q}}) dt + \sigma dW(t) \right) + \right. \\
& \quad \left. + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} (S(t^-))^2 \sigma^2 dt + (F(t, S(t)) - F(t, S(t^-))) \Delta N(t) \right) \\
& = e^{-rt} \left(-rF + \frac{\partial F}{\partial t} + \frac{\partial F}{\partial S} \cdot S(t^-) (r - q - (\mu_Y - 1)\lambda^{\mathbb{Q}}) + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} (S(t^-))^2 \sigma^2 \right) dt + \\
& \quad + e^{-rt} (F(t, S(t)) - F(t, S(t^-))) \Delta N(t) + e^{-rt} \frac{\partial F}{\partial S} \cdot S(t^-) \sigma dW(t)
\end{aligned}$$

In order to compensate the jump component

$$e^{-rt} (F(t, S(t)) - F(t, S(t^-))) \Delta N(t)$$

we evaluate its conditional expectation

$$\mathbb{E}_{t^-}^{\mathbb{Q}} [e^{-rt} (F(t, S(t)) - F(t, S(t^-))) \Delta N(t)].$$

We observe that its argument is different from zero only when $\Delta N(t) = 1$ i.e. when a jump arrives at t . Moreover N and Y_j are independent. Thus

$$\begin{aligned}
& \mathbb{E}_{t^-}^{\mathbb{Q}} [F(t, S(t)) - F(t, S(t^-)) \Delta N(t)] = \\
& = \mathbb{E}^{\mathbb{Q}} \left[F(\tau_j^-, S(\tau_j^-) Y_j) - F(\tau_j^-, S(\tau_j^-)) 1 \right] \cdot \mathbb{Q}_{t^-} [\text{a jump arrives at } t] \\
& = \mathbb{E}^{\mathbb{Q}} [F(t^-, S(t^-) Y_j) - F(t^-, S(t^-)) 1] \cdot \lambda^{\mathbb{Q}} dt
\end{aligned}$$

Hence $\mathbb{E}_{t^-}^{\mathbb{Q}} [d(F(t, S(t))e^{-rt})] = 0$ rewrites as

$$\begin{aligned}
& e^{-rt} \left(-rF + \frac{\partial F}{\partial t} + \frac{\partial F}{\partial S} \cdot S(t^-) (r - q - (\mu_Y - 1)\lambda^{\mathbb{Q}}) + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} (S(t^-))^2 \sigma^2 + \right. \\
& \quad \left. + \mathbb{E}^{\mathbb{Q}} [F(t^-, S(t^-) Y_j) - F(t^-, S(t^-)) 1] \cdot \lambda^{\mathbb{Q}} \right) dt = 0
\end{aligned}$$

since $e^{-rt} ((F(t, S(t)) - F(t, S(t^-))) \Delta N(t) - \mathbb{E}^{\mathbb{Q}} [F(t^-, S(t^-) Y_j) - F(t^-, S(t^-)) 1] \cdot \lambda^{\mathbb{Q}} dt)$ and $e^{-rt} \frac{\partial F}{\partial S} \cdot S(t^-) \sigma dW(t)$ have zero conditional expectation. We therefore obtain

$$\begin{aligned}
& -rF(t; S(t^-)) + \frac{\partial F(t; S(t^-))}{\partial t} + \frac{\partial F(t; S(t^-))}{\partial S} \cdot S(t^-) (r - q - (\mu_Y - 1)\lambda^{\mathbb{Q}}) + \\
& + \frac{1}{2} \frac{\partial^2 F(t; S(t^-))}{\partial S^2} (S(t^-))^2 \sigma^2 + \mathbb{E}^{\mathbb{Q}} [F(t^-, S(t^-) Y_j) - F(t^-, S(t^-)) 1] \cdot \lambda^{\mathbb{Q}} = 0
\end{aligned}$$

to be verified for any t and for any $x = S(t^-) > 0$. The last expected value becomes

$$\mathbb{E}^{\mathbb{Q}} [F(t^-, xY_j) - F(t^-, x)] = \int_0^{+\infty} (F(t, xy) - F(t, x)) f_Y(y) dy$$

where $f_Y(y)$ denotes the probability density of the random variable $Y_j > 0$.

Finally the PDE becomes

$$\begin{aligned} -rF(t; x) + \frac{\partial F(t; x)}{\partial t} + \frac{\partial F(t; x)}{\partial x} x \left(r - q - (\mu_Y - 1)\lambda^{\mathbb{Q}} \right) + \frac{1}{2} \frac{\partial^2 F(t; x)}{\partial x^2} x^2 \sigma^2 + \\ + \lambda^{\mathbb{Q}} \int_0^{+\infty} (F(t, xy) - F(t, x)) f_Y(y) dy = 0 \end{aligned}$$

to be verified for all $t \in (0, T)$ and for all $x > 0$.

5.6 Exercises

Let $B(t) = e^{rt}$ be the riskless bond and S be the risky security driven by

$$dS(t) = S(t^-)(\mu dt + \sigma dW(t) + dJ(t))$$

where J is a *jump process* given by

$$J(t) = \sum_{j=1}^{N(t)} (Y_j - 1)$$

with Y_j i.i.d. *positive* random variables and N is a *Poisson process* with intensity λ under the historical probability measure P . All the processes W , N and Y_j 's are assumed to be independent.

Suppose that $S(0) = 10$, $r = 6\%$, $\mu = 12\%$, $\sigma = 20\%$, $\lambda = 0.25$, $\mathbb{E}^P[Y] = \mu_Y = 0.8$, and $\text{Var}^P[Y] = 20\%$.

1. Compute the expected value of $S(t)$ at $t = 3$ under the historical probability measure P .
2. How many equivalent martingale measure Q do exist? Compute the expected value of $S(t)$ at $t = 3$ under each equivalent martingale measure Q . Does such expected value change, by changing Q ?

3. Find the no-arbitrage prices at time $t = 0$ for a European semiannual *power call* option on S with strike zero and maturity $T = 0.5$, whose terminal payoff is

$$X(T) = (S(T)^2 - 0)^+ = (S(T))^2$$

4. Suppose that the derivative of previous step is traded at the initial price $S_X(0) = 108.65$. Is the market arbitrage-free? What is the *risk-neutral* intensity of the Poisson process? And what is the risk premium for the diffusive risk W ?

Solution.

1. The expectation of the jump-diffusive risky asset S under the historical probability measure P is

$$\mathbb{E}^P[S(t)] = S(0)e^{(\mu + \lambda(\mu_Y - 1))t} = 10 \cdot \exp((0.12 + 0.25(0.8 - 1))3) = 12.337$$

Note that without jumps this expectation would be $10 \cdot \exp(0.12 \cdot 3) = 14.333$.

2. There exist infinitely many EMMs Q . The expectation of S under each EMM Q is

$$\mathbb{E}^Q[S(t)] = S(0)e^{rt} = 11.972$$

for *each* EMM Q : such expectation does not change by changing the EMM. In fact, we can see this also recalling that for each EMM Q the risky asset S is given by

$$S(t) = S(0)e^{\left(r - (\mu_Y - 1)\phi\lambda - \frac{\sigma^2}{2}\right)t + \sigma W^Q(t)} \left(\prod_{j=1}^{N(t)} Y_j \right)$$

where $\phi\lambda$ represents the intensity of the Poisson process under Q and W^Q is a standard Brownian motion under Q . Performing the same computations we made to derive the expectation of $S(t)$ under P we obtain

$$\begin{aligned} \mathbb{E}^Q[S(t)] &= S(0)e^{(r - (\mu_Y - 1)\phi\lambda)t} \cdot \mathbb{E}^Q \left[\prod_{j=1}^{N(t)} Y_j \right] = \\ &= S(0)e^{(r - (\mu_Y - 1)\phi\lambda)t} \cdot e^{\lambda\phi t(\mu_Y - 1)} = \\ &= S(0)e^{rt} \end{aligned}$$

for any $\phi > 0$, i.e. for any EMM Q .

3. The no-arbitrage price of the European derivative at time $t = 0$ is

$$S_X(0) = \mathbb{E}^Q \left[(S(T))^2 e^{-rT} \right]$$

Hence we have to compute the second moment of S under Q . We can reproduce the same steps we followed to derive $\mathbb{E}^P \left[(S(T))^2 \right]$. But there is also a shorter way to get the result: moving from P to Q , the drift μ of the risky asset S is replaced with $r - (\mu_Y - 1)\phi\lambda$, the volatility σ is unchanged, and the intensity of the Poisson process is multiplied by ϕ . The mean and the second moment of Y are unchanged. Hence, plugging the proper quantities into the formula for the second moment of $S(T)$ we get

$$\mathbb{E}^Q \left[(S(T))^2 \right] = S^2(0) \exp \left(\left(\underbrace{2(r - (\mu_Y - 1)\phi\lambda)}_{\text{replaces } \mu} + \sigma^2 \right) T + \underbrace{\lambda\phi}_{\text{replaces } \lambda} T (s_Y - 1) \right)$$

where $s_Y = \mathbb{E}^Q [Y^2] = \mathbb{E}^P [Y^2] = \text{Var}^P [Y] + \mu_Y^2 = 0.2 + 0.8^2 = 0.84$. Discounting, we get the no-arbitrage price of the derivative:

$$\begin{aligned} S_X(0) &= S^2(0) \exp \left\{ (r - 2(\mu_Y - 1)\phi\lambda + \sigma^2 + \lambda\phi(s_Y - 1)) T \right\} = \\ &= S^2(0) \exp \left\{ (r + \sigma^2 + \lambda\phi(s_Y - 2\mu_Y + 1)) T \right\} \end{aligned}$$

In the formula, all the parameters but ϕ are known. As ϕ varies among positive real numbers, we get the no-arbitrage prices of the derivative. The interval of no-arbitrage prices for the derivative is therefore

$$\left] S^2(0) e^{(r+\sigma^2)T}; +\infty \right[=]105.13; +\infty[$$

Inside the interval, we can find the no-arbitrage price determined under Merton's assumption of no risk premium for the jumps, i.e. $\phi = 1$. In particular in this case the price of the derivative is

$$10^2 \exp \left\{ (0.06 + 0.2^2 + 0.25(0.84 - 2 \cdot 0.8 + 1)) 0.5 \right\} = 108.33.$$

4. The price 108.65 for X at time zero does not allow for arbitrage opportunities in the market, because it is inside the interval of no-arbitrage prices: hence the market is still arbitrage-free. By inverting the formula previously deduced for the price of X :

$$\begin{aligned} S_X(0) &= S^2(0) \exp \left\{ (r - 2(\mu_Y - 1)\phi\lambda + \sigma^2 + \lambda\phi(s_Y - 1)) t \right\} \\ &= 108.65 \end{aligned}$$

we get $\phi = 1.1$. Hence the risk-neutral intensity is $\phi\lambda = 0.275$. Consequently, the risk premium for the diffusive risk W is

$$\begin{aligned}\theta &= \frac{\mu - r + (\mu_Y - 1)\lambda\phi}{\sigma} = \\ &= \frac{0.12 - 0.06 + (0.8 - 1)0.275}{0.2} = \\ &= 0.025\end{aligned}$$

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Dulcis in fundo

